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Ministry of Higher Education
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EpilepTech

**From Binary to Trinary: Enhancing Seizure Classification with AI for Epileptic,
Non-Epileptic, and Psychogenic Detection**

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ABSTRACT

Epilepsy is a complex neurological condition that impacts millions globally, often posing significant diagnostic challenges due to its clinical resemblance to Psychogenic Non-Epileptic Seizures (PNES). This project seeks to design an advanced diagnostic system that harnesses the capabilities of Deep Learning (DL) and Generative AI models to accurately distinguish between epileptic seizures and PNES through the analysis of EEG data. By automating EEG signal interpretation and integrating patient diagnostic histories, the proposed system aims to improve diagnostic precision while minimizing the likelihood of misdiagnosis. This innovative solution holds promise for enhancing patient care, optimizing healthcare workflows, and alleviating the strain on medical resources.

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LIST OF ABBREVIATIONS

Table 1: List of Abbreviations

AI	Artificial Intelligence
AUROC	Area Under The Receiver Operating Characteristic
CNN	Convolutional Neural Network
DL	Deep Learning
DTI	Diffusion Tensor Imaging
EEG	Electroencephalogram
iEEG	Intracranial Electroencephalogram
Iot	Internet of Things
KNN	K-Nearest Neighbors
LSTM	Long Short Term /Memory
ML	Machine Learning
MRI	Magnetic Resonance Imaging
PNES	Psychogenic Non-Epileptic Seizures
RF	Random Forest
RNN	Recurrent Neural Network
SOZ	Seizure Onset Zones
SVM	Support Vector Machine
VEP	Virtual Epileptic Patient
SDS	Software Design Specification
Se	sensitivity
Sp	specificity
SPMP	Software Project Management Plan
SPP	Spatial Pyramid Pooling
SRS	Software Requirement Specification
STP	Software Testing Plan
SVM	Support Vector Machine
LLM	Large Language Model

GAN	Generative Adversarial Network
WHO	World Health Organization

Chapter 1: Introduction

The following chapter gives a clear overview, focusing on the key aspects, and highlighting both the challenges and potential. It also dives into the complexities of why it's so important to accurately differentiate between these seizure types and the ongoing need to improve diagnosis. The chapter also lays out the main objectives, such as reducing misdiagnoses and enhancing accuracy. Finally, it walks the reader through the project's structure and methodology, offering insight into how the system is designed and developed.

1.1 Introduction

Neurological disorders, including epilepsy, hinder over 50 million individuals globally, making it one of the most common neurological conditions according to the World Health Organization (WHO) [1]. Recurrent seizures, which vary in frequency, intensity, and type, are the hallmarks of epilepsy and make it a truly challenging condition for diagnostics and treatments. Additionally, non-epileptic psychogenic seizures (PNES) which resemble epileptic seizures in their clinical representations are also a type that many people suffer from. These seizures are often triggered by psychological causes rather than abnormal brain functionalities. Misdiagnosis epileptic seizures for psychogenic seizures lead to inappropriate potential wellness-threatening treatments, causing increased healthcare costs and patient suffering.

Electroencephalogram (EEG) data is crucial for diagnosing epilepsy, their recordings capture electrical activity in the brain detect abnormal patterns to identify seizure episodes. Despite utilizing EEG, manual interpretation of its data is time-consuming and greatly susceptible to human error, as demonstrated by studies showing that misdiagnosis rates spike up to 20% in epileptic cases when solely relying on manual analysis [2]. These complications spurred the development of automatic solutions using progressive machine learning (ML) and deep learning (DL) techniques to improve diagnostic accuracy.

Recent Advancements in both DL and ML have performed exceptionally in medical diagnostics, particularly in analyzing large data, and complex datasets such as EEG recordings. A study by Acharya et al. [3] reported accuracy rates of up to 88% in detecting epileptic seizures using convolutional neural networks (CNNs) applied to EEG data, highlighting the viability of DL for detection. Additionally, studies by Hussein et al. [4] and Zhou et al. [5] have displayed that combining DL models with ML techniques such as support vector machines (SVM), achieve classification accuracy ranging over 90% for applied seizure detection.

Despite these advancements, it is still difficult to differentiate between PNES and epileptic seizures using the diagnostic methods available today. PNES and epilepsy are frequently mistaken as having similar clinical signs and seizure features. According to a 2019 study by LaFrance et al. stated that PNES was eventually discovered in 25–30% of patients who had been misdiagnosed with epilepsy. This emphasizes the alarming need for a more reliable diagnostic tool that can perform analysis on EEG data and Diagnosis History to differentiate between the two disorders.

This project seeks to fill the gap by developing and accessing a system that integrates deep learning (DL) and generative AI models to identify and classify epileptic, PNES and non-epileptic seizures using EEG data. Building on prior research, such as Liu et al. [7], that employed recurrent neural networks (RNNs) to analyze temporal EEG data with a 92.5% accuracy in seizure detection, this system aims to further enhance diagnostic precision. Additionally, it plans to utilize language models to incorporate clinical notes and Diagnosis History, potentially improving the distinction between different types of seizures. By automating EEG data analysis and integrating contextual information via language models, this system could enhance the reliability and speed of diagnosing seizure disorders, facilitating timely interventions. Deep learning techniques offers the potential to drastically cut down on the time needed for manual analysis while boosting diagnostic accuracy. This approach builds on findings from Thodoroff et al. [8], who achieved a 94% accuracy rate in epileptic seizure detection using deep learning models, demonstrating the promise of such technologies in medical diagnostics.

1.2 Problem Statement

Epilepsy is a complex brain condition that causes seizures. Around 50 million people around the world suffer from epilepsy. The disease affects not only the patients but also their families; the stigma and discrimination lead to social isolation that makes pursuing education and having a career more difficult. Many doctors and professional consultants have trouble determining epilepsy seizures via electroencephalography (EEG), a procedure that mandates training and is usually time-consuming. Reading an EEG complex operation can lead to incorrect diagnosis and treatment. This challenge restricts quick patient response and adds to medics' frustration since they may not have the essential resources and expertise. Using AI solutions to simplify this identification helps to improve patient outcomes and provides successful treatment. Creating a large language model (LLM) or short language model (SLM) can assist in professional consultation with doctors, patients, and anyone curious about epilepsy. We want to build a user-friendly platform that simplifies EEG reading and provides second opinions on treatment and diagnoses. Our goal is to develop an effective AI solution that overcomes the challenges of reading EEGs, allowing for faster recognition of epilepsy and improving patient care through trustworthy second opinions.

1.3 Motivation

- supporting Saudi Arabia's Vision 2030 by promoting medical technology innovation, which will enhance societal well-being and spur economic growth.
- giving medical personnel access to cutting-edge diagnostic equipment to enable more accurate and efficient disease identification to deliver exceptional patient care.
- lowering the burden of disease by promptly diagnosing conditions and taking appropriate action to stop them from getting worse and to improve general health.
- making efficient use of patient data to minimize problems, lower healthcare costs, and support the expansion of the national economy.
- enhancing diagnostic capacities to guarantee that all people have fair access to high-quality healthcare services and to improve the standard of patient care.

1.4 Justification

Epilepsy is one of the most prevalent neurological disorders, affecting millions globally, including a growing population in Saudi Arabia. The unpredictability and severity of seizures pose significant challenges for patients and healthcare providers alike. Early detection and real-time monitoring are crucial for improving patient outcomes. This project, aligning with Saudi Arabia's Vision 2030, aims to harness advanced Deep Learning, including Generative AI (GenAI), for early seizure detection. Traditional diagnostic methods often fail due to the difficulty of constant brain activity monitoring. By leveraging AI technology, the project seeks to reduce the social and healthcare burden of epilepsy, enhance patient care, and improve the quality of life for individuals with this chronic condition.

1.5 Aims & Objectives

The purpose of this research is to create an enhanced deep learning model for real-time detection of epileptic, PNES and non-epileptic seizures using EEG data, with the incorporation of Gen AI to improve seizure classification and resolve the imbalance data. This approach is intended to help medical professionals detect seizure patterns early and provide a second opinion, minimizing diagnostic complications. The model will be trained, verified, and deployed on a web-based platform for real-time monitoring. It will adapt to patient profiles to provide individualized seizure control, increasing accuracy over time while improving treatment results. The system intends to enhance the national healthcare infrastructure by offering comprehensive epilepsy monitoring solutions that enhance patient care.

1.6 Scope / Limitation of the Study

This study will focus on the clinical and demographic characteristics related with epilepsy and PNES in adults. The scope involves evaluating the patient's EEG data to detect any evidence of epilepsy or PNES and distinguishing between the two types. However, the study's limitations include potential recollection bias and epilepsy subcategories such as generalized seizures. Another drawback is data size, as PNES is uncommon and epilepsy seizure data is sparse when compared to non-epilepsy data. To preserve emphasis, genetic factors, social impacts, and pediatric cases are eliminated, potentially missing important variables impacting epilepsy management.

1.7 Social, Professional, Legal, and Ethical Implications

The anticipated result of the project is to increase consciousness and enable more early diagnoses and detection in order to avoid challenging scenarios, impacting many hospitals in the provision of their services. Our initiative is in line with Saudi Arabia's 2023 aim as well, where the country also has taken a specific interest in the implementation of new technologies for healthcare. The Saudi MOH advocates for this effort. In short, healthcare providers can provide clients with accurate diagnoses much faster and enhance the health outcomes in general by integrating technologies. Second, our project is not only in line with national targets but also sets Saudi Arabia as a leader in healthcare at regional level. It is a project that complies with regulations on privacy and works within established legal frameworks. We never use data

without consent, we make sure that data is de-identified (that no personal information could be extracted from it) to protect patient data. This way ensures the patient's privacy. Our intention with this approach is to provide the best and safest tools for the healthcare community.

1.8 Project Organization

The essential project management aspects are encompassed within the project structure, which is organized into eight main sections. Each section provides detailed information about specific parts and decisions of the process. The following are the key points of each phase:

❖ Chapter 1: Introduction

The chapter starts by defining the problem statement, which outlines the main issue being tackled. It then proceeds to provide the reasons and rationale behind the project. The goals and objectives are outlined, specifying what is aimed at being achieved. The study's scope and limitations are then examined. The chapter closes with a summary of the project's organization.

❖ Chapter 2: Background and Review of Literature

This chapter explores the literature related to epilepsy research, especially studies that utilize artificial intelligence techniques. Each piece examined describes the datasets used, the procedures used, and the results obtained.

❖ Chapter 3: Software Project Management Plans (SPMP)

This chapter describes the system's planning components, including its primary aims, scope, and objectives, as well as underlying assumptions, potential hazards, and limits. It also offers insights into the project's deliverables, timing, budget, and evaluation plans.

❖ Chapter 4: Methodology/Software Requirement Specification (SRS)

In this chapter, we examine the project's methodology and go into detail about the system. It defines the quantifiable requirements, including functional processes, data requirements, system interfaces, and non-functional or operational criteria.

❖ Chapter 5: Software Design Specification (SDS)

This chapter demonstrates our software architecture and structure. It shows how the components in the system interact with each other. It also explains how the system features can be used by users.

❖ Chapter 6: Implementation

This chapter discusses the project's execution, including related research, datasets used, and deep learning, machine learning approaches. It focuses on experiments conducted with datasets, followed by a discussion of the findings. The chapter closes with details on the final user interface.

❖ **Chapter 7: Software Testing Plan (STP)**

This chapter describes the testing strategy, including methods for ensuring the project's reliability and functionality.

❖ **Chapter 8: Conclusion**

This final chapter offers a recap of the entire document, highlighting key findings, contributions, and lessons. It also addresses any limitations and suggests directions for future improvements.

Chapter 2: Background and Review of Literature

This chapter provides an in-depth review of existing literature on the use of Artificial Intelligence (AI) techniques for the prediction and diagnosis of epilepsy. Each section offers a detailed summary of the methodologies, datasets, evaluation criteria, and findings of the reviewed studies. The literature spans various AI models, such as deep learning, machine learning, wearable technology, and large language models, highlighting recent advancements in AI-assisted healthcare. Furthermore, this review identifies critical gaps in current research, underscoring areas that require further investigation and development within the field.

2.1 Deep Learning-Based Seizure Detection and Prediction

2.1.1 Introduction

Deep learning has significantly transformed the detection of epileptic seizures by autonomously extracting features from raw EEG and video data, leading to more precise predictions compared to conventional approaches. Its ability to process and analyze large datasets with considerable flexibility has made it an indispensable tool in medical research, particularly for the early diagnosis and prediction of epilepsy. This technological advancement underscores its growing importance in improving the accuracy and efficiency of seizure detection methods.

2.1.2 Literature Review

A range of deep learning techniques has been introduced to enhance the detection and prediction of epileptic seizures. The subsequent section examines recent studies that employ deep learning models for these purposes, highlighting the innovations, methodologies, and outcomes of current research in this rapidly evolving area.

Hussein et al Look at the challenges in EEG signal classification, especially with noisy conditions [9]. They advance using feature extraction methods in frequency and time domains to get better accuracy. The data used is from Bonn University, and a deep recurrent neural network (LSTM) was implemented to achieve 99% accuracy under ideal conditions. However, the accuracy declines with noise conditions, mostly white noise.

USMAN et al, presented a deep-learning method to analyze abnormal brain activity to predict seizures [10]. They classify EEG into interictal and preictal states using CNN and SVM using the CHBMIT dataset and bandpass filtering to remove white noise to achieve 92.7% sensitivity and 90.8% specificity.

Yang et al. described a deep-learning video classification to detect tonic-clonic seizures. Having a video dataset of videos that were segmented into non-overlapping 5-second clips for examination [11]. Two methods were used (CNN and LSTM), and the best performance was executed with the CNN and LSTM model with 88% sensitivity and 92% specificity. The results indicate a possibility for real-time seizure detection in hospital and home settings.

Epileptic seizures are primarily monitored through EEG signals, and recent advances in machine learning (ML) and deep learning (DL) have significantly improved seizure detection accuracy [12]. Traditional ML models, such as Support Vector Machines (SVM) and k-Nearest Neighbors (KNN), have proven effective but face challenges with large, sequential EEG datasets. Long Short-Term Memory (LSTM) networks, designed for time-series data, offer superior performance by capturing long-term dependencies in EEG signals. In a study using the UCI Epileptic Seizure Recognition Dataset, an LSTM model achieved 97% accuracy, outperforming logistic regression, KNN, and SVM in both precision and recall. The study also employed a confusion matrix to validate the LSTM model's robustness in distinguishing between seizure and non-seizure events. The workflow involved EEG data acquisition, preprocessing, feature extraction, and classification, all contributing to accurate seizure detection. Future research aims to enhance real-time detection using larger datasets and wearable technology, positioning LSTM networks as a promising tool for automating epilepsy diagnosis and improving patient care outcomes.

A Systematic Review- by Sobhana Jahan et al. Provides a thorough analysis of the application of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT) technologies to detect and predict epileptic seizures. The study states that EEG (Electroencephalography) signals, a non-invasive method measuring electrical activity in the brain, are used extensively for seizure detection and prediction because of their great ability to monitor brain activity [13]. The paper reviews 56 research articles that cover different AI and IoT frameworks for epilepsy management published from 2010 to 2021. Various methodologies are used broadly classified under EEG-based machine learning models, deep learning models, and IoT frameworks. Among the machine learning models, including Support Vector Machines (SVM), Random Forest (RF), and k-nearest Neighbors (k-NN) that use feature extraction like Fast Fourier Transform (FFT) and Wavelet Transform (WT). Deep learning and IoT-based frameworks also contribute significantly to intelligent seizure detection and prediction, particularly in large-scale EEG datasets as well as real-time remote monitoring using connected devices transmitting EEG signals for continuous analysis. The datasets driving these frameworks are also derived from sources such as the CHB-MIT Scalp EEG Database and are comprised of seizure and non-seizure data. Our results indicate that CNNs and RNNs were able to detect seizures, a few minutes in advance of their occurrence at an overall (combined preictal, ictal, and post-ictal) sensitivity of around 81% and run at over 95 %accuracy on unseen data. Unfortunately, the study also highlights some areas that need work,

including issues with data quality, model generalization, explainability of AI models, and scalability of IoT frameworks. The authors call for future research to further standardize datasets, increase the interpretability of AI models, and tackle scalability issues to capitalize on these tools in a clinical setting.

The paper by Zurdo-Tabernero et al. [14] provides a comprehensive review of machine learning (ML) and deep learning (DL) techniques for epileptic seizure prediction, emphasizing the importance of accurate predictions in enhancing patient quality of life. EEG data, which captures brain activity, serves as the primary input for these predictive models. The review highlights critical steps in EEG signal analysis, such as feature extraction and selection, noting that irrelevant features can adversely affect model performance. Traditional feature selection methods like Principal Component Analysis (PCA) are contrasted with DL models, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, which autonomously extract features. The paper also examines studies involving both animal models and human subjects, with some models achieving up to 99% sensitivity. A key challenge discussed is the trade-off between prediction timing and accuracy, as earlier predictions tend to reduce sensitivity. The authors suggest that combining multiple techniques may enhance results, though it introduces added computational complexity. While ML and DL show considerable promise, significant challenges remain in their clinical application [14].

This paper presents a deep learning approach for predicting epileptic seizures, providing a valuable tool for patients with drug-resistant epilepsy [15]. Unlike traditional models that depend on hand-crafted features from EEG data and classifiers like Support Vector Machines (SVM), the authors introduce an end-to-end convolutional neural network (CNN) that processes raw EEG data directly. This method overcomes the limitations of manual feature extraction by enabling the model to capture both temporal and spatial patterns. Utilizing one-dimensional and two-dimensional convolutional kernels, CNN effectively extracts both local and global features. The model was tested on the Kaggle intracranial EEG and CHB-MIT scalp EEG datasets, achieving notable sensitivities of 93.5% and 98.8%, along with low false prediction rates of 0.063/h and 0.074/h. With area under the curve (AUC) scores of 0.981 and 0.988, the model demonstrated strong predictive accuracy. The end-to-end CNN structure significantly outperforms conventional methods by capturing a comprehensive representation of the pre-seizure state, enhancing prediction performance. This research highlights the potential of deep learning in advancing epileptic seizure prediction, reducing false positives, and improving clinical management for epilepsy.

Epilepsy, a common neurological disorder characterized by abnormal brain activity, is typically detected through EEG data, but current diagnostic methods face limitations in real-time seizure detection. In a study by Saric et al. [16], an FPGA-based system utilizing a feed-forward multi-layer neural network (MLP ANN) was developed to classify generalized and focal epileptic episodes. The system processed 822 EEG samples from the Temple University Hospital Seizure Detection Corpus, achieving a classification accuracy of 95.14% using 70% of the data for training and 30% for testing. The process involved EEG signal preprocessing, feature extraction through Continuous Wavelet Transform (CWT), and model implementation in Very High-Speed Integrated Circuit Hardware Description Language (VHDL). The FPGA design is flexible and reconfigurable, showing promise for application-specific integrated circuits (ASIC) and large-scale production. This method represents significant progress in real-time seizure detection, improving accessibility and diagnostic accuracy for epilepsy patients, although further work is needed to address hardware challenges and enhance performance for broader clinical use.

The reviewed studies highlight significant progress in epileptic seizure detection and prediction using deep learning techniques. However, many models face challenges in real-time applications and in managing noisy EEG data. Hussein et al. [9] achieved the highest accuracy at 99%, but performance declined in noisier environments. This study aims to address these limitations by enhancing model robustness for real-world conditions and integrating wearable technologies for continuous monitoring. Moreover, we emphasize the importance of explainability in deep learning models to promote clinical adoption.

2.2 Machine Learning-Based Seizure Detection

2.2.1 Introduction

Machine learning models have been widely employed for the detection of epileptic seizures. This section examines recent studies that have utilized machine learning techniques for seizure detection, focusing on the analysis of EEG signals. These studies explore various algorithms, methodologies, and datasets to improve the accuracy and efficiency of detecting seizure events, contributing to advancements in the field of epilepsy diagnosis and management.

2.2.2 Literature Review

The study explores the use of AI-enhanced wearable devices to detect epileptic seizures, offering a non-invasive alternative to traditional EEG systems [17]. The devices collected data like electrodermal activity (EDA), accelerometry (ACC), and blood volume pulse (BVP) from 166 patients across 13,254 hours, recording over 900 seizures. AI models, including CNN and CNN-LSTM, were employed to classify 28 seizure types, with the CNN-LSTM model showing strong results, particularly for generalized tonic-clonic seizures with 95% sensitivity. The fusion of ACC and BVP data achieved an overall sensitivity of 83.9%. Personalized models using autoencoders further improved detection for certain seizure types, demonstrating the system's adaptability. Despite promising results, further refinement and larger datasets are needed to reduce false positives and enhance accuracy, particularly for real-world applications in continuous monitoring.

The paper "Machine Learning for Precision Epilepsy Surgery" by Lara Jehi [18] investigates how machine learning (ML) can enhance the outcomes of epilepsy surgery, particularly for patients with drug-resistant epilepsy. While brain surgery offers the highest likelihood of long-term seizure freedom, only about 50% of patients remain seizure-free following the procedure. The study highlights how ML can optimize surgical decision-making by analyzing complex presurgical data, allowing for a more personalized approach. Traditional ML techniques, such as Support Vector Machines (SVM), are applied to structured data like EEG and imaging, while deep learning models process unstructured data, such as raw images, though they require larger datasets to be effective. The research incorporates scalp EEG, structural MRI, functional MRI (fMRI), and diffusion tensor imaging (DTI) to form a comprehensive multimodal evaluation for surgical planning. SVM classifiers using MRI and DTI data achieved 85% sensitivity and specificity in predicting surgical outcomes, while neural networks showed high accuracy in predicting outcomes for temporal lobe epilepsy (TLE) patients. Nonetheless, the study identifies key challenges, including small dataset sizes and the limitations of single-modality models. It advocates for future strategies such as the development of multimodal ML models, the application of transfer learning, and the establishment of multicenter collaborations to collect more extensive datasets, which could enhance the generalizability and reliability of the models. The study underscores ML's potential to revolutionize epilepsy surgery but emphasizes the need for integrated multimodal data and broader datasets to ensure robust clinical predictions.

Yazdanpanah et al. [19] discuss the use of big data and AI in epilepsy research, focusing on advancements in seizure detection, seizure onset zone localization, and prediction. The paper highlights how common data models (CDMs) and AI/ML techniques (e.g., CNNs, LSTMs) are applied to EEG datasets to enhance diagnostic and predictive accuracy. The study explores two key technological areas: 1) big data analytics through CDM standardization of EMRs and 2) AI/ML models applied to continuous EEG for improved seizure detection and forecasting. Using datasets like CHB-MIT and Freiburg Hospital, CNN-based models achieved up to 99.02% accuracy for seizure detection, and CNN+LSTM models reached 98% sensitivity.

Seizure forecasting models showed up to 81% sensitivity with low false-positive rates. Despite these promising results, challenges such as data variability, false alarms, and CDM scalability remain significant barriers to clinical adoption. The paper suggests that integrating EEG data with AI could revolutionize epilepsy care, though more work is needed to address these obstacles.

This research paper "Application of Artificial Intelligence (AI) and Machine Learning (ML) in Pediatric Epilepsy: A Narrative Review by Kim, and Hwang 2022 investigated the potential use-case application of AI and ML technologies for pediatric epilepsy which is a neurological disease that could impact variously on the severity aspect from one patient to another [20]. Artificial intelligence and machine learning provide pediatricians with cutting-edge tools for clinical decision support, more accurate forecasting of seizures, and treatment that is more specific to each patient. In this paper, we provide a review of different methods such as supervised learning in seizure detection or classification based on labeled EEG and MRI data, and unsupervised learning for the innovation discovery from huge datasets of EEG. Deep Learning Models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are used in image, time-series data etc tasks. Further, the paper discusses models that combine different data types -EEG, MRI, and clinical record- to improve diagnosis and predictions of outcomes in pediatric epilepsies. These data sets being analyzed in the reviewed studies were EEG data for seizure detection, MRI scans to detect brain lesions, and Electronic Health Records (EHR) to classify epilepsy types and treatment outcomes prediction. The results demonstrate that deep learning models trained on EEG data can accurately detect seizures with over 95% accuracy and forecast seizures many minutes before they begin, therefore increasing the safety of the patients. In addition, patients with epilepsy are a particularly high-priority population to use these AI models trained on MRI data for identifying the epileptogenic zone and aiding in surgical planning as well. While this is a step in the right direction, issues around data quality, model generalization, and interpretability persist. The review states that while AI and ML show promise in changing how pediatric epilepsy is managed, better quality data sets and model transparency are necessary to increase clinical acceptance of these models and improve patient outcomes.

The main aim of this paper is to present a survey on various concepts, techniques, and challenges in the field of electroencephalogram (EEG) based seizure detections relating to epilepsy and the future trend of EEG Seizure detection [21]. The review explores such classic (traditional) Machine Learning (ML) and deep learning (DL) methods for seizure detection in an automatic manner with the superiority of automated systems in classification speed and accuracy to that of manual EEG assessment by clinicians. An insight into traditional ML techniques such as SVM, RF, and k-NN with feature extraction methods like FFT and DWT. We also examined modern DL approaches such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) and applied 1D-CNN on raw EEG data, and 2D-CNN on spectrogram data that can take care of spatial features. Mostly suited for temporal pattern recognition, hybrid architecture such as SeizureNet and Temporal Graph Convolutional Networks (TGCN) were introduced to integrate CNNs with other approaches in searching for

better spatial and temporal processing. We review datasets like the CHB-MIT Scalp EEG Database, Freiburg, Kaggle Epilepsy Dataset, and Bonn EEG Databases. DL models, in particular CNNs and RNNs, outperform classical ML methods at high accuracy (up to 98%), but challenges still arise when minimizing noise/errors in the datasets or handling imbalance among classes; and when translating results from one dataset of patients to another. Improvements in the future will be made by using better preprocessing techniques, and data augmentation to alleviate imbalance and construct real-time, light-weight models for wearable devices. The study concludes with a potential way forward that mixing DL into IoT platforms for remote monitoring and using hybrid models can improve seizure detection systems employed in clinical practice.

The paper "*Application of Machine Learning in Epileptic Seizure Detection*" by Tran et al. [22] explores the use of machine learning (ML) techniques to detect epileptic seizures through EEG data, with a focus on improving diagnostic accuracy and computational efficiency. Using a public EEG dataset from the University of Bonn, the authors extracted features with discrete wavelet transform (DWT) and applied binary particle swarm optimization (BPSO), reducing data dimensionality by 75%. Among several ML models evaluated, the random forest classifier performed best, achieving 98.4% accuracy. The study highlights other seizure detection approaches, such as Sharma et al.'s use of analytic time-frequency flexible wavelet transforms (ATFFWT) with least-squares SVM, which achieved 100% accuracy, and Siuly et al.'s clustering with LS-SVM, reaching 94.18% accuracy. Tran et al. emphasize the importance of optimizing feature selection and hyperparameters to develop fast and reliable models suitable for clinical use, potentially aiding neurologists and improving early seizure detection [22].

Dweiri Y. et al. [23] focus on developing a machine learning (ML) model for real-time epileptic seizure detection using single-channel EEG systems. Utilizing the CHB-MIT dataset, which includes EEG data from 22 epilepsy patients, the study extracted features such as kurtosis, energy, and differential entropy from the EEG signals. The XGBoost algorithm was employed to classify seizures from 1-second EEG segments, achieving an accuracy of 82%, a sensitivity of 89.21%, and a false negative rate (FNR) of 10.79%. This model shows promise for portable, wearable systems designed for continuous seizure monitoring, especially in home settings. It is particularly useful for detecting temporal lobe seizures through single-channel EEG placed near the ear, emphasizing the potential for real-time seizure detection outside of traditional clinical environments using wearable, near-ear devices [23].

The reviewed studies demonstrate substantial advancements in applying machine learning for epileptic seizure detection, particularly using wearable devices and multimodal data integration. However, these models struggle with generalizing across diverse datasets and delivering real-time predictions. Tran et al. [22] achieved the highest accuracy at 98.4%, but this was based on highly curated datasets, which may not capture real-world variability. This study seeks to address these challenges by enhancing data quality and model generalization, enabling real-time applications in wearable devices for continuous patient monitoring.

2.3 Explainability, Interpretability, and AI Ethics

2.3.1 Introduction

Explainability and interpretability are vital for using AI in healthcare, especially in seizure detection. This section reviews studies aimed at making AI models more transparent, focusing on providing patient-specific explanations. These efforts help clinicians better understand AI predictions, enhance trust and personalized care.

2.3.2 Literature Review

This paper presents a CNN-based model for predicting seizures and distinguishing between ictal and interictal phases using EEG signals [24]. The model was chosen over RNN due to better performance in previous studies, using Stochastic Gradient Descent (SGD) with a learning rate of 0.005. The hardware setup included AMD EPYC processors and a Nvidia Tesla T4 GPU, with training taking 99 minutes for 50 epochs. Post-processing used Bayesian and different filter methods to detect seizures, with optimal settings of 12.5-15 seconds for the signal window and a log-odds ratio of 0.8-0.9. Visualization techniques like gradient ascent were employed to understand how the network responded to EEG frequency patterns. The model performed comparably to human specialists and showed potential for use in wearable devices, though further research is needed to explore frequency components and decision probabilities for different patient sub-populations [24].

The paper "Estimating Patient-Level Uncertainty in Seizure Detection Using Group-Specific Out-of-Distribution Detection Technique" by Wong et al. [25] addresses the issue of patient-specific variability in seizure detection using machine learning (ML) models. While ML models have shown promise in identifying seizures from EEG data, variability among patients and limited, diverse training datasets often lead to unreliable predictions. To overcome this, the authors propose an ensemble method that quantifies uncertainty at the patient level, focusing on out-of-distribution (OOD) detection to flag uncertain predictions, especially with unfamiliar seizure patterns. Combining Deep Support Vector Data Description (SVDD) with convolutional neural networks (CNNs), the model reduces the risk of misdiagnosis by identifying when it is unsure. Achieving 87% accuracy, 89% specificity, and 75% sensitivity, the method outperforms traditional uncertainty techniques like Monte Carlo Dropout. Tested on the CHB-MIT EEG dataset, it effectively manages patient variability, offering a more reliable system for seizure detection. This approach enhances the safety and reliability of ML-based seizure detection, providing a valuable tool for clinical use [25].

The reviewed studies reveal notable progress in enhancing the interpretability of AI models for seizure detection, including CNN-based models with visualization techniques [24] and uncertainty quantification methods [25]. Despite these advancements, many models continue to struggle with patient-specific variability, resulting in inconsistent predictions. Wong et al. [25] achieved the highest accuracy at 87%, underscoring the need for models that can effectively manage diverse patient populations and seizure patterns. This study aims to overcome these limitations by improving model interpretability and robustness in varied clinical settings.

2.4 EEG-Based Seizure Detection

2.4.1 Introduction

EEG-based detection remains the most reliable method for diagnosing epileptic seizures. The following section reviews recent studies that apply machine learning and deep learning techniques to analyze EEG data for seizure detection, highlighting advancements in accuracy and efficiency.

2.4.2 Literature Review

The paper "Epileptic Seizure Detection by Using Interpretable Machine Learning Models" by Zhao et al. [26] explores the use of machine learning (ML) models for detecting epileptic seizures from EEG data, with a focus on interpretability for clinical applications. The study evaluates deep learning models such as LeNet, VGG, ResNet, and Vision Transformer (ViT), employing techniques like Grad-CAM and attention layers to enhance transparency in decision-making. A key issue of data imbalance between seizure and non-seizure events is addressed through a novel Random Channel Ordering (RCO) method, which improves dataset quality and model performance. ResNet emerged as the top-performing model, with Vision Transformer also demonstrating strong results. The incorporation of Grad-CAM and attention layers allows clinicians to better understand the model's decisions, mitigating concerns about deep learning models being "black boxes." The study demonstrates that interpretable ML models, paired with techniques like RCO, can not only detect seizures effectively but also provide valuable insights for clinical use [26].

The study introduces Epi-AI, a machine learning algorithm designed to detect seizures in EEG recordings from various epilepsy mouse models, including kainic acid-induced, Dravet syndrome, and pilocarpine-induced models. Epi-AI, which is based on the XGBoost algorithm, outperformed traditional methods like TKEO and TKEO-DWT, achieving sensitivities ranging from 91.4% to 98.8% and specificities between 93.1% and 98.8% across the three models. Even when tested on an unseen model, Epi-AI maintained a sensitivity of 76.3% and specificity of 98.1%, demonstrating superior generalization compared to conventional methods. This algorithm offers a fast, scalable solution for analyzing long-duration EEG recordings in preclinical studies, though further refinements are needed for broader clinical applications [27].

The reviewed studies highlight the promise of interpretable machine learning models in EEG-based seizure detection, with models such as ResNet and Epi-AI demonstrating strong performance [26, 27]. However, challenges persist in addressing data imbalances between seizure and non-seizure events and in ensuring models generalize effectively to unseen datasets. Epi-AI achieved the highest sensitivity at 98.8% [19], but further refinement is needed for widespread clinical application. This study aims to enhance both generalization and interpretability for real-world use.

2.5 Wearable and Real-Time Seizure Detection

2.5.1 Introduction

Wearable technologies have revolutionized the real-time monitoring of epileptic patients. This section reviews studies that leverage AI techniques in wearable devices for continuous, real-time seizure detection, highlighting advancements in accuracy, efficiency, and patient care through these innovations.

2.5.2 Literature Review

This study investigates the application of wearable devices enhanced with artificial intelligence (AI) for detecting various types of epileptic seizures [28]. Traditional seizure detection using EEG is often invasive and unsuitable for continuous monitoring, so the research focuses on wrist- and ankle-worn devices that measure electrodermal activity (EDA), accelerometry (ACC), and blood volume pulse (BVP). AI models, such as convolutional neural networks (CNN) and a hybrid CNN-long short-term memory (LSTM) network, were trained on data from 166 patients over 13,254 hours, with over 900 recorded seizures across 28 types. The combination of ACC and BVP data provided the highest sensitivity at 83.9% with a false positive rate of 35.3%, and the CNN-LSTM model excelled in detecting generalized tonic-clonic (GTC) seizures with 95% sensitivity. Additionally, the study explored the use of personalized detection models with autoencoders, showing improvement in detecting specific seizures like hyperkinetics and automatisms. These results highlight the potential of AI-driven wearables for non-invasive, continuous seizure detection. However, the researchers emphasize the need for larger datasets and further optimization to reduce false alarms and improve the system's real-world applicability [28].

Rai P. et al in this paper evaluate a contactless Artificial Intelligence (AI) based video system, Nelli, capable of detecting epileptic seizures [29]. Involving 230 patients with a total of 1,114 video-captured seizure events with 334 motor seizures with an effect lasting longer than 10 seconds, the system was tested across various environments. Nelli demonstrated accurate sensitivity, specifically for tonic-clonic seizures with an accuracy as high as 95.2% and hyperkinetic seizures with an accuracy of 92.9%, false detection rates (FDR) were measured to be 0.09 per hour for tonic-clonic seizures. Nelli was also assessed across various motor seizure types, psychogenic, and non-epileptic seizures. The system's performance

yielded persistent results with a total accuracy of 88% and an FDR of 6.48 per hour for the optimal measures threshold. However, the algorithm's detection sensitivity was challenged by environmental factors such as light conditions, and blanket occlusions. With the advancements provided by AI-based solutions, seizure monitoring could significantly improve in both clinical and home settings, reducing the workload on clinicians and improving patient treatment by facilitating real-time detection alerts for major seizures [29].

The reviewed studies demonstrate promising outcomes for AI-enhanced wearable and video systems in detecting epileptic seizures, achieving sensitivity rates of up to 95% for generalized tonic-clonic seizures [28, 29]. Despite these advances, real-world applications remain challenging due to high false-positive rates, sensitivity to environmental factors, and limited data availability. This study seeks to overcome these obstacles by enhancing the accuracy and robustness of wearable systems for continuous seizure monitoring.

2.6 Large Language Models (LLM) and Natural Language Processing (NLP) in Epilepsy

2.6.1 Introduction

Large language models and natural language processing (NLP) techniques are increasingly being applied in epilepsy research. The following section examines how these models aid in analyzing medical text data, enhancing the understanding of seizure disorders, and improving diagnosis and treatment outcomes.

2.6.2 Literature Review

Zhao et al explore the fine-tuning of LLMs for epilepsy knowledge in Japanese., with LLaMA (7B) and LLM-JP. Their model EpilepsyLLM addresses the problem of LLMs in the specialized medical field, by having epilepsy-related information gathered from Japanese sources using Stanford's Alpaca method. The study found that the model improved performance, especially in LLaMA (7B), but LLaMA (30B) did not excel despite adding more parameters. When comparing the model to medical-specific LLM such as BioGPT, they did not perform better than EpilepsyLLM due to lack of Japanese support. EpilepsyLLM showed promising results in both general and medical specific tasks [30].

Dani et al studied the use of NLP in the treatment of epilepsy, focusing on how it can improve patient identification and treatment options. The used dataset is Semio2Brain to evaluate the LLM capability to distinguish seizure onset zones (SOZ) from instructed text descriptions. The model achieved a performance level matching clinical evaluations with prompt engineering such as Self-Consistency. Drawbacks included citation errors and overconfidence. NLP and LLM are promising improvements in any medical diagnosis, especially for epilepsy. Careful implementation and detailed evaluation are essential for their full potential in clinical environments [31].

Xie K. et al. undertook this work to test a large language model (LLM) designed for epilepsy. The model was tested on a total of 25,612 patients, concentrating on characteristics such as race, ethnicity, gender, income, and insurance. The results revealed no substantial underlying bias in prediction accuracy and confidence across demographic groupings. The multivariable analysis found that female patients (OR 1.33, $P \leq 0.001$), those with public insurance (OR 1.53, $P \leq 0.001$), and those from lower-income locations (OR 1.22, $P \leq 0.007$) had worse seizure outcomes. Black patients (OR 1.03, $P=0.66$) had worse univariable outcomes than white patients, but the difference was not significant in the multivariable analysis. The findings underline the necessity for addressing inequities in epileptic care [32].

The study investigates EEG-GPT, a large language model designed to classify and interpret EEG signals, specifically distinguishing normal from abnormal patterns. Using the Temple University Hospital Abnormal Corpus, which contains 1140 hours of EEG data, EEG-GPT achieved an AUROC of 0.86 with only 2% of the training data, outperforming traditional machine learning models. The model also uses a Tree-of-Thought framework, allowing it to mimic the step-by-step analysis of epileptologists, improving interpretability. While EEG-GPT shows promise, the study highlights challenges such as its tendency to generate errors. Human oversight and structured verification are suggested to ensure clinical reliability. Further development is necessary to improve its accuracy, particularly in rare disease applications where data is limited [33].

This study examines the effectiveness of using large language models (LLMs) to simulate and analyze randomized controlled trials (RCTs), focusing on epilepsy treatment. The authors built a simulated RCT based on the anti-seizure drug cenobamate, utilizing 240 patients divided into drug and placebo arms [34]. Clinical notes were generated by the CHOCOLATES simulator and summarized using LLMs. The AI pipeline, consisting of multiple LLMs, closely mirrored human analysis, with less than 3% difference in detecting treatment efficacy and reported symptoms. LLMs like Llama2 and Mistral generated clinical notes and summaries, while Claude 2 synthesized the data into a structured table for statistical evaluation. The AI's assessment of seizure reduction closely matched that of human experts, demonstrating a marginal 1% difference in the 50% responder rate and median percentage change between drug and placebo arms. These results highlight the potential of AI in streamlining clinical trial analysis, offering faster and scalable alternatives to traditional methods. Despite these promising outcomes, challenges such as the introduction of noise and LLM hallucinations remain, requiring human oversight for accurate clinical assessments. Future improvements include expanding LLM capabilities for broader clinical applications and addressing limitations in detecting rare side effects. In conclusion, the study demonstrates the potential of LLMs to simulate clinical trials, accurately replicate treatment outcomes, and aid in inductive learning from medical records, which could revolutionize clinical research and decision-making processes [34].

The reviewed studies emphasize the promising potential of large language models (LLMs) and natural language processing (NLP) in epilepsy research, particularly for specialized medical tasks and analyzing patient demographics [30, 31, 32, 33, 34]. However, challenges such as LLM hallucinations, citation errors, and difficulty handling rare clinical cases persist. EEG-GPT achieved the highest AUROC at 0.86 [33], yet further refinement is necessary for broader clinical adoption. This study focuses on addressing these limitations by improving LLM accuracy and enhancing their interpretability for use in clinical settings.

2.7 General AI Applications in Epilepsy

2.7.1 Introduction

Artificial intelligence has been employed across multiple areas of epilepsy research. This section explores the wide-ranging applications of AI in the field, including advancements in diagnostics, seizure prediction, and patient phenotyping, demonstrating its transformative potential in improving epilepsy care.

2.7.2 Literature Review

The paper "Artificial Intelligence and Computational Approaches for Epilepsy" written by An, Kang, and Lee, provides an understanding of how machine learning (ML) and computational modeling have been integrated with epilepsy diagnosis and treatment, indicating the great progress in the precision of diagnosis, seizure type classification, as well as prognosis prediction due to AI-based techniques. The researchers focused on two main computational strategies: machine learning and biophysical modeling. Traditional supervised learning algorithms from ML such as SVM, k-NN, and Random Forest for classifying types of seizure using labeled EEG data while more advanced models from DL like CNN and RNN analyze time-series EEG Data to detect patterns in the data such as Seizure onset zones (SOZ). He discusses unsupervised learning, focusing on new ways of clustering seizure onset patterns devoid of prior knowledge. This talk is followed by research on biophysical modeling based on the Virtual Epileptic Patient (VEP) framework to predict the outcome of surgical intervention through simulation of seizure propagation in patient-specific LFP data streams. Using EEG data (and especially intracranial EEG [iEEG]) and integrating together other neuroimaging modalities like MRI and DTI—the study reveals models for seizure detection and prediction, and brain connectomics to predict where treatment will work best. CNNs could classify seizures with up to 97% accuracy; and RNNs, primarily Long Short-Term Memory (LSTM) networks, were able to predict seizures nearly hourly prior to the actual onset of a seizure with over 90% sensitivity. As also shown in our results, biophysical methods, such as VEP, had the best SOZ localizing accuracy and correlation with good surgical outcomes. The study successfully demonstrated these abilities, however, it also identified challenges coming from data heterogeneity, transparency, and generalizability by models. Moving forward, the authors recommend that work concentrates on increased generalization which happens when multiple modalities are combined (EEG data, MRI, and genetic data) to more comprehensive models. Collectively, the study indicates a momentous promise of AI and computational models for the effective reinventing of epilepsy diagnostics and therapeutics; however, future

progress critically relies on the overcoming of current limitations in data variation and model interpretability [35].

AI and machine learning (ML) have significantly advanced epilepsy management, enhancing diagnosis, seizure detection, classification, and personalized treatment. AI processes large datasets from EEG, MRI, and other tools to aid diagnosis and seizure prediction. For instance, Liu et al. (2021) used a hybrid CNN-RNN model to analyze EEG data with high accuracy, while Amarreh et al. (2021) employed diffusion tensor imaging and SVMs to distinguish between active epilepsy and remission. Daoud and Bayoumi (2019) developed a deep-learning model achieving 99.6% accuracy in seizure prediction, and Soriano et al. (2021) used neural networks and MEG data to classify seizures with 88% accuracy. AI also plays a role in personalized treatment, with Aslan et al. (2010) using ANN to predict seizure freedom with 91.1% accuracy and Grigsby et al. (1998) predicting outcomes of anterior temporal lobectomy. Wearable devices, such as the FDA-approved Neuropace, use AI to provide real-time seizure alerts and therapeutic interventions (Thomas & Jobst, 2015). Despite these advances, challenges remain, such as the need for large datasets and concerns about the "black box" nature of deep learning models (Nair et al.). Nevertheless, AI is revolutionizing epilepsy care by improving diagnostic and treatment outcomes, though further progress in data collection and computational techniques is necessary for broader adoption [36].

Knight A. et al. reviews the use of artificial intelligence in electro-clinical phenotyping on both animal and human epilepsy models. AI was able to reliably identify behavioral states in animal models, as well as the relationship between brain activity and interictal and ictal behaviors. In a clinical setting, artificial intelligence is utilized to automatically evaluate audio and video records for epileptic episodes, resulting in significant data reduction and accurate detection of big motor seizures. AI can also detect electrographic biomarkers including spikes, seizure patterns, and high-frequency oscillations. Integrating artificial intelligence with electroencephalography, behavioral, and clinical data may improve epilepsy therapy and prevention [37].

The reviewed studies show that artificial intelligence and computational models have made notable advances in epilepsy management, particularly in improving diagnostic accuracy, seizure prediction, and phenotyping [35, 36, 37]. Despite this progress, challenges persist related to data heterogeneity, model transparency, and generalizability. The highest accuracy, 99.6%, was reported by Daoud and Bayoumi [36], but further refinement is needed for these models to be effectively applied in diverse clinical settings. This study aims to address these limitations by enhancing model generalizability and integrating multimodal data for more comprehensive epilepsy care.

2.8 Other Approaches or Techniques

2.8.1 Introduction

Beyond mainstream AI techniques, novel methods emerge in epilepsy research. This section reviews alternative approaches, such as extreme value statistics, which are being applied to improve epilepsy diagnosis and seizure prediction, offering new insights and enhancing the accuracy of current models.

2.8.2 Literature Review

In this paper, Luca S. et al [38]. propose an unsupervised algorithm for detecting hypermotor epileptic seizures in children by accelerometers attached to extremities. Based on an extreme value theory, the method modeled normal behavior from unlabeled data, detecting abnormal movements as potential seizures. It was tested on data from 7 pediatric patients and was able to detect all hypermotor seizures in 5 of 7 patients, with a sensitivity score up to 90% and a positive predictive value (PPV) with a range of 21% to 89%. The proposed approach avoids the cost and computational complexity of supervised models that require expert analysis. In addition, it is comparable to supervised models but is more efficient for long-term at home monitoring, specifically for pediatric patients where nocturnal seizures are frequent. The system promises to develop patient specific in-home detection tools that require minimal to zero human intervention [38].

The reviewed study introduces an unsupervised algorithm for detecting hypermotor seizures in pediatric patients using accelerometers, showing promising results with sensitivity scores of up to 90% [38]. However, despite reducing the need for expert supervision, the positive predictive value (PPV) varies significantly across patients, ranging from 21% to 89%. This study aims to enhance the efficiency and predictive accuracy of unsupervised models for long-term at-home seizure monitoring, especially for pediatric patients. Table X provides a summary of the literature review outcomes.

2.9 Generative Adversarial Networks (GANs) and Epileptic Seizures

2.9.1 Introduction

Epileptic seizures, resulting from sudden electrical disruptions in the brain, are primarily monitored through electroencephalogram (EEG) data. To analyze and predict seizure activity effectively, machine learning models, including deep learning techniques, require large, high-quality datasets. However, such datasets are often limited in size, imbalanced, or fail to adequately represent certain seizure types. Generative Adversarial Networks (GANs) have emerged as a solution to these challenges by generating synthetic data to augment training datasets and capturing complex, subtle patterns within real EEG signals. This section examines four studies that leverage GANs and other advanced methodologies for improving seizure detection and prediction, focusing on their approaches, outcomes, and broader implications for the field.

2.9.2 Literature Review

Ahmed et al. [39] introduced a semi-supervised algorithm for epileptic seizure forecasting that leverages both labeled and unlabeled EEG data using Generative Adversarial Networks (GANs). EEG signals were transformed into two-dimensional matrices via Short-Time Fourier Transform (STFT), and a Deep Convolutional GAN (DCGAN) with three deconvolution layers was employed for feature extraction. This approach was tested on the CHB-MIT, Freiburg Hospital, and EPILEPSIAE datasets, achieving Area Under the Curve (AUC) scores of 77.68%, 75.47%, and 65.05%, respectively. Notably, the application of an up-sampling technique enhanced AUC scores by 2% and 0.7% on the EPILEPSIAE dataset. The study highlighted the model's scalability and particular efficacy for younger patients with focal-aware seizures, presenting a viable option for long-term monitoring with minimal labeled data.

Lee et al. [40] tackled the challenge of imbalanced preictal and interictal EEG data in seizure prediction through a novel data augmentation method called DiffEEG. This technique applies a two-step diffusion process: introducing noise to EEG samples and subsequently denoising them to generate synthetic data. By broadening data distribution and minimizing cluster distances, DiffEEG significantly improved prediction accuracy on the CHB-MIT dataset, which predominantly contains pediatric EEG data. Results demonstrated that DiffEEG outperformed existing augmentation techniques, providing an effective solution to imbalanced data challenges.

Kumar et al. [41] investigated the use of Graph Neural Networks (GNNs) for EEG data analysis, achieving an impressive 96% accuracy in seizure classification. GNNs excelled in modeling non-Euclidean relationships inherent in EEG data, enabling strong generalization to datasets from unseen patients. This capability underscores the potential of GNNs for real-time diagnostic applications, offering robust and reliable methods for both seizure classification and prediction.

Craley et al. [42] proposed a hybrid model combining Convolutional Neural Networks (CNNs) and Probabilistic Graphical Models (PGMs) for epileptic seizure detection using multichannel EEG data. By integrating a Coupled Hidden Markov Model (CHMM) to identify temporal patterns with CNNs for raw EEG signal processing, this approach achieved superior performance compared to standalone CNN models. The model demonstrated higher true positive rates, AUC, and F1 scores, although it exhibited a slightly increased false positive rate, highlighting the need for further optimization.

The reviewed studies illustrate the potential of advanced deep learning techniques in addressing significant challenges in epilepsy research. GANs effectively mitigate data scarcity and imbalance, with methods like DiffEEG further enhancing dataset quality and model performance. GNNs provide exceptional accuracy and generalization, while hybrid models like CNN-CHMM capture both spatial and temporal dynamics, improving classification metrics. Collectively, these innovations underscore the transformative role of machine learning in epilepsy research and point toward opportunities for further development and refinement.

Deep Learning-Based Seizure Detection and Prediction

Table 2:Deep Learning-Based Seizure Detection and Prediction Summary

N	Author/s	Year	Dataset	Technique/s	Results
[9]	Hussein et al.	2018	Bonn University EEG Dataset: Contains 5 sets of EEG signals (A to E) from healthy subjects and epilepsy patients. The data includes 100 single-channel EEG segments, each 23.6 seconds long.	LSTM	Achieved 99% accuracy in controlled environments with clean data, but performance dropped significantly with noise interference, highlighting the model’s sensitivity to real-world conditions.
[10]	Usman et al.	2020	The CHB-MIT EEG Dataset consists of 23 cases of scalp EEG data from pediatric epilepsy patients, recorded at the Children’s Hospital Boston. Each case includes EEG data featuring multiple seizure events, making it a valuable resource for developing and testing seizure detection algorithms.	CNN, SVM, Bandpass filtering	The model achieved 92.7% sensitivity and 90.8% specificity in detecting interictal and preictal phases, with bandpass filtering playing a key role in effectively reducing white noise interference, thereby improving the accuracy of seizure phase detection.
[11]	Yang et al.	2021	The Video Dataset consists of recorded videos of tonic-clonic seizures, which are segmented into non-overlapping 5-second clips. These clips are used for training and testing machine learning models aimed at improving seizure detection and analysis.	CNN, CNN-LSTM	The CNN-LSTM model achieved 88% sensitivity and 92% specificity, demonstrating significant potential for real-time seizure detection in both hospital and home environments. Its combination of convolutional and recurrent layers makes it particularly

					effective for processing time-series data like EEG recordings.
[12]	Jahan et al.	2023	The CHB-MIT Scalp EEG Database has been integrated with Internet of Healthcare Things (IoT) frameworks, allowing pediatric patients' EEG recordings to be continuously monitored. This integration facilitates real-time seizure detection and management, improving patient care through constant, remote observation.	CNN, RNN, IoT frameworks	The model achieved a sensitivity of 81% and an accuracy exceeding 95% on unseen data, demonstrating strong performance. However, scalability and real-time application in clinical settings still require further development to ensure broader practical use.
[13]	Kunekar et al.	2024	The UCI Epileptic Seizure Recognition Dataset contains 11,500 EEG samples, each lasting 23.6 seconds, collected from both healthy subjects and epilepsy patients. This dataset is widely used for developing and testing seizure detection models.	LSTM, Logistic Regression, KNN, SVM	The LSTM model achieved 97% accuracy, surpassing traditional machine learning techniques in seizure detection. However, challenges persist in adapting these models for integration with wearable technology to enable real-time monitoring and detection.
[14]	Zurdo-Tabernero et al.	2023	Multiple datasets combine EEG data from both animal and human epilepsy studies, incorporating intracranial and scalp EEG recordings from various sources. This diverse collection enhances the ability to develop more comprehensive models for seizure detection and	CNN, LSTM, PCA	The model achieved up to 99% sensitivity in seizure prediction, demonstrating high accuracy. However, challenges remain in balancing early prediction accuracy with the

			prediction across different experimental and clinical settings.		computational complexity required for real-time application and practical use.
[15]	Xu et al.	2020	Kaggle Intracranial EEG Dataset, CHB-MIT Dataset: Includes intracranial EEG recordings for seizure prediction tasks.	End-to-end CNN	The model achieved a sensitivity of 93.5% on the Kaggle dataset and 98.8% on the CHB-MIT dataset, with false prediction rates below 0.074 per hour. It also attained an area under the curve (AUC) score of 0.981, indicating strong predictive performance across both datasets.
[16]	Sarić et al.	2020	The Temple University Hospital Seizure Detection Corpus contains 822 EEG samples that have been processed using FPGA-based hardware for seizure classification. This dataset is utilized to enhance the efficiency and accuracy of seizure detection models.	Feed-forward Neural Network (MLP ANN)	Achieved 95.14% classification accuracy, with challenges related to hardware implementation noted for broader clinical application.

Machine Learning-Based Seizure Detection

Table 3:Machine Learning-Based Seizure Detection Summary

N	Author/s	Year	Dataset	Technique/s	Results
[17]	Vieira et al.	2023	CHB-MIT EEG Dataset: EEG recordings from pediatric epilepsy patients at Boston Children's Hospital, comprising 24 EEG cases of continuous EEG monitoring.	Explainable AI (XAI), Random Forest (RF), SVM	The model achieved 90% sensitivity and 87.3% specificity, with explainable AI (XAI) methods such as SHAP enhancing model transparency, making it easier for clinicians to interpret the model's decisions and improve its clinical applicability.
[18]	Jehi	2023	Multimodal EEG, MRI, and DTI Data: Pre-surgical data from epilepsy patients, integrating EEG, MRI, and diffusion tensor imaging (DTI) from temporal lobe epilepsy (TLE) cases.	SVM, CNN, RNN, Multimodal Learning	The model achieved 85% sensitivity and specificity in predicting post-surgical seizure freedom, with Convolutional Neural Networks (CNNs) performing particularly well for temporal lobe epilepsy prediction. However, challenges related to the limited size of available datasets remain an obstacle for broader application.

[19]	Chung et al	2021	CHB-MIT EEG, Freiburg Hospital Datasets: Large-scale EEG datasets including pediatric and adult epilepsy data, processed using a common data model (CDM).	CNN, LSTM, Data Analytics	CNN-based models achieved an impressive 99.02% accuracy for seizure detection, while LSTM-based forecasting models demonstrated 81% sensitivity for seizure prediction. The integration of Cross-Domain Modeling (CDM) further improved the scalability of these models, enhancing their potential for broader clinical use.
[20]	Kim and Hwang	2021	Pediatric EEG, MRI, and EHR Data: Pediatric patient datasets including EEG, MRI scans, and electronic health records (EHR) from pediatric epilepsy cases.	CNN, RNN, ML- based classifiers	Deep learning models achieved 95% accuracy in seizure detection, but issues with data generalization and transparency hinder widespread adoption. Integration of clinical data showed promise for personalized treatment.
[21]	Tran et al.	2022	Bonn University EEG Dataset: Single-channel EEG data from both healthy individuals and patients with epilepsy, used to extract temporal and frequency features.	Random Forest, BPSO, ATFFWT	Random forest models reached 98.4% accuracy, with dimensionality reduction through Binary Particle Swarm Optimization (BPSO) significantly improving model efficiency. Additionally, feature extraction using the Adaptive Time-Frequency Feature Weighted Transformation (ATFFWT) method

					further enhanced classification performance.
[22]	Dweiri and Al-Omary	2024	CHB-MIT Scalp EEG Dataset: Pediatric epilepsy patients' EEG data, analyzed using single-channel EEG devices to enable portable seizure detection.	XGBoost	The model achieved 82% accuracy, with 89.21% sensitivity and a false negative rate (FNR) of 10.79%. It demonstrated strong potential for use in wearable, near-ear EEG devices designed for seizure detection in home environments, providing a promising solution for continuous monitoring outside clinical settings.
[23]	Chung et al.	2021	CHB-MIT and EMR data (Common Data Model Analysis): The study standardized EEG data with electronic medical records (EMR) in a common data model to enhance seizure detection and forecasting.	CNN, LSTM, Big Data Analysis	CNN models achieved 98% sensitivity and 81% accuracy in seizure forecasting. The integration of big data further enhanced the model's robustness and scalability, making it more effective for real-world applications in seizure detection and prediction.

Explainability, Interpretability, and AI Ethics

Table 4:Explainability, Interpretability, and AI Ethics Summary

N	Author/s	Year	Dataset	Technique/s	Results
[24]	Gabeff et al.	2021	The TUH EEG Seizure Corpus is a large-scale dataset containing over 1,000 hours of EEG recordings, encompassing a wide range of seizure types. This dataset is a valuable resource for developing and testing seizure detection models due to its extensive coverage and variety of seizure events.	CNN, Grad-CAM, Attention Mechanisms	Achieved 93% accuracy in detecting seizure events. Grad-CAM and attention layers provided visual explanations for model decisions, improving interpretability for clinical use.
[25]	Wong et al.	2023	The CHB-MIT Scalp EEG Dataset contains EEG recordings from pediatric patients, with a particular emphasis on patient-specific variability in seizure events. This dataset is commonly used for studying individualized seizure patterns and developing personalized seizure detection models.	Deep SVDD, CNN, Out-of-Distribution Detection	Achieved 87% accuracy, with 89% specificity and 75% sensitivity. The model successfully flagged out-of-distribution seizure patterns, reducing misdiagnosis risk through improved uncertainty quantification.

EEG-Based Seizure Detection

Table 5:EEG-Based Seizure Detection Summary

N	Author/s	Year	Dataset	Technique/s	Results
[26]	Zhao et al.	2023	The TUH EEG Seizure Corpus is an extensive EEG dataset containing both seizure and non-seizure events. It is widely used for developing interpretable models, aiming to enhance seizure detection accuracy and improve clinical understanding of model decision-making.	ResNet, Vision Transformer (ViT), Grad-CAM	ResNet achieved the highest performance with 93% accuracy. The inclusion of Grad-CAM and attention layers enhanced model interpretability, enabling clinicians to gain clearer insights into the model's decision-making process and increasing its practical utility in clinical settings.
[27]	Wei et al.	2021	The Multiple Mouse Models of Epilepsy EEG dataset includes EEG recordings from various mouse models of epilepsy, such as kainic acid-induced and Dravet syndrome models. This dataset is used to study seizure mechanisms and to develop seizure detection algorithms that can be applied in both preclinical and clinical research.	XGBoost, Spontaneous Seizure Detection	XGBoost demonstrated sensitivities ranging from 91.4% to 98.8% and specificities between 93.1% and 98.8%. The model showed strong generalization to unseen data and enabled rapid analysis of long-duration EEG recordings, making it well-suited for real-time seizure detection.

Wearable and Real-Time Seizure Detection

Table 6: Wearable and Real-Time Seizure Detection Summary

N	Author/s	Year	Dataset	Technique/s	Results
[28]	Yu et al.	2023	Wearable Sensor Data includes electrodermal activity (EDA), accelerometry (ACC), and blood volume pulse (BVP) data collected from wrist and ankle sensors worn by 166 epilepsy patients. Over 13,254 hours of data were recorded, capturing more than 900 seizures. This dataset is valuable for developing real-time seizure detection models using multimodal physiological signals.	CNN, CNN-LSTM, Autoencoders	The CNN-LSTM model achieved 95% sensitivity for detecting generalized tonic-clonic seizures. When accelerometry (ACC) and blood volume pulse (BVP) data were fused, the overall sensitivity reached 83.9%. Additionally, personalized models utilizing autoencoders significantly improved the detection of specific seizure types, such as hyperkinetic seizures.
[29]	Rai et al.	2021	Video Recordings Dataset contains 1,114 video-captured seizure events from 230 patients, focusing on multiple motor seizures that last over 10 seconds. This dataset is utilized for developing and testing models for seizure detection based on video analysis.	AI-Based Video Analysis (CNN)	The system achieved 95.2% accuracy for tonic-clonic seizures and 92.9% for hyperkinetic seizures. False detection rate (FDR) was measured at 0.09 per hour, with some challenges due to environmental factors like lighting.

Large Language Models (LLM) and NLP in Epilepsy

Table 7: Large Language Models (LLM) and NLP in Epilepsy Summary

N	Author/s	Year	Dataset	Technique/s	Results
[30]	Zhao et al.	2024	The Japanese Epilepsy Dataset consists of medical data gathered from epilepsy patients in Japan. The dataset was fine-tuned using the LLaMA (7B) model, enhancing its applicability for epilepsy-related research and improving the performance of AI models in analyzing region-specific epilepsy data.	EpilepsyLLM, LLaMA (7B), BioGPT, Alpaca Method	EpilepsyLLM outperformed BioGPT on epilepsy-specific tasks, showing significant improvements in Japanese-language medical tasks. However, the use of LLaMA (30B) did not yield further performance gains, suggesting that the smaller model was sufficient for these specialized tasks without the need for larger architectures.
[31]	Dani et al.	2024	The Semio2Brain Dataset consists of epilepsy-related clinical text descriptions, designed to identify seizure onset zones (SOZ) based on semiological information. This dataset is used to develop models that map clinical symptom descriptions to brain regions associated with seizure activity.	LLM, NLP, Self-Consistency, Prompt Engineering	The model achieved clinical-level performance in identifying seizure onset zones (SOZ). However, challenges such as overconfidence in predictions and citation errors were noted, highlighting the need for improvements in reliability and accuracy for broader clinical adoption.

[32]	Xie et al.	2024	Clinical and Demographic Data includes epilepsy-related information from over 25,000 patients, covering key factors such as race, gender, and socioeconomic status. This dataset is essential for analyzing disparities in epilepsy diagnosis, treatment outcomes, and personalized care across diverse patient populations.	LLM, Multivariable Analysis	The analysis found no significant bias across demographic groups; however, disparities in seizure outcomes were identified based on insurance type, income, and gender. These findings highlight the need for equity improvements in epilepsy care to ensure better outcomes for all patients, regardless of socioeconomic factors.
[33]	Kim et al.	2024	Temple University EEG Dataset consists of 1,140 hours of EEG recordings, which were used to train the EEG-GPT model for seizure classification. This dataset plays a crucial role in developing AI models aimed at improving the accuracy of automated seizure detection.	EEG-GPT, Tree-of-Thought Framework	Achieved an AUROC of 0.86 using only 2% of the training data, outperforming traditional ML models in interpretability and generalization, but requiring further error reduction.

[34]	Goldenholz et al.	2024	Simulated RCT Dataset represents a virtual clinical trial involving 240 patients, split into placebo and treatment arms. It leverages anti-seizure medication data to assess and compare the effectiveness of treatments in managing seizure frequency and severity under controlled conditions.	LLM, Simulated RCT, Clinical Summarization	LLMs successfully replicated the trial results, showing only a 1% difference in responder rates between the drug and placebo groups. This demonstrates the potential of LLMs for accurate and efficient analysis in clinical trial settings.
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General AI Applications in Epilepsy

Table 8: General AI Applications in Epilepsy Summary

N	Author/s	Year	Dataset	Technique/s	Results
[35]	An et al.	2020	EEG, MRI, iEEG, DTI Datasets: Used EEG, intracranial EEG (iEEG), MRI, and diffusion tensor imaging (DTI) from epilepsy patients.	SVM, k-NN, CNN, RNN, Virtual Epileptic Patient (VEP)	CNNs achieved up to 97% accuracy in seizure classification, while RNNs were able to predict seizure onset nearly an hour in advance with over 90% sensitivity. Biophysical modeling, particularly through the Virtual Epileptic Patient (VEP) framework, demonstrated strong performance in localizing seizure onset zones (SOZ).
[36]	Tran et al.	2022	Bonn University EEG Dataset: Single-channel EEG data used for seizure detection and feature extraction.	Random Forest, ATFFWT, BPSO	The random forest model achieved 98.4% accuracy, with the use of Adaptive Time-Frequency Feature Weighted Transformation (ATFFWT) for feature extraction and Binary Particle Swarm Optimization (BPSO) for dimensionality reduction enhancing both classification

					performance and computational efficiency.
[37]	Knight et al.	2023	Human and Animal Models (EEG) dataset includes electro-clinical phenotyping data from both human epilepsy patients and animal models. This dataset is instrumental in studying seizure mechanisms and developing models for seizure detection and prediction across species.	AI Phenotyping, Behavioral Analysis	AI identified behavioral states and linked them to electrographic biomarkers like spikes and high-frequency oscillations, improving phenotyping and therapy decisions.

Other Approaches or Techniques

Table 9:Other Approaches or Techniques Summary

N	Author/s	Year	Dataset	Technique/s	Results
[38]	Luca et al	2014	Pediatric Epilepsy Data (Accelerometry) consists of accelerometer recordings from pediatric epilepsy patients, used to detect hypermotor seizures. This dataset supports the development of models for real-time monitoring and seizure detection in children.	Extreme Value Theory, Unsupervised Learning	The model achieved up to 90% sensitivity for detecting hypermotor seizures in 5 out of 7 patients, with positive predictive values ranging from 21% to 89%. This unsupervised approach effectively reduced computational complexity compared to traditional supervised models, offering a more efficient solution for seizure detection.

Generative Adversarial Networks (GANs) and Epileptic Seizures

Table 10: Generative Adversarial Networks (GANs) and Epileptic Seizures

N	Author/s	Year	Dataset	Technique/s	Results
[39]	Truong et al.	2019	CHB-MIT, Freiburg Hospital, and EPILEPSIAE EEG datasets: EEG recordings from pediatric and adult patients.	GANs, STFT, DCGAN with deconvolution layers.	The model achieved AUC scores of 77.68%, 75.47%, and 65.05% on the CHB-MIT, Freiburg Hospital, and EPILEPSIAE datasets, respectively. The implementation of an up-sampling technique further improved the AUC by 2% on the EPILEPSIAE dataset, highlighting the method's potential for enhancing diagnostic accuracy.
[40]	Lee et al.	2023	CHB-MIT EEG Dataset: Scalp EEG recordings from 23 pediatric patients.	Data Augmentation, Diffusion and Denoising Processes.	The approach significantly enhanced seizure prediction performance by effectively addressing data imbalance challenges in EEG datasets, leading to more reliable and accurate model outcomes.
[41]	Kumar et al.	2022	General EEG datasets: Diverse EEG recordings for seizure classification and other neurological conditions.	Graph Neural Networks (GNNs).	Achieved 96% accuracy in seizure classification with strong generalizability across unseen patient datasets.

[42]	Craley et al.	2020	Multichannel EEG datasets: Large-scale EEG recordings processed with temporal and spatial patterns.	Hybrid CNN and Coupled Hidden Markov Models (CHMM).	The hybrid model outperformed standalone CNN models in terms of true positive rate, AUC, and F1 score, although it exhibited a slight increase in the false positive rate, indicating room for further refinement.
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Chapter 3: Software Project Management Plan (SPMP)

3.1 Project Overview

3.1.1 Purpose, Scope, & Objectives

3.1.1.1 Purpose

Early and precise diagnosis is essential for patients with epileptic seizures, non-epileptic events, and psychogenic non-epileptic seizures (PNES), all of which are severe medical diseases that can significantly impair a patient's quality of life. Inappropriate therapies, needless patient suffering, higher medical expenses, and even deterioration of the underlying ailment can all arise from misdiagnosing one of these disorders. The physical symptoms of PNES, in particular, resemble those of epileptic seizures, but the diagnosis is more difficult because of its non-neurological etiology.

This project's primary goal is to classify EEG data into three groups: epileptic, non-epileptic, and psychogenic non-epileptic seizures using generative AI, machine learning (ML), and deep learning (DL) models. Our goal is to develop an AI-powered system that helps physicians make quick, very accurate diagnoses by utilizing these cutting-edge technologies. Reducing diagnostic delays, encouraging prompt and appropriate actions, and eventually improving patient outcomes are the goals of this approach.

3.1.1.2 Scope

For this study, EEG data from local and public sources will be gathered and analyzed. Both epileptic and non-epileptic seizure data were taken from publicly accessible databases, such as the Temple University Hospital (TUH) EEG Seizure Corpus. We also worked with King Fahd University Hospital to get local clinical data for PNES cases. In order to categorize EEG signals into the three target categories and guarantee the system's applicability and dependability for both general and regional healthcare contexts, our study focuses on applying sophisticated neural networks, generative models, and machine learning approaches.

3.1.1.3 Objectives

This project's main goal is to use cutting-edge deep learning (DL) and machine learning (ML) techniques to create a dependable and accurate diagnostic model that can categorize EEG data into three different categories: epileptic, non-epileptic, and psychogenic non-epileptic seizures (PNES). The project's main goal is to increase classification accuracy and processing efficiency by optimizing and preprocessing EEG information. To make sure the system is flexible and applicable to various patient populations, publicly accessible datasets are combined with locally gathered clinical data from Saudi Arabia. Additionally, generative AI techniques are used to improve and supplement the existing EEG data, resulting in more robust and powerful model training. Additionally, a user-friendly web platform is created that enables physicians to quickly submit EEG data, obtain automatic trinary classification findings, and produce thorough medical reports produced by AI. Continuous input from medical experts is included into the development process to improve the model's clinical efficacy and guarantee its suitability for use in actual diagnostic situations.

3.1.2 Assumptions, Constraints, & Risks

3.1.2.1 Assumptions

This project is assumed on the following:

- Access to relevant EEG datasets from Saudi hospitals locally or from open-source databases.
- EEG patterns are effectively used to differentiate between PNES and epileptic patients using deep learning algorithms.
- Collaborating with medical professionals to verify that the diagnostic tool is accurate.
- Timely availability of computer power to conduct in-depth deep learning research.

3.1.2.2 Constraints

The following constraints should be considered during the execution of the project:

- **Resources:** We'll utilize the cash we've set aside for computational resources, model deployment, and cloud services.
- **Technological Restrictions:** Handling large EEG datasets may require high-performance processing resources for training complex neural networks.
- **Time Restrictions:** Academic deadlines must be met by research, and any delays in acquiring data may affect the timeline.
- **Data Privacy:** Because of the sensitive nature of the information at hand, compliance with data privacy laws (such HIPAA or its local equivalents) will be necessary.

3.1.2.3 Risks

The following points present the expected risks that might negatively influence the work progress of the project:

- **Limitations of the Dataset:** If Saudi Arabian hospitals refuse to provide us with local data, we may be forced to rely solely on publicly available information, which may not fully represent our target demography.
- **Requirement Modifications:** In response to feedback from the medical community or the discovery of new data processing requirements, the project's scope or specifications may change.
- **Overfitting:** When a model is overfitted, it may perform well on training data but poorly on unobserved patient data.
- **Extended Timelines:** Because of the complexity of DL/ML models and the volume of EEG datasets, training times may exceed expectations.
- **Issues with Data Preprocessing:** The model may have a harder time differentiating between different types of seizures if the data is incorrectly or insufficiently labeled.

3.1.3 Project Deliverables

Table 10 Highlights the deliverables and their deadlines for the project.

Table 11:Project Deliverables

Deliverable	To Whom	Delivery Media	Date
Project Proposal	Dr. Mustafa Youldash	Softcopy	25 th Aug. 2024
Literature Review	Dr. Mustafa Youldash	Softcopy	28 th Sep. 2024
Mid Semester Report	Dr. Mustafa Youldash	Softcopy	13 th Oct. 2024
Submit Software Project Management Plan (SPMP)	Dr. Mustafa Youldash	Softcopy	13 st Oct. 2023
Methodology	Dr. Mustafa Youldash	Softcopy	20 st Oct. 2023
Submit System Requirements Specification (SRS)	Dr. Mustafa Youldash	Softcopy	27 th Oct. 2024
Submit Software Design Specification (SDS)	Dr. Mustafa Youldash	Softcopy	17 th Nov. 2024
First Semester Final Report	Dr. Mustafa Youldash	Softcopy	15 th Dec. 2024
First Semester Final Presentation	Dr. Mustafa Youldash	Softcopy	29 th Dec. 2024
Submit Software Test Plan (STP)	Dr. Mustafa Youldash	Softcopy	4 th May 2025
Submit User Manual	Dr. Mustafa Youldash	Softcopy	10 th May 2025
Senior Project Source Code and Report	Dr. Mustafa Youldash	Softcopy	14 th May 2025
Senior Project Final Presentation	Dr. Mustafa Youldash	Softcopy	14 th May 2025

3.1.4 Schedule & Budget Summary

Because the project uses a lot of open-source software, it is considered inexpensive. The only expenses covered are those for any associated publication fees and the software required for system design and construction. Table 11 lists the project phases along with an estimate of how long each will take.

Table 12:The Project Schedule

The Project Phase	Duration
Proposal Discussion	3 days
Define Epileptic Seizures and PNES seizures	1 week
Datasets Epileptic Seizures and EEG	1 week
Literature Review	3 weeks

Mid Semester Report	2 weeks
Software Project Management Plan (SPMP)	2 weeks
Methodology	1 week
System Requirements Specification (SRS)	1 week
Software Design Specification (SDS)	1 week
First Semester Final Report	2 weeks
Implementing/Coding	11 weeks
Software Test Plan (STP)	2 weeks
User Manual	2 days
Final Project Report	1 week
Presentation	4 days

3.1.5 Evolution of the Plan

The Software Project Management Plan (SPMP) is described in this section of the paper in accordance with IEEE standards. The group will move forward with these suggested plans until changes are judged essential. Any modifications will only be released following extensive deliberation and consensus among all team members, with consent from the external advisors, Dr. Modhi Alkhaldi and Dr. Zakia Yasawy, and the project supervisor, Dr. Mustafa Youldash.

3.1.6 Definitions

Table 12 presents the definitions of the main terms included in the document.

Table 13: Terms and Definitions

Term	Definition
Epileptic Seizures	An epileptic seizure is a brief episode of abnormal brain activity that can cause various physical and sensory symptoms, such as convulsions or loss of consciousness. It is a key feature of epilepsy, a disorder characterized by recurrent seizures [1].
Psychogenic non-epileptic seizures (PNES)	Seizure-like events that are psychological rather than neurological, frequently brought on by stress or trauma, and not associated with aberrant brain activity are known as psychogenic non-epileptic seizures, or PNES [1].
Software Project Management Plan (SPMP)	A document that details the management approach, schedule, and resources necessary to successfully execute a software project. [43]. IEEE Std 1058-1998, "IEEE Standard for Software Project Management Plans," IEEE, 1998.

Software Requirements Specification (SRS)	A written document that lists the system's functional and non-functional needs is called a software requirements specification, or SRS. Software engineers can use it as a guide to ensure that the application meets both user needs and technological constraints [44]. IEEE, "IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications," 1998.
Software Design Specification (SDS)	Describes a software system's architecture, parts, interfaces, and design components. It describes the structure and operation of the program in accordance with the SRS's criteria. [45]. IEEE, "IEEE Std 1016-2009, IEEE Standard for Information Technology—System Design Descriptions," 2009.
Software Test Plan (STP)	A document that describes the goals, methods, timetable, and materials required for software testing. It guarantees that the system satisfies the functional and quality standards. [46]. IEEE, "IEEE Std 829-2008, IEEE Standard for Software and System Test Documentation," 2008.
Waterfall Model	a methodical approach to software development in which every stage—requirements, design, implementation, testing, and maintenance—must be finished before moving on to the next. It is a linear method that is frequently applied to projects with precise specifications. [47]. Royce, W. W., "Managing the Development of Large Software Systems," <i>Proceedings of IEEE WESCON</i> , 1970.
Work Breakdown Structure	A project is broken down into smaller, more manageable tasks or components using a hierarchical process called a work breakdown structure (WBS). It facilitates project planning, execution, and tracking by decomposing the entire task into distinct components [48]. IEEE, "IEEE Std 1233-1998, IEEE Guide for Developing System Requirements Specifications," 1998.

3.1.7 Document Structure

Overview of Document Structure:

1. **Project Overview:** The aims, objectives, and general scope of the project are succinctly summarized in this section. Along with describing crucial elements like deliverables, potential hazards, assumptions, and limits, it also contains important information regarding the project's budget and schedule.
2. **Project Organization:** This section describes the internal teams and their external interfaces, with an emphasis on the internal structure of the project. It also makes it clearer what duties and responsibilities each team member participating in the project's development and execution has been given.
3. **Managerial Process Plans:** In this document, the management techniques used for the project are explained in detail. It covers important topics like work plans, risk management, progress tracking, and start-up and close-out procedures.
4. **Technological Process Plans:** The technological infrastructure required for the project is thoroughly explained in this section. It describes the tools, procedures, and techniques used

throughout the project and contains the acceptance criteria for the finished result.

5. **Plans for supporting processes:** The documentation needed to ensure seamless operations during the project's development and management phases is finally compiled in this part.

3.2 Project Organization

3.2.1 External Interfaces

The project for differentiating between Epileptic seizures PNES is supervised by Dr. Mustafa Youldash and targets a few selected Saudi hospitals. Figure 1. illustrates the external interfaces.

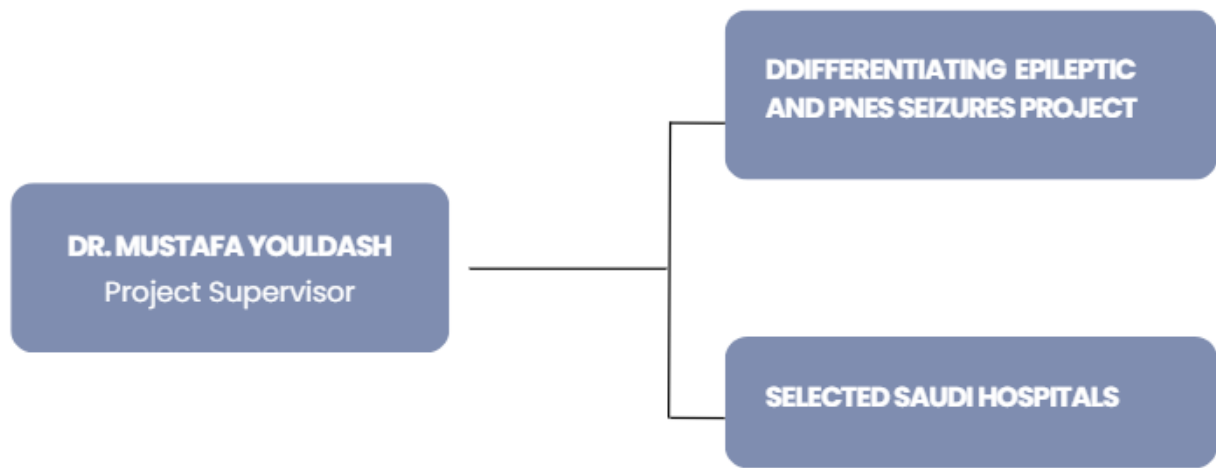


Figure 1: External Interface

3.2.2 Internal Structure

To ensure that the project is completed on schedule, Dr. Mustafa Youldash oversees it in conjunction with Dr. Modhi Alkahldi and Dr. Zakia Yasawy. Sana Araj, Shahad Adel, Deem Alrashidi, Saher Alshehri, and Sara Thaer are the members of the project team who will prepare it. Each team member is given a specialized position according to their skills, knowledge, and experience in order to provide the greatest results. Figure 2 shows the internal organizational structure of the project.

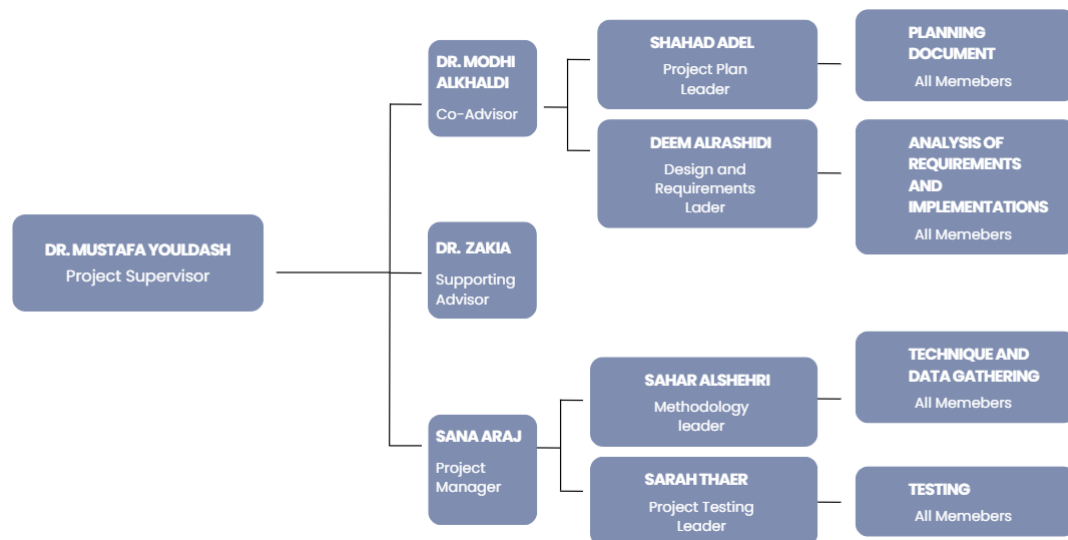


Figure 2:Internal Interface

3.2.3 Roles & Responsibilities

The team leader assigns tasks to all team members in an equitable way and supervises their completion. As a result, each team member participates in all phases of the development process. Table 13 summarizes each team member's tasks and responsibilities.

Table 14:Roles and Responsibilities

Role	Responsibilities	Member
Project Manager	- Direct contact with the supervisor. - Determine and assign the project's goals. - Make sure that everyone in the team shares the workload equally. - Make sure the project stays on course to fulfil deadlines.	Sana Araj
Project Plan Leader	- Support the project manager. - Create a thorough project plan that includes deadlines and completion dates. - Make a budget and oversee the project's funds. - Create a risk management strategy to recognize and address possible problems.	Leader: Shahad Adel All team members
Requirements and Design Leader	- Get the requirements ready and distribute them. - Consider the prerequisites. - Construct a prototype to verify design ideas. - Finish all of the design documentation.	Leader: Deem Alrashidi All team members

Methodology	• Data collection and strategy. • Make a plan for development. • Collecting and assessing the needs.	Leader: Sahar Alshehri All team members
Project Testing Leader	• Test the system and fix any issues. • Finish the test report. • Indicate the operations and test data.	Leader: Sara Thaer All team members

3.3 Managerial Process Plans

This section clarifies everything related to the project start-up, work, tracking, risk management, and close-out plans.

3.3.1 Start-up Plan

Express the resources and supplies required to begin the project, with a staffing plan, training and estimation; These plans could be merged with other plans, depending on the size and scope of the project.

3.3.1.1 Estimates

Accurate estimates play a crucial role in effective management and successful project implementation. The cost of the software depends on several factors including (Time, Cost and Resources) Estimates.

- **Time Estimate:** The entire senior year will be dedicated to finishing the project. The first semester will be documenting the project and the final presentation. By the end of the year, the project will be finished. Issues may be solved by the team and cover any additional cost.
- **Cost Estimates:** The cost of planning, designing, implementing and testing the project will be free. The cost of the document relies on the materials, the time the team spent drafting, editing, and finalizing the document, and other resources. The journal fee for publication will be included. Any finances spent on transport will be added as well.
- **Resources Estimates:** Hardware equipment such as laptops, storage devices and processing units. Software tools used include LaTeX, Notions and Visual Studio. Also, any education expenses will be included.

3.3.1.2 Staffing

Artificial Intelligence student at Imam Abdulrahman bin Faisal University will develop the project. Each member will be appointed specific tasks based on their skills and experiences to ensure the project is completed in a reasonable time. Skills in data analysis, Python programming and a basic understanding of the deep learning framework are fundamental technical skills for the entire team. Soft skills including critical thinking, communication, literature reviews analyzing and documenting are all required. Table 14 shows the duration and human resources to finish each phase.

Table 15:Human Resources

Project Phase	Human Resource	Duration
Project Initiation	All Team Members	2 weeks
Planning		10 days
Data Collection		1 month
Methodology		9 days
Architecture Design		1 week
Implementing		20 weeks
Evaluation, Testing & Optimization		3 weeks
Documentation		10 days

3.3.1.3 Project Staff Training

Table 15 lists the necessary skills for each team position. To ensure an effective project conclusion, a staged strategy or phased approach will be used. Each phase will require specific skills and experience.

Table 16:Skills Needed for Each Role

Position	Skills
Project Manager (Leader)	○ Leadership skills. ○ Communication skills. ○ Abilities in Critical Thinking. ○ Decision-making capabilities.

	○ Negotiation skills. ○ Teamwork. ○ Problem-solving skills. ○ Technical writing talents. ○ Time management.
Project Plan Manager	○ Decision-making capabilities. ○ Analytical skills. ○ Communication skills. ○ Technical writing talents. ○ Time management. ○ Organizational skills. ○ Attention to detail.
Requirements Analysts	○ Technical skills. ○ Analytical skills. ○ Communication skills. ○ Time management. ○ Problem-solving skills.
Data Scientist	○ Programming skills. ○ Statistical understanding. ○ Data Visualization ○ Data manipulation
Project Design Manager	○ Design knowledge.

	○ Reasoning skills. ○ Adaptability. ○ Creativity.
Project Testing Manager (Tester)	○ Testing planning and execution. ○ Risk management. ○ Analytical skills. ○ Time management. ○ Teamwork. ○ Decision-making capabilities. ○ Abilities in Critical Thinking.
Stakeholder Management	○ Analytical skills. ○ Networking. ○ Empathy intelligence. ○ Emotional intelligence. ○ Communication skills. ○ Cultural grasp.
Technical Writing	○ Documenting. ○ Proofreading. ○ Editing. ○ Research skills. ○ Audience Awareness.

3.3.2 Work Plan

This section outlines the project's activity, timetables, assets, and finances.

3.3.2.1 Work Breakdown Structure

Table 16 displays the task breakdown structure for this project.

Table 17: Breakdown Structure

NO	Work Activity	Resources		Duration	Deliverable	Status
		Software	Hardware			
1	Project Initiation.			3 weeks	Mid-Semester Report	Completed
1.1	First meeting for the senior project.	Google Docs and Notion	Personal Laptops	2 hours		
1.2	Searching for Epilepsy seizure detection methods.	Google Docs, Notion, and Google Scholar	Personal Laptops	2 days		
1.3	Choosing AI generative methods to detect epilepsy.	Google Docs and Google Scholar	Personal Laptops	2 days		
1.4	Searching for dataset.	Browser and Excel	Personal Laptops	1 week		
1.5	Updating the chosen disorder.	Google Docs and Browser	Personal Laptops	1 week		
2	Introduction Chapter.			1 week	Mid-Semester Report	Completed
2.1	Writing the chapter.	Google Docs	Personal Laptops	4 days		

2.2	Reviewing the chapter	Google Docs	Personal Laptops	2 days		
2.3	Editing the chapter	Google Docs and Microsoft Word	Personal Laptops	1 days		
3	Literature Review Chapter.			1 week	Mid-Semester Report	Completed
3.1	Reading and Summarizing literature reviews.	Google Scholar, Notion and Google Docs	Personal Laptops	4 days		
3.2	Writing literature review chapter	Google Docs and Notion	Personal Laptops	1 day		
3.3	Reviewing literature review.	Notion	Personal Laptops	1 day		
3.4	Editing literature review chapter.	Microsoft Word	Personal Laptops	1 day		
4	Software Project Management Plan.			4 days	SPMP	Completed
4.1	Planning, and identifying project goals.	Google Docs	Personal Laptops	2 hours		
4.2	Writing the SPMP.	Google Docs	Personal Laptops	2 day		
4.3	Reviewing the SPMP.	Google Docs	Personal Laptops	1 day		
4.4	Editing the SPMP.	Google Docs and Microsoft Word	Personal Laptops	1 day		
4.5	Submitting the SPMP.	Google Docs, Microsoft	Personal Laptops	1 hour		

		Word and One Drive				
5	Submitting the Mid-Semester Report.			1 hour	Mid-Semester Report	Completed
6	Methodology.			5 days	Methodology	Completed
6.1	Writing the methodology.	Google Docs	Personal Laptops	3 days		
6.2	Reviewing the methodology.	Google Docs	Personal Laptops	1 day		
6.3	Editing the methodology.	Microsoft Word	Personal Laptops	1 day		
6.4	Submitting the methodology.	Google Docs, Microsoft Word and One Drive	Personal Laptops	1 hour		
7	Software Requirement Specification.			1 week	SRS	Completed
7.1	Gather the requirements.	Google Docs	Personal Laptops	2 hours		
7.2	Analysis.	Google Docs	Personal Laptops	1 day		
7.3	Specifying the requirements.	Google Docs	Personal Laptops	2 days		
7.4	Submitting the requirements.	Google Docs and One Drive	Personal Laptops	1 day		
7.5	Updating and editing the requirements.	Google Docs	Personal Laptops	2 days		
7.6	Submitting the requirements.	Google Docs and One Drive	Personal Laptops	1 hour		

8	Software Design Specification			1 week	SDS	Completed
8.1	Designing data and initial interface.	Google Docs	Personal Laptops	2 days		
8.2	Developing the interfaces.	Google Docs	Personal Laptops	2 days		
8.3	Analyzing the interface.	Google Docs	Personal Laptops	1 days		
8.4	Completing SDS.	Google Docs	Personal Laptops	1 day		
8.5	Review SDS.	Google Docs	Personal Laptops	1 day		
8.6	Submitting SDS.	Google Docs and One Drive	Personal Laptops	1 hour		
9	Final Report			22 days	Final Report (First Semester)	Completed
9.1	Reviewing the report.	Google Docs	Personal Laptops	15 days		
9.2	Preparing for the presentation.	Canva	Personal Laptops	7 days		
9.3	Submitting the report and presentation.	Google Docs, Canva and One Drive	Personal Laptops	1 hour		
10	Implementation			20 weeks	Progress Implementation	Completed
10.1	Preprocessing the dataset.	Python, Vscode, Google colab.	Personal Laptops	2 weeks		

10.2	Exploratory Data Analysis.	Python, Vscode, Google colab.	Personal Laptops	3 weeks		
10.3	Fine-tuning models.	Python, Vscode, Google colab.	Personal Laptops	2 weeks		
10.4	Model Evaluation.	Python, Vscode, Google colab.	Personal Laptops	1 week		
10.5	Building Deep Learning models.	Python, Vscode, Google colab.	Personal Laptops	5 weeks		
10.6	Deployment.	Python, Vscode, Google colab.	Personal Laptops	4 weeks		
10.7	Monitoring	Google AI Platform.	Personal Laptops	2 weeks		
10.8	Documenting	Google Docs and Notion	Personal Laptops	1 week		
11	Software Testing Plan			2 weeks	STP	Completed
11.1	Develop test cases.	Google Docs and Notion	Personal Laptops	4 days		
11.2	Unit testing.	Vscode and server.	Personal Laptops	4 days		
11.3	Integration testing.	Vscode and server.	Personal Laptops	2 days		
11.4	Testing the system as a whole.	Vscode and server.	Personal Laptops	4 days		
11.5	Reviewing the STP.	Vscode and server.	Personal Laptops	2 days		

11.6	Submitting the STP.	One Drive	Personal Laptops	1 hour		
12	Submitting the Final Report			1 week	Final Report	Completed
12.1	Reviewing the final report.	Google Docs	Personal Laptops	3 days		
12.2	Editing the final report.	Google Docs and Microsoft Word	Personal Laptops	2 days		
12.3	Submitting the final report.	Google Docs and One Drive	Personal Laptops	1 hour		
12.4	Creating the user manual.	Google Docs	Personal Laptops	2 days		
12.5	Preparing the final presentation.	Google Docs and Canva	Personal Laptops	1 hour		

3.3.2.2 Schedule Allocation

The timelines for each activity, including the start and end dates, are in Table 17 for this academic year.

Table 18: Schedule Allocation

NO	Work Activity	Duration	Start Date	End Date
1	Project Initiation.	3 weeks	1/8/2024	22/8/2024
1.1	First meeting for the senior project.	2 hours	28-8-2024	28-8-2024
1.2	Searching for Epilepsy seizure detection methods.	2 days	29/8/2024	31/8/2024
1.3	Choosing AI generative methods to detect epilepsy.	2 days	2/9/2024	4/9/2024
1.4	Searching for dataset.	1 week	6/9/2024	13/9/2024
1.5	Updating the chosen disorder.	1 week	14/9/2024	21/9/2024
2	Introduction Chapter.	1 week	14/9/2024	21/9/2024
2.1	Writing the chapter.	4 days	14/9/2024	17/9/2024
2.2	Reviewing the chapter	2 days	18/9/2024	20/9/2024
2.3	Editing the chapter	1 days	21/9/2024	21/9/2024
3	Literature Review Chapter.	1 week	22/9/2024	29/9/2024
3.1	Reading and Summarizing literature reviews.	4 days	22/9/2024	26/9/2024
3.2	Writing literature review chapter	1 day	27/9/2024	27/9/2024
3.3	Reviewing literature review.	1 day	28/9/2024	28/9/2024
3.4	Editing literature review chapter.	1 day	29/9/2024	29/9/2024
4	Software Project Management Plan.	4 days	30/9/2024	3/10/2024
4.1	Planning, and identifying project goals.	2 hours	30/9/2024	30/9/2024
4.2	Writing the SPMP.	2 days	1/10/2024	3/10/2024
4.3	Reviewing the SPMP.	1 day	4/10/2024	4/10/2024
4.4	Editing the SPMP.	1 day	5/10/2024	5/10/2024
4.5	Submitting the SPMP.	1 hour	5/10/2024	5/10/2024

5	Submitting the Mid-Semester Report.	1 hour	6/10/2024	6/10/2024
6	Methodology.	5 days	6/10/2024	10/10/2024
6.1	Writing the methodology.	3 days	6/10/2024	8/10/2024
6.2	Reviewing the methodology.	1 day	9/10/2024	9/10/2024
6.3	Editing the methodology.	1 day	10/10/2024	10/10/2024
6.4	Submitting the methodology.	1 hour	10/10/2024	10/10/2024
7	Software Requirement Specification.	1 week	11/10/2024	18/10/2024
7.1	Gather the requirements.	2 hours	11/10/2024	11/10/2024
7.2	Analysis.	1 day	12/10/2024	12/10/2024
7.3	Specifying the requirements.	2 days	13/10/2024	14/10/2024
7.4	Submitting the requirements.	1 day	15/10/2024	15/10/2024
7.5	Updating and editing the requirements.	2 days	16/10/2024	17/10/2024
7.6	Submitting the requirements.	1 hour	18/10/2024	18/10/2024
8	Software Design Specification	1 week	19/10/2024	26/10/2024
8.1	Designing data and initial interface.	2 days	19/10/2024	20/10/2024
8.2	Developing the interfaces.	2 days	21/10/2024	22/10/2024
8.3	Analyzing the interface.	1 days	23/10/2024	23/10/2024
8.4	Completing SDS.	1 day	24/10/2024	24/10/2024
8.5	Review SDS.	1 day	25/10/2024	25/10/2024
8.6	Submitting SDS.	1 hour	26/10/2024	26/10/2024
9	Final Report	22 days	27/10/2024	17/11/2024
9.1	Reviewing the report.	15 days	27/10/2024	10/11/2024
9.2	Preparing for the presentation.	7 days	11/11/2024	17/11/2024
9.3	Submitting the report and presentation.	1 hour	17/11/2024	17/11/2024
10	Implementation	20 weeks	18/11/2024	7/3/2025
10.1	Preprocessing the dataset.	2 weeks	18/11/2024	1/1/2025
10.2	Exploratory Data Analysis.	3 weeks	1/1/2025	22/1/2025
10.3	Fine-tuning models.	2 weeks	22/1/2025	5/2/2025

10.4	Model Evaluation.	1 week	5/2/2025	12/2/2025
10.5	Building Deep Learning models.	5 weeks	12/2/2025	19/3/2025
10.6	Deployment.	1 weeks	19/3/2025	26/3/2025
10.7	Monitoring	1 weeks	27/3/2025	4/4/2025
10.8	Documenting	1 week	5/4/2025	12/4/2025
11	Software Testing Plan	2 days	13/4/2025	15/4/2025
11.1	Develop test cases.	2 days	15/4/2025	17/4/2025
11.2	Unit testing.	2 days	17/4/2025	19/4/2025
11.3	Integration testing.	2 days	19/4/2025	21/4/2025
11.4	Testing the system as a whole.	4 days	21/4/2025	25/4/2025
11.5	Reviewing the STP.	1 day	26/4/2025	26/4/2025
11.6	Submitting the STP.	1 hour	27/4/2025	27/4/2025
12	Submitting the Final Report	2 days	28/4/2025	30/4/2025
12.1	Reviewing the final report.	1 day	1/5/2025	1/5/2025
12.2	Editing the final report.	1 day	2/5/2025	2/5/2025
12.3	Submitting the final report.	1 hour	2/5/2025	2/5/2025
12.4	Creating the user manual.	1 day	3/5/2025	3/5/2025
12.5	Preparing the final presentation.	1 hour	10/5/2025	10/5/2025

3.3.2.3 Resource Allocation

The human resources for this project consist of the team members from our project group. The allocation of tasks will be distributed among the team based on individual expertise and project needs to ensure smooth progress and timely completion. In Addition, non-human resources have the following plan:

Table 19:Resource Allocation

Infrastructure Plan	
Hardware	Personal computers and servers for training and deploying the AI models. Mobile devices for testing any mobile integration features.

Software	• Python and AI libraries like TensorFlow and PyTorch for model development. • Microsoft Office for documentation tasks. • Overleaf for collaborative document editing. • Colab for running and testing machine learning models in the cloud.
Written resources	• Research papers on AI-driven epileptic seizure detection. • Online AI courses for team training on specialized techniques.

3.3.2.4 Budget Allocation

We will be utilizing a Colab subscription, which costs around 150 SR, to access cloud computing services that will significantly enhance our model training capabilities. Additionally, when it comes time to publish our research, we're looking at publication fees that could range anywhere from 4000 to 6000 SR, depending on the journal we select. This will allow us to share our findings with the wider scientific community while factoring in the costs associated with high-quality publication platforms.

3.3.3 Project Tracking Plan

The following subsection outlines key aspects of the project's requirements management, schedule control, quality control, reporting, and metrics.

3.3.3.1 Requirements Management

The requirements management process for the Epileptic Detection using AI project focuses on gathering, documenting, and tracking the necessary features and performance expectations. We will work closely with key stakeholders such as healthcare professionals, users, and the development team to ensure the system meets both clinical and technical needs. This includes addressing crucial capabilities like real-time seizure detection, as well as ensuring the system's accuracy, reliability, and adherence to privacy regulations.

To stay organized, we'll track any changes and make sure all requirements are accounted for. Regular reviews with stakeholders will help confirm that we're on the right path, and any adjustments will go through a formal approval process. Finally, we will validate the system through thorough testing and user feedback, making sure it fully meets its intended goals.

3.3.3.2 Schedule Control

To ensure tasks stay on track while addressing any delays or challenges. Progress will be monitored using tools like Gantt charts and platforms such as Trello or Jira, with clear deadlines for each phase. Regular meetings will help identify issues early, allowing for adjustments like resource redistribution or timeline changes. Any schedule modifications will go through an approval process to ensure proper documentation. This approach keeps the project running smoothly while staying flexible to handle unexpected challenges.

3.3.3.3 Quality Control

The quality control process for the AI project is all about making sure everything works as it should and meets the highest standards of performance. To do this, we'll run thorough tests at every level unit testing, integration testing, and system testing to make sure the system functions smoothly and accurately. We will evaluate the AI model based on key performance factors like accuracy, precision, and how quickly it processes data.

Regular code reviews will help us keep the code efficient and free of bugs, and we will also gather feedback from healthcare professionals to ensure the system is easy to use and effective in real-world clinical settings. If any issues come up, we will address them quickly to ensure the final product is of the highest quality.

3.3.3.4 Reporting

For reporting, here's how we will keep everyone in the loop:

Regular check-ins will be held with the team to track progress. These meetings will happen both online and in person, depending on what works best at the time.

Every week, we also meet with our advisors to go over project updates and discuss any new developments.

After each weekly meeting, detailed meeting minutes will be shared to ensure everyone is on the same page and has a clear record of what was discussed.

This keeps communication open and ensures we're all aligned as the project moves forward.

3.3.3.5 Project Metrics

The metrics to be considered when assessing the project's success are set out in Table 19.

Table 20:Project Metrics

Metrics	Comment
Time	We will check the project's timeline, continuously monitor progress and compare it to the planned schedule to make sure everything stays on track.
Value	Throughout the project, we regularly assess whether the work meets the set requirements, ensuring all necessary specifications are fulfilled as we move forward.
Scope	After each task is completed, any changes in scope are tracked to make sure they align with the original plan or any approved adjustments, keeping us on course.
Quality	Once tasks are completed, we perform thorough quality checks to catch and fix any issues, making sure the system functions smoothly and meets performance standards.

3.3.4 Risk Management Plan

A critical component of the project involves identifying all potential risks. Table 20 outlines various risks, detailing their likelihood, prevention strategies, potential impact, and the corresponding mitigation measures to be implemented.

Table 21: Risk Management Plan

No.	Potential risk	Probability	Prevention	Action	Impact
1	Changes in meeting times	Low	Ensure that all members are informed well in advance before any meetings are scheduled.	Schedule a convenient time for regular meetings that suit the availability of all team members.	Time
2	Tight project timeline	Moderate	Stick closely to the established plan and collaborate effectively to meet deadlines.	Consider allocating additional hours to work and completing tasks on time.	Time
3	Delay in task submissions	High	Work longer hours to stay on track with deadlines.	Implement a well-organized management plan to avoid last-minute submissions.	Time
4	Changes in project requirements	Moderate	Dedicate enough time to fully understand the new requirements and explain their effects to the team.	Increase collaboration and divide tasks efficiently to accommodate changes.	Time
5	One team member leaves the group	Moderate	Motivate the remaining team to continue and complete the project.	Redistribute tasks to cover the extra work created by the member's departure.	Time

3.3.5 Project Closeout Plan

Before the deadline, make sure everyone checks these key steps to wrap things up smoothly. Double-check that every feature and requirement of the system is good to go. Get all the source code, documents, and project materials neatly organized and saved in the right spots. Do a final review to make sure all the latest versions of everything documents and the system themselves are in tip-top shape before we submit. Lastly, hand in both a physical and digital copy of everything to the supervisor so they can give it the green light.

3.4 Technical Process Plans

3.4.1 Process Model

The primary software process model will be the waterfall model. All deliverables will be carefully and individually planned as the processes depend on each other, ultimately contributing to the primary project work activities.

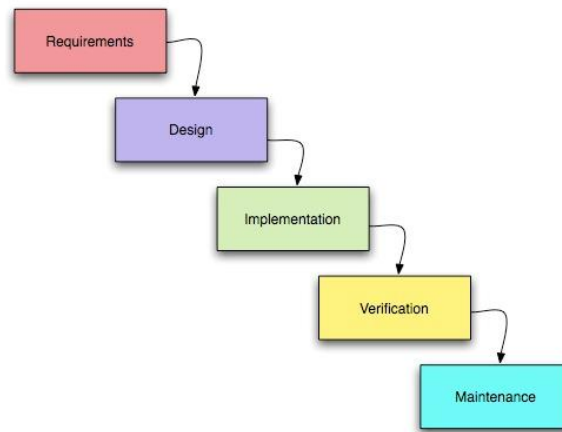


Figure 3: Waterfall model

Table 22: Waterfall Phases

Phase	Goal	Due Date	Deliverable
Requirements Analysis and Definition	Collecting, analyzing, and documenting all possible requirements of the system.	October 15 th	SRS document
System and Software Design	Designing the system's overall architecture, its functionalities, and user interactions.	October 29 th	SDS document
Implementation and Unit Testing	Implementing and testing the program units practically using Python programming language.	June 21 st	Application program
Integration and System Testing	Testing and debugging the system as a whole and deploying it in the client environment	June 21 st	• Application program • STS document
Operation and Maintenance	Fixing any issues customers may face in future versions.	—	This is out of the project's scope

3.4.2 Methods, Tools, and Techniques

Throughout the phases of system development, a variety of methodologies, tools, and techniques will be employed.

Table 23:Used Tools

Phase	Tool	Method	Standards
Planning	· Microsoft Teams · Google Docs · Google Sheets · Google Drive · Microsoft Word · WhatsApp	Waterfall Model	IEEE Std 1058 – 1998
Requirements Analysis and Definition	· Google Docs · Microsoft Word		
System and Software Design	· Google Docs · Microsoft Word · Axure		
Implementation and Testing	· Python Programming Language · VS Code · GPUs · Google Colab · GitHub · Jupyter Notebook		

3.4.3 Infrastructure

The following table outlines the necessary hardware, operating systems, networks, software, and facilities for the development environment of the team members.

Table 24:Infrastructure Required

Hardware	· A personal laptop for each team member · GPUs
Software	· Microsoft Teams · Microsoft Word · Google Doc · Google Drive · Python · VS code · Google Colab · GitHub · Kaggle · Jupyter Notebook
Operating System	· Windows 10 or later · MacOS
Network	· High speed internet. · Local Area Network Connection
Facilities	· An office space for team members. · Resources such (printers, scanners, projectors, etc.) · King Fahad University Hospitals.

3.4.4 Product Acceptance

The primary factor determining customer satisfaction is achieving the highest possible accuracy in the deliverables using the most effective machine learning techniques. The SRS document must include all necessary specifications to capture the clients' interest. Additionally, we engaged a neuroscientist to formally evaluate the client's approval at each phase. To ensure successful completion of all project stages, Dr. Mustafa Youldash, Dr. Modhi Alkhaldi, and Dr. Zakia Yasawy will provide continuous feedback and approval for advancing to the next stage. Before release, the deep learning model will undergo testing using a labeled EEG testing set, and neuroscientists will assess its performance using their own data.

3.5 Supporting Process Plans

3.5.1 Documentation

Different documents will be delivered at different points in the timeline of the project. All team members will write and then review everyone's docs. The paper to be produced is listed on the table below.

Table 25: System Documentation

Document Type	Format Standard	Prepare Document	Review Document
Introduction and literature review	Provided by the supervisor	All team members	Dr. Mustafa Youldash, Dr. Modhi Alkhaldi and Dr. Zakia Yasawy.
Mid semester report	Provided by the supervisor		
Software Project Management Plan (SPMP)	IEEE Standard 1058-1998		
Software Requirements Specification (SRS)	IEEE Standard 1058-1998		
Methodology	Provided by the supervisor		
Software Design Specification (SDS)	IEEE Standard 1058-1998		
First Semester Final Report	Provided by the supervisor		
Software Test Plan	IEEE Standard 1058-1998		
User Manual	IEEE Standard 1058-1998		
Senior Project Source Code and Report	Provided by the supervisor		

Chapter 4: Methodology / Software Requirements Specification (SRS)

4.1 Methodology

4.1.1 Operational Framework

The project's goal is to develop a technique for accurately detecting epilepsy seizures and classifying the type of seizure using deep learning. Working with Convolutional Neural Networks (CNNs), Bidirectional Long Short-Term Memory Networks (BLSTMs), Graphic Neural Network (GNNs), ShallowFBCSPNet (Shallow Filter Bank Common Spatial Pattern Network), DeepEEGNet, and CNN-PGM (Convolutional Neural Network with Probabilistic Graphical Modeling). The strategies shown together make an effective system to identify epilepsy. In addition, we may investigate additional approaches, such as the Short-Time Fourier Transform (STFT) utilized by Truong et al [39] and other pretrained CNNs.

The description of the development of the intelligence systems in 4.1.1, and section 4.1.2 provides a full explanation of the model architecture. Section 4.1.3 outlines the performance metrics, and section 4.1.4 summarizes the whole chapter.

4.1.2 Step 1: Data Gathering

The model's effectiveness is dependent on the dataset; hence we want to use data that has been improved by specialists. In addition, we can include publicly available datasets that were used in many lectures reviews we read, such as CHB-MIT [43] and Siena Scalp EEG Database [50]. Initially, we will collect a variety of EEG recordings and export them as numpy matrix. We take care to curate this dataset to include a wide range of seizures, allowing the model to adapt and perform well across many conditions.

4.1.3 Step 2: Data Pre-processing

Because EEG signals are often noisy, pre-processing is necessary to improve dataset quality. Wavelet decomposition is a method for removing noise and artifacts, including eye blinks. Data augmentation helps to have more diversity in the dataset, allowing the model to recognize patterns in several versions of the same image.

4.1.4 Step 3: Model Development

According to prior studies, CNN-PGM, ShallowFBCSPNet, DeepEEGNet and BLSTM are optimal for EEG data. Other approaches, such as the Graphic Neural Network (GNN), can capture non-Euclidean correlations in EEG data. Furthermore, it outperformed other methods such as CNN and displayed high performance across unseen patient data, demonstrating strong generalization based on the work of Abadal et al. [44]. The model will be trained and tested using preprocessed images. The performance of these models will be evaluated to determine their acceptability, which will inform the enhancement or substitution process in the following phase.

4.1.5 Step 4: Model Training and Testing

The dataset will be trained using a variety of deep learning models, each with its own architecture and hyperparameters, we will be modifying parameters such as learning rates, dropout rates, and architectural configurations. The dataset will also be divided into three sets: a training set, a validation set and a testing set, to assess the performance of each model.

4.1.6 Step 5: Models Prediction

This phase will be used to employ the final version of the model, after completing the previous phases.

4.1.7 Step 6: Result Interpretation

This phase will go over the specific performance metrics described in section 4.1.3. They will be utilized to evaluate and comprehend the prediction results.

4.1.8 The Proposed Individual Computational Intelligence Approaches

This section describes the computational intelligence approaches utilized in the study, including ShallowFBCSPNet, DeepEEGNet, CNN-PGM, BLSTM and GNN

4.1.8.1 ShallowFBCSPNet

ShallowFBCSPNet is a deep learning model from btaindecode designed for EEG, brain computer interface and epilepsy detection task. It uses the Filter Bank Common Spatial Pattern (FBCSP) methods in combining with a neural network architecture to extract information such as discriminative spatial and frequency properties from raw EEG data. The network's shallow design focuses on learning bandpower features from the temporal convolution, spatial filtering, and a logarithmic activation function, making it excellent for EEG data with small sample numbers and noise levels.

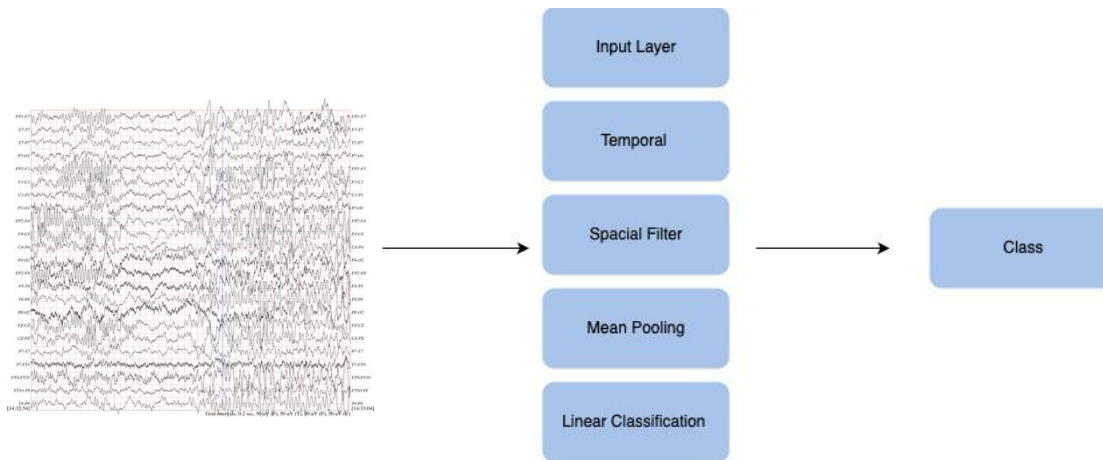


Figure 4:Architecture of ShallowFBCSPNet.

4.1.1.1 BLSTM

LSTM networks are typically used in time-series analysis [47]. which is perfect given that the EEG signals are time series data. The network operates by treating the hidden layer as a memory unit, and it can deal with correlation within time series in both the short and long term [47]. Because the network operates in a sequential way, it can assist in predicting an impending seizure. To have more advantage we used BLSTM to run our input in 2 ways (Past to future) and (Future to past), this approach is suitable because using information from the future is easier for the network to understand.

In our study, we have the choice to utilize BLSTM alone because it can accept raw EGG data and keep a memory cell to indicate an oncoming seizure, or we can combine it with CNN to use both models' benefits make use of the other models. BLSTM are capable of detecting tiny changes in signals, detecting upcoming seizures, and processing data from numerous EGG channels. However, the network has numerous problems, ranging from high training costs to overfitting. One of the most effective methods to use BLSTM in a project is to combine it with other models.

4.1.1.2 Graphic Neural Networks

Kipf and Welling proposed the GNN war in 2016. [48], the features of a given node in each layer are averaged with those of its neighbors, weighted by edge features. The collected features are then multiplied by a weight matrix and processed using a nonlinear activation function. The entire operation is performed as many times as layers [44]. Based on the figure presented by [48], we may optimize the model's information propagation on graphs by conditioning it on both the data X and the underlying graph structure's adjacency matrix A [48].

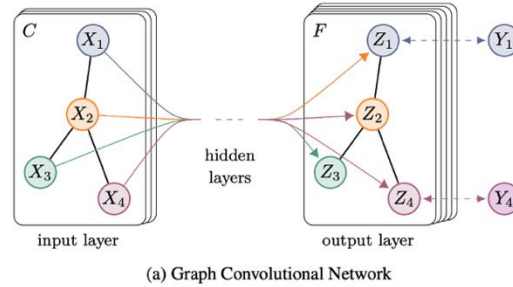


Figure 5: Graph Convolutional Network ([44])

In our work, we believe that the GNN model is superior in the long term because it can capture non-Euclidean correlations in EEG data, whilst the generalized model can handle enormous amounts of EEG. However, there is a drawback. Building the model can be time-consuming and expensive, and it requires enough data to learn useful graph structures. We can also use RNN as an option. When compared to CNN+LSTM, GNN clearly outperforms them. Also, we can use other GNN architectures such as Graph Convolutional Networks (GCN), which extract features from neighboring nodes.

4.1.1.3 DeepEEGNet

A deep learning model designed for EEG classification from Braindecoder. Using hierarchical convolutional neural network architecture to learn spatiotemporal features from multi-channel in the EEG raw data eliminating the need for manual feature extraction. The model incorporates information such as temporal and spatial convolutions to capture frequency-specific patterns while minimizing overfitting. DeepEEGNet can detect both seizure and non-seizure events with high accuracy, making it ideal for a binary classification model and real-time monitoring and clinical assistance.

4.1.1.4 CNN-PGM

A hybrid deep learning approach that combines CNN's robust feature extraction capabilities with Probabilistic Graphical Models' hierarchical logic. In EEG-based epilepsy detection, the CNN component

extracts spatial and temporal information from raw EEG data, recognizing complex patterns associated with seizure activity. These extracted characteristics are then sent into a PGM, which simulates connections and uncertainty between the features or across EEG channels. This two-stage approach improves interpretability and classification performance by combining learnt descriptions with probability judgment.

4.1.2 Performance Evaluation of the Proposed Models

The percentage of accurate predictions compared to all observations was known as accuracy. The precision metric quantified the percentage of accurate positive predicts among all positive predictions. The ratio of accurately identified positive predictions to the number of positive observations is often referred to as recall, sensitivity, or True Positive Rate. Ultimately, the F-score achieved a compromise between precision and recall by combining the two into a single measure.

Equation 1 : Accuracy.

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Equation 2: Precision.

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Equation 3: F1 Score.

$$F1 - Score = 2x \frac{\text{Precision} x \text{Recall}}{\text{Precision} + \text{Recall}}$$

4.1.3 Integration with Large Language Models

The desired outcome is an LLM that will assist a thorough interpretation of the diagnostic information derived from the EEG dataset, which will benefit many specialists. This integration will help to construct a sophisticated tool that blends image interpretation with textual insights based on medical experience.

4.1.4 Summary

In summary, we suggest combining processing and computational intelligence methodologies for model construction. These strategies will be applied to the dataset we require from Saudi hospital that have been improved by specialists. Our first solution is ResNet18, which was designed to address the performance degrading issue. The model is more efficient and faster since it skips layers, however it can only analyze 2D data, whereas our EEG data is 1D. The second method is LSTM, which uses time-series analysis, which is perfect for EEG data, and hidden layers as memory units to predict seizure, although it can be expensive. Finally, GNNs can catch complicated patterns, process enormous amounts of EEG data, and are extremely generalizable. Combining GNN with LSTM can enhance its effectiveness

4.2 Software Requirements Specification (SRS)

4.2.1 Introduction

This section presents a comprehensive analysis of the system, outlining its overarching perspective, principal functionalities, and the defining attributes of its target user base.

4.2.1.1 Overview

The SRS chapter is organized into six comprehensive sections that collectively address all project requirements:

- **Section 1 Introduction:** This section introduces the SRS, outlining the chapter's structure and objectives.
- **Section 2 Overall Description:** This section elaborates on the system's perspective and core functionalities while defining the characteristics of its intended users.
- **Section 3 Specific Requirements:** This section details the requirements for external interfaces and the functional needs of system users.
- **Section 4 Performance Requirements:** This section specifies the system's performance expectations, including response time, availability, fault tolerance, recovery mechanisms, and capacity.
- **Section 5 Design Constraints:** This section outlines design constraints, such as the implementation language, as well as software and hardware limitations.
- **Section 6 Software System Attributes:** The final section explores critical software attributes, including usability, reliability, availability, safety, information security, privacy, maintainability, and portability.

4.2.2 Overall Description

This section aims to offer a detailed overview of the system, highlighting its product perspective, primary functionalities, and the attributes of its target users.

4.2.2.1 Product Perspective

The proposed system is a stand-alone application developed to assist in diagnosing epileptic and psychogenic seizures. By incorporating advanced technologies such as CNN-PGM, BLSTMs, GNNs, ShallowFBCSPNet, DeepEEGNet and LLMs, it delivers precise and efficient diagnostic solutions using EEG data. While the current focus is on seizure classification, the system is designed with future scalability in mind, allowing for the inclusion of additional neurological conditions. Drawing on a carefully curated dataset, it builds robust classification models to enhance diagnostic accuracy and improve patient outcomes. The system supports user such as Doctors and patients, fostering a collaborative and comprehensive diagnostic process.

4.2.2.2 User Roles and Responsibilities

The system supports 2 distinct user roles, each playing a critical part in its functionality and the

diagnostic process:

- **Doctors**

Doctors utilize diagnostic models to input and analyze patient EEG data, review diagnostic histories, and generate detailed patient reports. They provide valuable feedback to enhance the accuracy and performance of the AI models.

- **Registered Users (Patients)**

Patients have the option to book appointment with doctors, view medications, have access to view and download their EEG reports, and chat with AI assistant empowering them to stay informed and take an active role in their treatment journey.

4.2.2.3 Product Functions

The primary objective of the system is to provide accurate diagnosis and differentiation between epileptic and psychogenic seizures while delivering robust functionalities to support its user base. Each user role is equipped with customized features designed to enable a seamless and efficient diagnostic process. The key functions of the system are as follows:

❖ Account Management

- Users can log in to their accounts, manage their profiles, and log out when finished.
- Doctors can update their email and password to maintain account security.
- Registered users have the flexibility to create or delete their accounts as needed.

❖ Diagnosis and Prediction:

- Doctors can input patients' EEG data into the model to generate diagnostic predictions for epileptic, non-epileptic or psychogenic seizures.
- Users can download EEG report for record-keeping or consultation purposes.

❖ Viewing Patient Diagnostic History

- Doctors can access diagnostic histories for informed decision-making and reference:
 - Doctors can review the complete diagnostic history of patients, including model accuracy and relevant details.
 - Registered users have access to their own diagnostic history for personal reference and tracking.

❖ Generated AI:

- Doctors can generate a full patient report after uploading the EEG case.
- Patient can access an AI assistant chatbot to ask health-related questions.

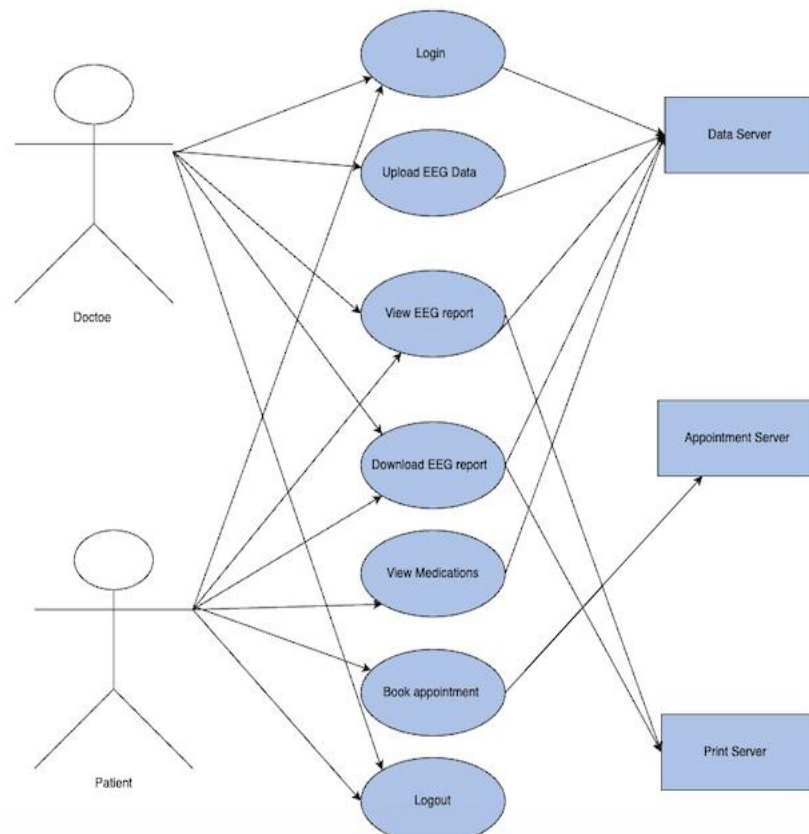


Figure 6: System use-case diagram

4.2.2.4 User Characteristics

The system's user base consists of 2 distinct groups: admins, Doctors and Patients. Table 26 provides a detailed overview of the general characteristics and roles of each user group.

Table 26: User Characteristics

User	General Characteristics	
	Education Level	Technical Expertise
Doctor	Holds a technical bachelor's degree in medicine.	• Has basic computer literacy. • Trained to operate the system. • Experienced in imaging and diagnostic tools. • Knowledgeable in diagnosing and managing Epilepsy & PNES.
Registered user	Any educational background is sufficient.	• Basic understanding of computer usage.

4.2.3 Specific Requirements

The section contains a detailed description of the system's requirements, including both external interfaces and features needed by each user of the system.

4.2.3.1 External Interfaces Requirements

This section describes the external interfaces used in our system, namely users, hardware systems, and software interfaces.

4.2.3.1.1 User Interface

The user interface is the main gateway for interaction, where one can seamlessly communicate and effortlessly execute functions with ease. A consistent structure anchors navigation while rigorous security safeguards sensitive information.

4.2.3.1.2 Login Interface

The login interface serves as the gateway allowing authorized users into the system. It requests that the user provide verified credentials for authentication purposes. Should the supplied username and password match what is stored securely in the system, access will be granted. Otherwise, an error message will prompt the user to re-enter or reset login details. A concise breakdown of the primary input fields appearing on the login interface can be found in Table 27 below

Table 27: The Main Fields of the Login Interface

Field Name	Format	Level	Input/Output	Comment
Username	Integer	Required	Input	Unique
Password	Encrypted Text	Required	Input	The entered password must be at least 8 alphanumeric

4.2.3.1.3 Dashboard Interface

Users are brought to their dashboard with a prominent profile tab upon logging in. Selecting this tab unveils the profile interface containing personalized details. Patient can have easily view their EEG analysis, track seizure trends, medications and upcoming appointment.

Key profile fields and their uses are outlined in Table 21 for reference. Distinct permissions preserve data privacy and security, while empowering user autonomy over personal medical records through a versatile interface. The main fields of the Profile Interface are briefly described in Table 28.

Table 28: The main fields in dashboard interface

Field Name	Format	Level	Input/Output	Comment
Name	Text	Required	Input/Output	-
Role	Text	Required	Output	-

4.2.3.1.4 EEG Report Interface

This interface enables the users to easily view their EEG reports. Table 29 gives a description of the fields in the changed email interface.

Table 29: The Main Fields of the EEG Report

Field Name	Format	Level	Input/Output	Comment
ReportID	Text	Required	Input	Unique
Date	Date	Required	Input	-
Result	Text	Required	Input	Class output

4.2.3.1.5 Appointment Interface

This interface enables the users book appointment with doctors. Table 30 gives a description of the fields in the changed password interface.

Table 30: The Main Fields of the appointment interface

Field Name	Format	Level	Input/Output	Comment
Select Doctor	Dropdown	Required	Input	-
Data & Time	Date & Time	Required	Input	-
Reason for visit	Text	Optional	Input	-

4.2.3.1.6 Contact Interface

This tab enables other user to contact us through the form. As outlined in Table 31, key details of the main fields in the Manage Users Tab.

Table 31: The Main Fields of the Manage Users Tab in the contact interface

Field Name	Format	Level	Input/Output	Comment
Email	Text	Required	Input/Output	Unique, and must include the @ symbol
Phone Number	Integers	Required	Input/Output	-
Email	Text	Required	Input/Output	-
Message from	Text	Required	Input/Output	-

4.2.3.1.7 Upload EEG Interface

Doctors have unique access to the Diagnosis Interface. This interface allows doctors to upload a patient's retina image to the model and start the diagnostic process by pressing the "Diagnose" button, which displays the results. Table 32 summarizes the major fields of the Diagnosis Interface.

Table 32: The Main Fields of the Upload EEG interface

Field Name	Format	Level	Input/Output	Comment
EEGID	Text	Required	Input	Unique auto generated
Date	Date	Required	Input	-
Patient Name	Text	Required	Input	-
Age	Integer	Required	Input	-
Gender	Dropdown	Required	Input	-
Notes	Text	Optional	Input	Add symptoms.

4.2.3.1.8 Hardware Interface

A hardware interface is insignificant for the proposed system because it does not require direct communication with hardware components. Instead, the system is designed to work fluidly with any operating system GPU and CPU.

4.2.3.1.9 Software Interface

The requirements for software interfaces include:

- The system must work with both macOS X and Windows operating systems.
- The system must be able to read from and write to a MongoDB database in order to manage patient records and EEG findings.
- The system must be able to read EEG data files in standard formats, such as. edf, as uploaded by the doctor. Communication Interface Using MongoDB Connector/Python, the communication interface guarantees effective data interchange between the application and the MongoDB database. When creating an account, automated username verification improves user authentication. Additionally, it enables smooth interactions for tasks like retrieving diagnostic findings, maintaining accounts, and uploading EEG data.

4.2.3.1.10 Communication Interface

Through the use of MongoDB Connector/Python, the communication interface guarantees effective data interchange between the application and the MongoDB database. When logging an account, automated username verification improves user authentication. Additionally, it enables smooth interactions for tasks like retrieving diagnostic findings, maintaining accounts, and uploading EEG data.

4.2.3.2 Functional Requirements

The different categories of end users and their roles—Doctors and Patients (registered users) are used to classify functional needs.

4.2.3.2.1 Common Functionalities

All users share several features, including the ability to log in examine their EEG report history, and print findings.

4.2.3.2.2 Login

Table 33 shows the “Login” functionality.

Table 33:Login Functionality

Actors	• Doctors • Patients
---------------	--

Description	• User will be able to log in using the username • For authentication, the system compares the supplied credentials to the database. • Invalid inputs result in clear error messages.
Data	• Username. • Password.
Stimulus	Every end user clicks the Login button to log in to the system.
Response	• The dashboard homepage indicates if the user's long-term password was successfully entered.
Abnormal condition	• The user will receive the message "Incorrect username or password" if they input either the password or their username incorrectly. You can try logging in again after seeing the "Please try again" prompt.

Figure 7 demonstrates the activity diagram of the “Login” functionality

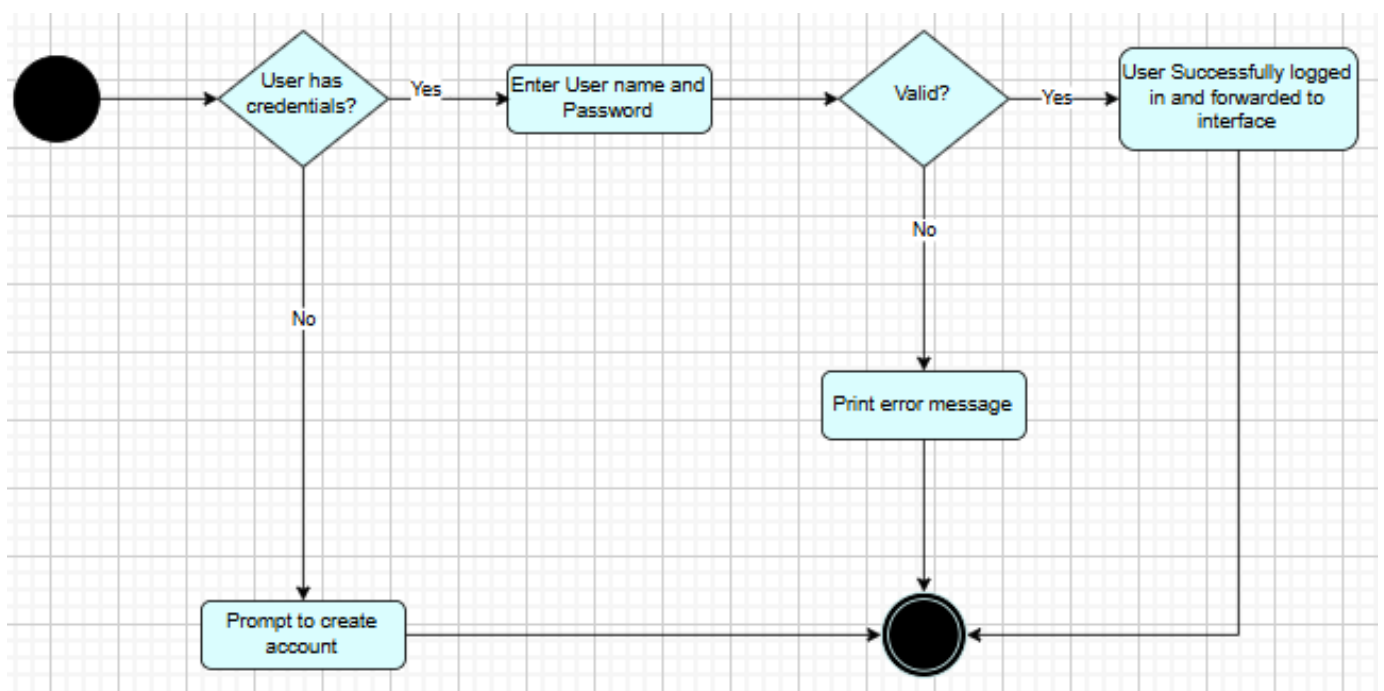


Figure 7: The Activity Diagram of the Login Functionality

Table 34 shows how to use "View Diagnosis." Patients and Doctors are able to view all diagnosis related data through this interface.

Table 34:View Diagnosis Functionality

Actors	• Doctors • Patients
Description	A patient's previous diagnosis must be accessible to both patients and doctors.
Data	Doctors: • Patient ID
Stimulus	Doctors: • By entering the patient's MRN and pressing the search button, the patient diagnosis is accessed via a table For users who have registered: • Clicking "EEG Reports" gives the user command.
Response	For nurses and Doctors: • It will be shown if the database contains a record of the prospective patient's prior diagnosis. For users who have registered: • A history of previous diagnoses will be retrieved if it is present in the database.
Abnormal conditions	For nurses and Doctors: • Invalid MRN. For users who have registered: • None

The activity diagram for the "View Diagnosis" feature is shown in Figure 8.

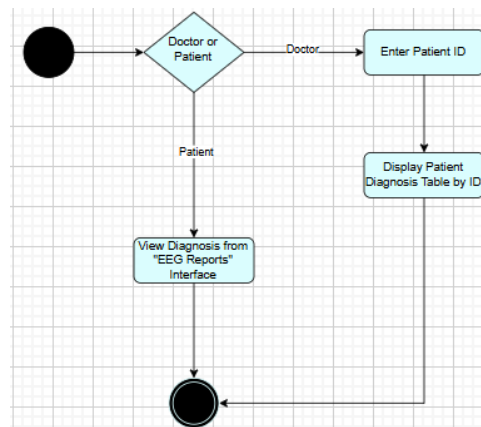


Figure 8: The Activity Diagram of the View Diagnosis History Functionality

4.2.3.2.3 Download and View Results

The "Download and View Results" functionality of the system allows doctors to view and download a PDF containing diagnosis results and model working summary Table 35 describes the “Download and View Results” functionality.

Table 35:Download and View Results Functionality

Actors	• Doctors.
Description	A PDF file report is created using a base LLM with all information about model analysis and final diagnosis.
Data	The diagnosis results.
Stimulus	When the "Print Result" button is clicked, the user command is sent.
Response	Once the diagnostic findings have been converted to PDF format, they will be forwarded to the interface to be accessed via the download button or the view button.
Abnormal conditions	LLM downtime.

The activity diagram for the "Download and View Results" feature is shown in Figure 9.

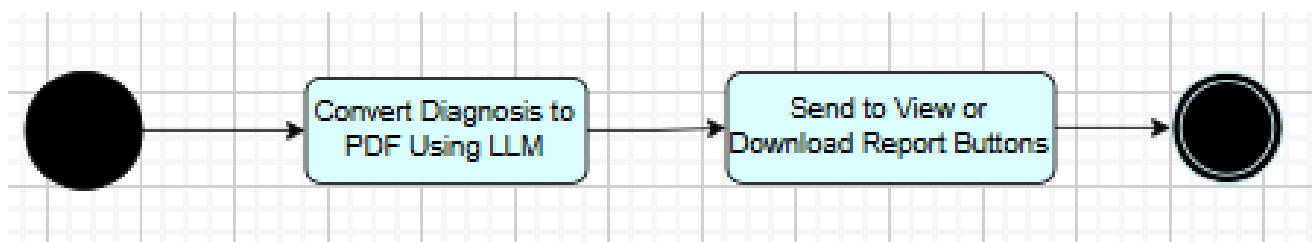


Figure 9:The Activity Diagram of the Download and View Results Functionality

4.2.3.3 User Class 1: Doctor

The functionalities of medical specialists are covered in this section.

4.2.3.3.1 EEG Upload & Diagnose

Table 36 shows the “EEG Upload & Diagnose” functionality.

Table 36: EEG Upload Diagnose Functionality

Actors	Doctors
Description	• Medical specialists should be able to distinguish between epileptic seizures and psychogenic non-epileptic seizures (PNES) by analyzing the patient's EEG data and clinical history. • The diagnosis result will indicate the seizure type, confidence level. • The diagnosis result will be stored along with the model accuracy.
Data	EEG data (electroencephalogram), and patient information
Stimulus	The user command is issued by clicking on the “Diagnose” button after uploading the necessary data.
Response	• The uploaded EEG data is processed using the classification model. • The system identifies the seizure type (epileptic or PNES) and displays the result along with a confidence score. • Diagnostic results, accuracy levels, and timestamp are stored in the diagnosis report for each patient.
Abnormal conditions	• Missing or corrupted EEG data files. • High levels of noise in the EEG signal that prevent accurate classification.

Figure 10 illustrates the activity diagram of the “EEG Upload & Diagnose” functionality.

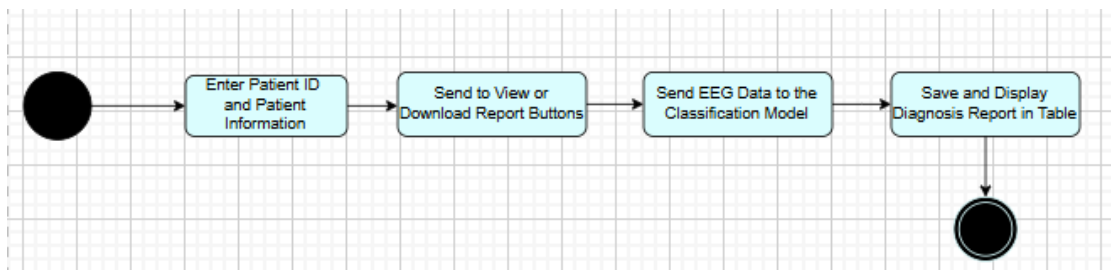


Figure 10: The Activity Diagram of the Diagnose Functionality

4.2.3.4 User Class 2: Registered User (Patient).

This section explains the functions of a registered user.

4.2.3.4.1 Book Appointment

Table 37 section displays the description of the “Book Appointment” functionality.

Table 37: Book Appointment Functionality

Actors	Patients
Description	• Patient is able to book an appointment with the institution by choosing appointment time, day, etc...
Data	• Doctor • Appointment Type • Date • Time • Reason to visit
Stimulus	The user action is initiated by filling out all needed fields and clicking the "Book Appointment" button.
Response	An account is created if all required information is filled.
Abnormal condition	• Doctor Name Invalid • Reason to visit invalid characters

Figure 11 shows the activity diagram of the “Book Appointment” functionality.

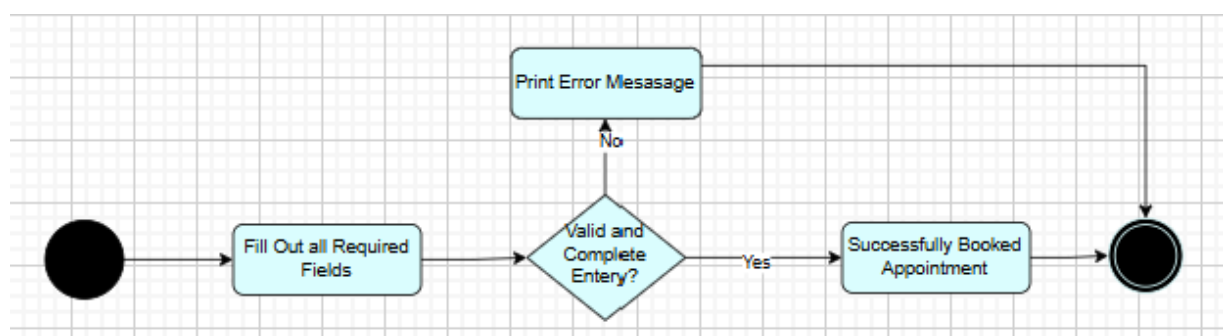


Figure 11: The Activity Diagram of Book Appointment Functionality.

4.2.4 Performance Requirements

The employed performance requirements are based on certain criteria to approximate the Response Time, Availability, Fault Tolerance Recovery, and Capacity.

4.2.4.1 Response Time

Generating the diagnostic report for the model takes up the most time due to the heavy LLM approximately around 20 seconds. Accordingly, the entire system must respond within a maximum of 30 seconds.

4.2.4.2 Availability

Availability determines the accessibility of the system. The proposed system should be available 24/7 for diagnosis.

4.2.4.3 Fault Tolerance Recovery

To ensure the seamless operation of the **EpilepTech** system, it is essential to promptly identify and address any errors caused by user input, hardware malfunctions, or operating system failures. The system incorporates robust fault tolerance and recovery mechanisms to maintain its normal functionality under such circumstances. The following steps outline how the system manages faults:

- **User Faults:** The system validates all user inputs to prevent errors during operation. If invalid or erroneous input is detected, the system generates an appropriate warning message, notifying the user of the issue. Clear instructions are provided to guide the user in correcting the input, ensuring smooth and error-free interaction.
- **System Faults:** In the event of a system failure, the system employs recovery protocols to restore functionality. This includes reverting to the last known stable state to minimize disruptions. The system logs error details for further analysis and ensures that diagnostic processes resume seamlessly, minimizing impact on user operations and maintaining reliability.

4.2.4.4 Capacity

A computer must have minimum available storage of 60 MB to utilize the full potential of the system's functionalities and uses.

4.2.5 Design Constraints

The **EpilepTech** system is a web-based application designed with the following tools and technologies to support its processes:

- **Python:** Used for preprocessing EEG datasets and for the development, training, and testing of the diagnostic models, including ShallowNet, DeepNet, CNN-PGM, GNN, and BLSTM frameworks.
- **MongoDB:** Serves as the database to store patient data, diagnostic history, and model configurations efficiently and securely.
- **Fine-Tuned LLMs:** Specifically tailored to enhance communication with medical specialists by delivering detailed insights and clear explanations of diagnostic results in a downloadable PDF, ensuring accurate interpretation and informed decision-making. Also, provide patient communication with a chatbot to serve as a base for question answering.

4.2.6 Software System Attributes

4.2.6.1 Usability

The system utilizes universal design principles to create intuitive and user-friendly interfaces, allowing for faster user onboarding and lowering the risk of operational errors.

4.2.6.2 Reliability

System reliability is assured by the use of sophisticated input validation techniques. Error messages are clear and helpful, alerting users to any system faults. The system's reliability is further proved by the confidence of its predictive diagnosis model.

4.2.6.3 Availability

To ensure user trust and reliability, the system must be available 24 hours a day, seven days a week. Any downtime may have a negative impact on the user experience and ability to successfully use the system. Thus, ensuring ongoing availability is critical for maintaining the system's functionality and dependability.

4.2.6.4 Safety Requirements

The system must be built to function without causing any damage to the hosting server or interfering with other software or internal components.

4.2.6.5 Security and Privacy

Patient data is securely maintained within the system, and comprehensive security procedures have been deployed to protect critical information. The key security measures include:

- To provide safe login credentials, passwords are hashed before being saved in the database. During authentication, the system validates user-entered passwords by comparing them to previously saved hashed passwords.
- To improve session security, the system automatically logs out users who have been idle for longer than 15 minutes.

4.2.6.6 Maintainability

The system maintains extra medical data for Epilepsy and PNES by allowing the administrator to update the diagnostic model. This assures that the model is still accurate and effective in diagnosing Epilepsy and PNES in the present medical environment.

4.2.6.7 Portability

The system can be operated from any browser connected to a server that contains Python and medical data.

Chapter 5: Software Design Specification (SDS)

The Software Requirements Specification (SRS) document outlines the various components and interfaces discussed in this chapter. It provides a thorough overview of the system architecture, emphasizing its evolution to assure the establishment of an efficient system that meets the client's needs.

The SDS is carried out in two distinct stages:

- **The Initial Design Phase** provides a high-level overview of the system architecture and data components.
- **The Detailed Design Phase** This phase defines the system's overall design. It offers thorough information about the project's objective, scope, acronyms, and references.

5.1 System Overview

Early diagnosis of epilepsy and psychogenic non-epileptic seizures (PNES) is essential for improving patient outcomes and addressing related problems. The approach is intended to aid in the timely diagnosis of various illnesses, resulting in major advantages for both healthcare professionals and society. Each user is assigned specific tasks within the system, which helps to achieve the goal of early diagnosis and successful management of Epilepsy and PNES.

5.1.1 System Functionality

The system's functionality is intended to give users access to specified regions based on their given privileges. Each user has certain tasks within the system. There are two different user roles: doctor, and patient. The functionalities associated with each role are described in the following subsections.

5.1.2 Common Functionalities

Users can log in to use the system's features and log out when they finish their assigned duties.

5.1.2.1 Doctor

- Upload EEG files.
- View past and present EEG reports.
- View patient upcoming appointments.
- Update the medical information for Patient.
- Generate and provide diagnoses based on available medical information.
- Print the diagnosis results for your records or future use.

5.1.2.2 Patient

- Create and print diagnostic results for reference or documentation.
- Access and read the history of past diagnoses.
- Book an appointment.
- Access a medical chatbot.
- Update your personal profile information, including account information.

5.2 Design consideration

The design solution entails defining and fixing the system's existing challenges. Some of the potential concerns will be covered in the following sections.

5.2.1 Assumptions and Dependencies

In this subsection, we will look at the system's assumptions and dependencies, as well as their applications. The sections that follow address a variety of topics.

5.2.1.1 Related Software or Hardware

The system is intended to work seamlessly with many web browsers, including Microsoft Edge, Safari, and Chrome. It works with a MongoDB database to efficiently process, save, and retrieve data.

5.2.1.2 Operating System

Our system is specifically developed and optimized to run well on Windows 11 and macOS 12. To execute the system with Python, users must first download the Python runtime environment. This will allow for seamless and hassle-free use of our system, giving users a smooth and delightful experience.

5.2.1.3 End-User Characteristics

The system comprises two end users: doctor, and patient. Section 4.2.2 of the SRS (Chapter 4) specifies the roles and privileges of each user.

5.2.1.4 Possible and/or Probable Changes in Functionalities

Specific changes are permissible if they improve the project's overall quality. Otherwise, large adjustments are not possible due to the project's time constraints, and the most significant alterations are time-consuming.

5.2.2 General Constraints

This section identifies critical constraints and limitations that have a substantial impact on the overall architecture of the software system.

5.2.2.1 Hardware or Software Environment

To function properly, the suggested system must be compatible with both Windows 11 and macOS 12. The system also requires MongoDB to support database connectivity and manage data storage and retrieval activities.

5.2.2.2 Interface Requirements

Interfaces should be simple enough for people to comprehend how the operating system works. When developing a system, consider the following design considerations:

- Whenever possible, choose the simplest and least error-prone input components.
- Provide detailed error and warning messages to address user errors.
- Use readable fonts, labels, and colors.

5.2.2.3 Data Repository and Distribution Requirements

The system's data should be stored in a MongoDB database. The system should handle the data with care to avoid accidental data loss.

5.2.2.4 Security Requirements

Users will be granted access to the system based on their positions, ensuring confidentiality.

5.2.2.5 Performance Requirements

A thorough examination and testing of all performance requirements described in Section 4.2.4 of Chapter 4 of the SRS is required to ensure compliance with the prescribed criteria.

5.2.2.6 Verification and Validation Requirements (Testing)

System validation and verification are undertaken to ensure that all specifications and requirements have been met. For a test to be considered successful, every requirement must be implemented within the system. Any variation from the set parameters will result in the test being deemed failed.

5.3 User Interface Design

This section presents a detailed review of user interface design, with an emphasis on design concepts, screen elements, and user activities.

5.3.1 Overview of User Interface

The login page is the system's primary interface, allowing users to access their accounts with varied capabilities to maintain system security.

5.3.1.1 Doctor Interface

The doctors have three tabs with the following options:

❖ Diagnosis History:

- View patients' Diagnosis History.
- Print the outcomes of diagnosis.

❖ Diagnose:

- Perform diagnoses for patients.

❖ Profile:

- Update profile information, including email and password.

5.3.1.2 Patient Interface

The Patient must have account to access the system. Then, an interface is displayed with the following tabs:

❖ Diagnosis Results:

- View a history of past diagnoses.
- Print the diagnostic results for future reference.

❖ Profile:

- Edit their profile information, such as name, date of birth, and gender.
- Change email and password.

5.3.2 Interface Design Rules

To properly meet user needs, interfaces must be intuitive, efficient, and consistent. A crucial part of the design is to reduce user confusion while increasing usability. To achieve universal usability, Shneiderman's

Eight Golden Rules of Interface Design [49], which include the following principles:

<https://www.cs.umd.edu/users/ben/goldenrules.html>

- **Strive for consistency:** Consistency is critical in helping users become acquainted with the interface. Predictable patterns should be consistent across menus, color schemes, and fonts. Furthermore, the language used in prompts, error messages, and menus should be clear and consistent, assuring simplicity of use.
- **Seek universal usability:** Interfaces should be user-friendly for both beginners and experts, with clear explanations and many navigation options. This strategy ensures that all users have a convenient and high-quality experience.
- **Offer informative feedback:** The system must provide users with relevant and human-readable feedback after major actions or sequences. This feedback allows users to comprehend the current condition of the system and confirms the effectiveness of their activities.
- **Design dialogs to yield closure:** Interfaces should clearly define the procedures necessary to execute an action. Users should be able to identify the start and finish of a process and validate its effective completion, which improves user happiness and comprehension.
- **Prevent Errors:** Designers must anticipate probable user errors and include tools to address them efficiently. When erroneous input is detected, the system should clearly show where the issue occurred and provide a succinct, intelligible message to help the user remedy it. For example, the interface should prevent inadvertent selections or impose input limits, such as demanding numeric entries when alphabetic letters are not permitted. By implementing these design principles, the system can dramatically reduce the risk of user errors while improving overall usability.
- **Permit easy reversal of actions:** Users should be able to undo single or multiple actions with ease. This feature fosters discovery and lowers the fear of making mistakes, increasing user confidence and enjoyment.

- **Support internal locus of control:** The interface should empower users by giving them unambiguous authority over system operations, making them feel in command of their interactions and the system's results.
- **Reduce short-term memory load:** Designers should create interfaces that reduce cognitive strain by allowing users to rapidly access and retrieve data from past interactions, minimizing the reliance on memory to complete tasks.

5.3.3 Screen Images

In this section, we will describe the interfaces of the system, using React frontend screenshots.

5.3.3.1 System Access Interfaces

5.3.3.1.1 Home Page Interface

The “Home Page” interface will be displayed to all users. It contains the project logo and slogan, and a navigation bar to access the services, AI, resources and contact, and a login button for patient and doctor to access the website. Figure 20 shows the home page interface.

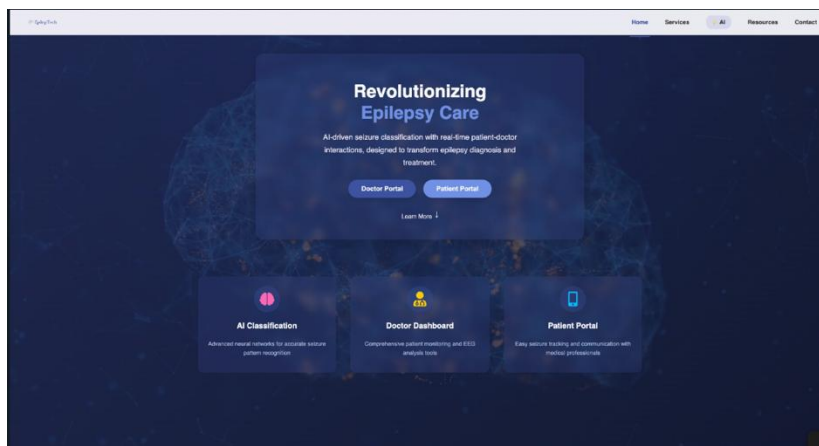


Figure 12: Home Page Interface

5.3.3.1.2 Services Interface

The services interface starts with 6 labels each relating to the services the website offers to the system, “EEG Analysis with AI”, “Doctor-Patient Consultations”, “Automated Epilepsy Diagnosis”, “Medical Report Generation”, “Seizure Classification” and “Treatment Efficacy Analysis”.

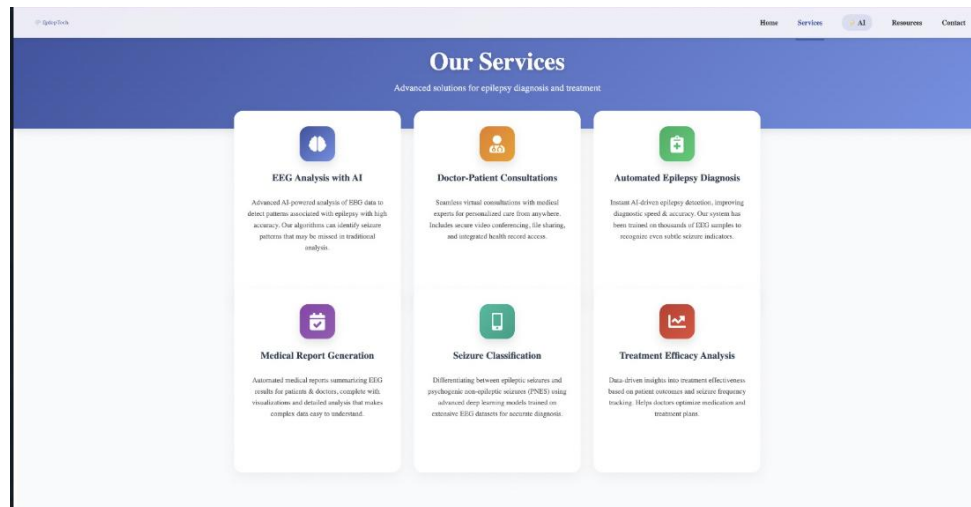


Figure 13: Services Interfaces

5.3.3.1.3 AI Interface

The AI interface provide the basic concept underlying EpilepTech's AI survives, which is to improve epilepsy diagnosis and patient treatment by utilizing cutting-edge artificial intelligence models, to provide precise, efficient, and scalable healthcare solutions using deep learning. Also, showcasing handpicked medical articles and scientific inputs. Including “Deep Learning Analysis” to use precise EEG-based seizure classification, especially discriminating between epileptic and psychogenic seizures (PNES). “AI Chatbot” for users to use it anytime to ask about medical questions. “High Accuracy” meaning the model has a very high accuracy with great generalization. “Neural Network” using them to train on several datasets to achieve generalizability. "Extensive Training" ensures that the models are carefully tuned for excellent diagnostic accuracy. Also, “Human-AI Collaboration” guaranteeing that clinical expertise is integral to decision-making while harnessing AI for increased efficiency and scalability.

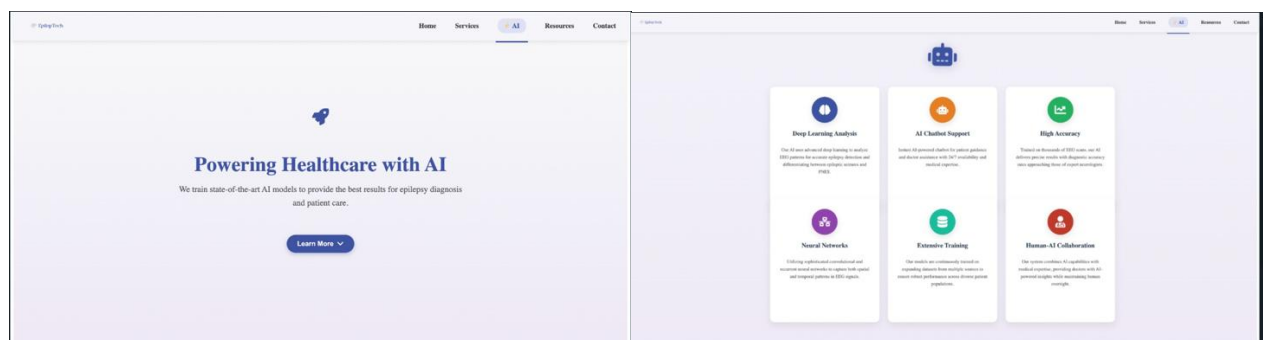


Figure 14: AI Interface.

5.3.3.1.4 Research & Resources Interface

It includes crucial information including research articles on AI-powered EEG diagnoses, insights into how deep learning improves seizure classification, and official WHO recommendations for epilepsy therapy. These tools help doctors and researchers use the platform to learn more and make more evidence-based decisions.

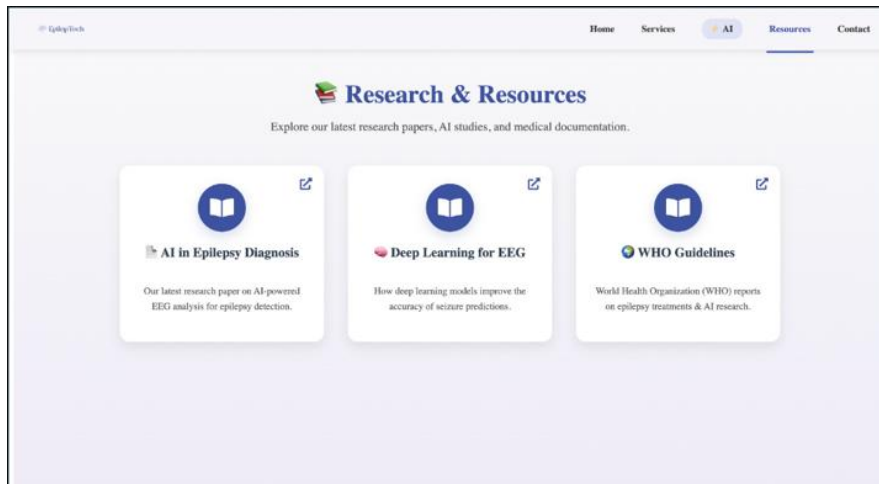


Figure 15: Research & Resources Interfaces

5.3.3.1.5 Contact

The contact interface offers customers with a direct and easy way to contact the epilepsy care support staff. Whether visitors have queries regarding diagnostic services, want technical support, or want to collaborate, this area provides timely contact. Users can contact out by email, phone, or in person at the university campus at Imam Abdulrahman Bin Faisal University.

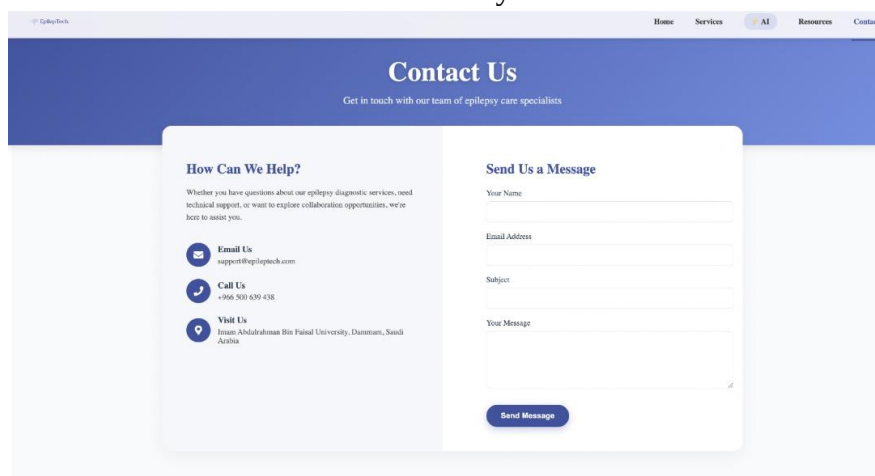


Figure 16: Contact Interface.

5.3.3.1.6 Patient Login Interface

The patient login interface provides access to the patient's own portal. After properly inputting the login and password, the user may access the patient services.

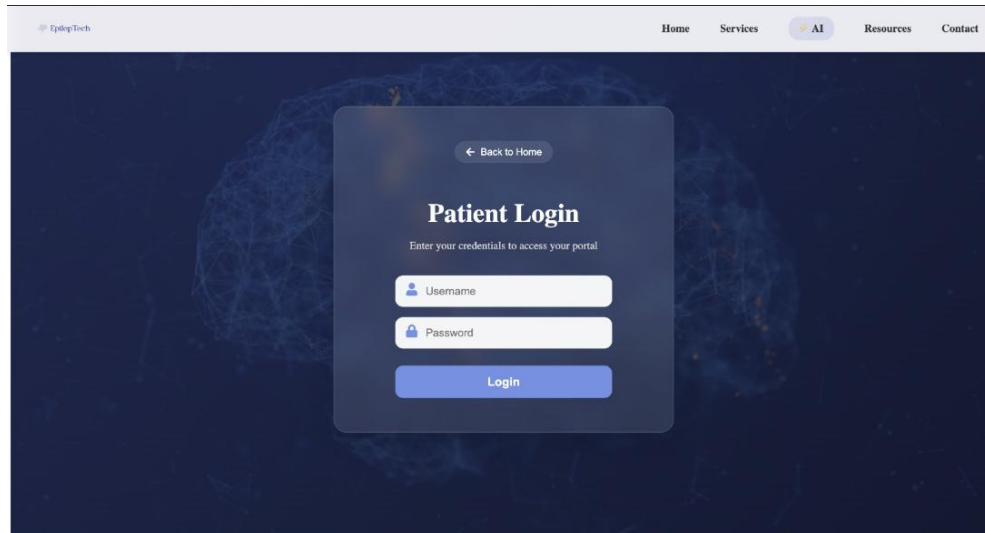


Figure 17: Patient Login Interface.

5.3.3.1.7 Patient EEG Report

After the user login successfully, the user can view the EEG report. Also, the user can search for a specific report.

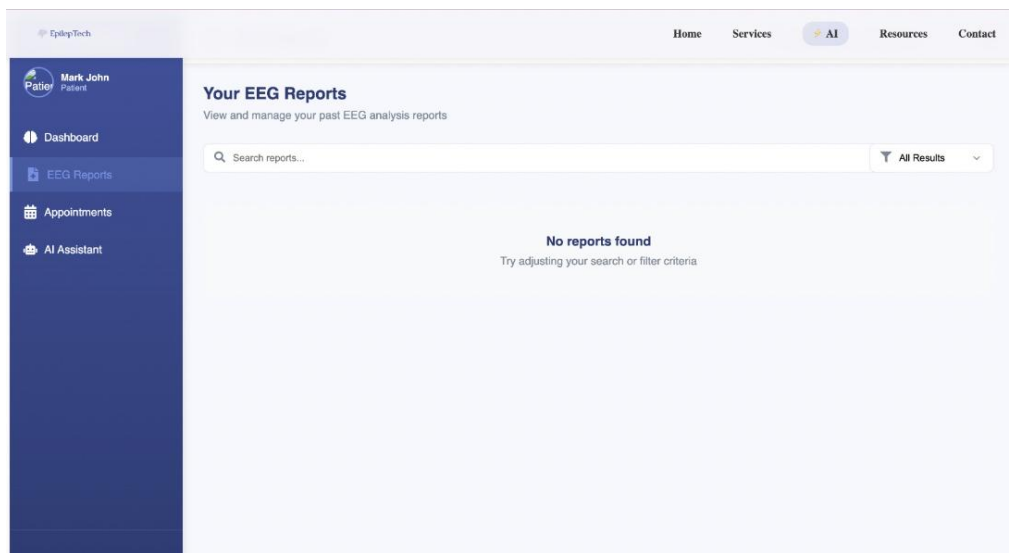


Figure 18: Patient EEG Report Interface.

5.3.3.18 Patient Dashboard Interfaces.

The “Patient Dashboard” provides a customized snapshot of the user's epilepsy management process. Patients may readily obtain their most recent EEG analysis with AI-generated insights, follow seizure trends, and evaluate treatment effectiveness using visual analytics. The dashboard also shows upcoming appointments, current medications, and a recent activity log with changes such as doctor notes or prescription refills. This consolidated and user-friendly interface keeps patients informed, coordinated, and engaged in their care plan.

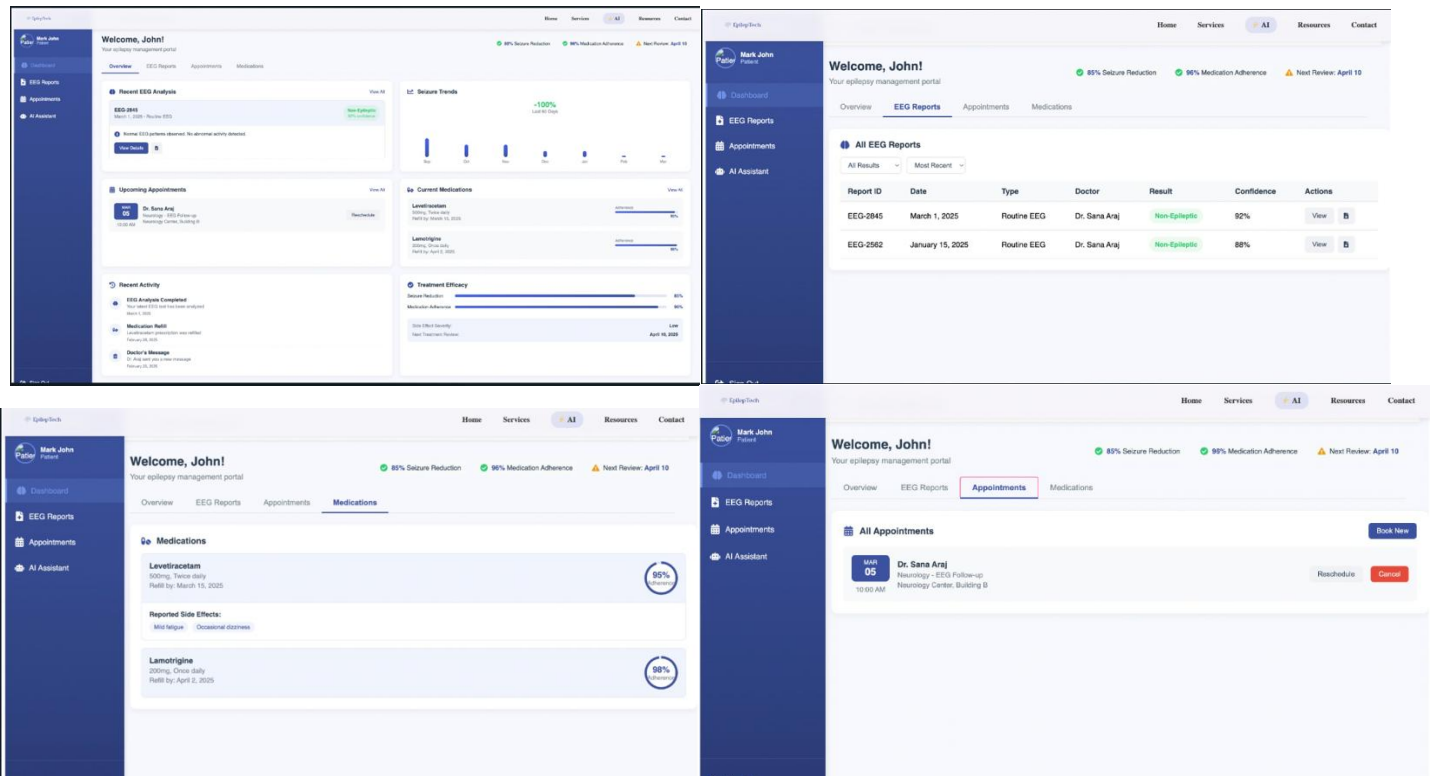


Figure 19: Patient Dashboard Interfaces.

5.3.3.1.9 Appointment Interfaces.

Patients may easily manage their epilepsy-related consultations. They may examine future or past appointments, make new ones by selecting a doctor, date, and purpose for the visit, and reschedule or cancel existing ones. The design is simple and helps patients keep organized and connected to their healthcare professionals.

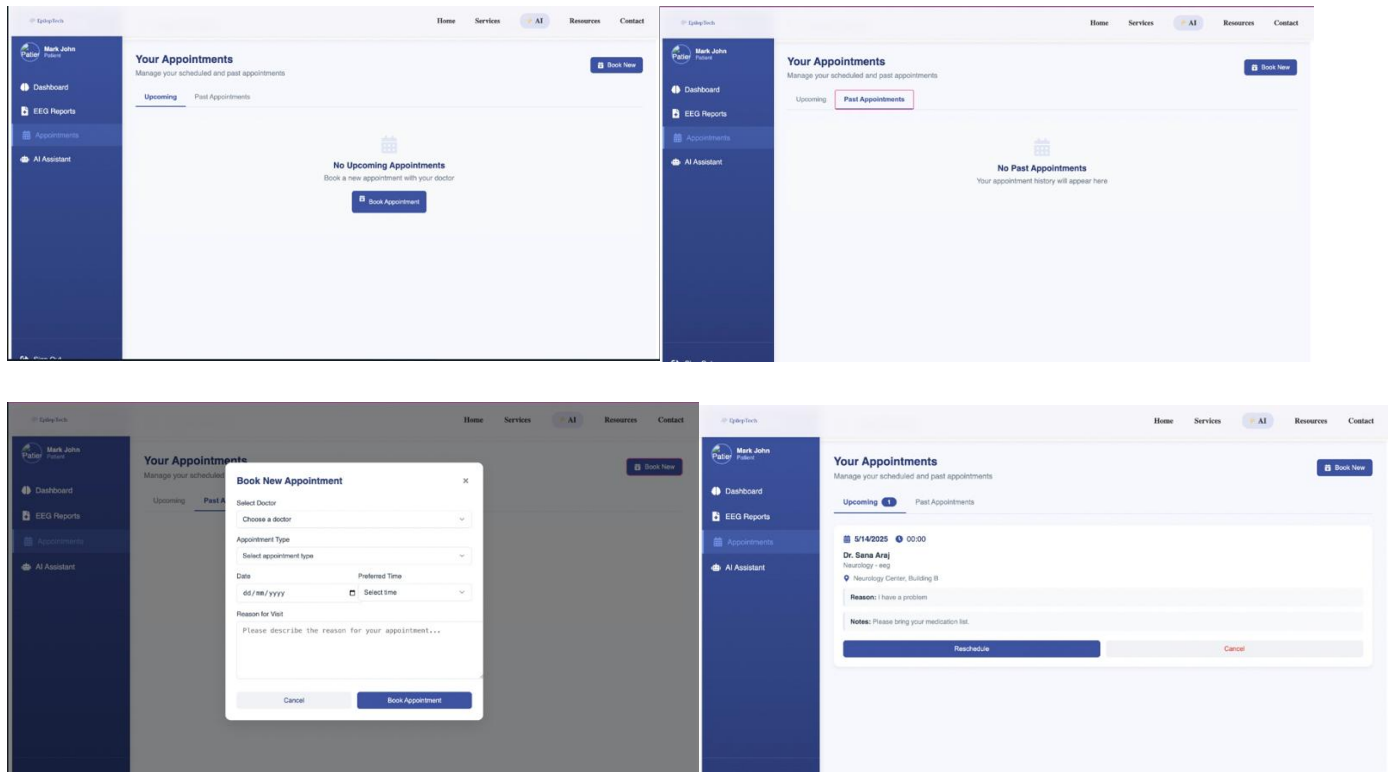


Figure 20: Appointment Interfaces.

5.3.3.1.10 AI Assistant Interface.

The interface provides support tool for controlling epilepsy and answering treatment-related inquiries. It gives immediate answers to typical questions concerning drugs, seizure kinds, lifestyle modifications, and appointment scheduling.

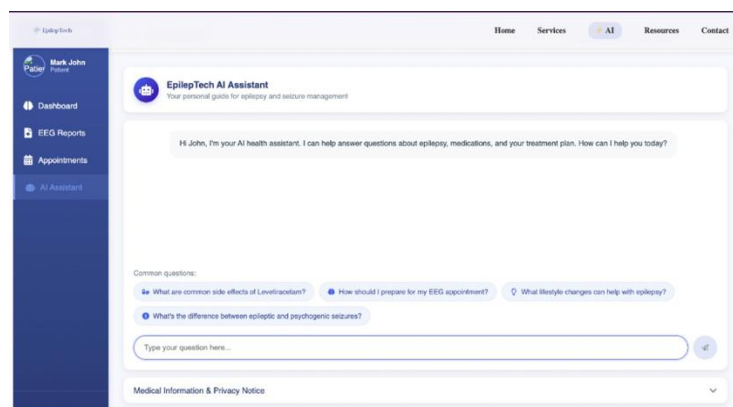


Figure 21: AI Assistant Interface

5.3.3.1.11 Doctor Login

The Interface gives medical professionals with safe access to the EpilepTech dashboard. With a clear, professional UI, using the same logic as the patient login.

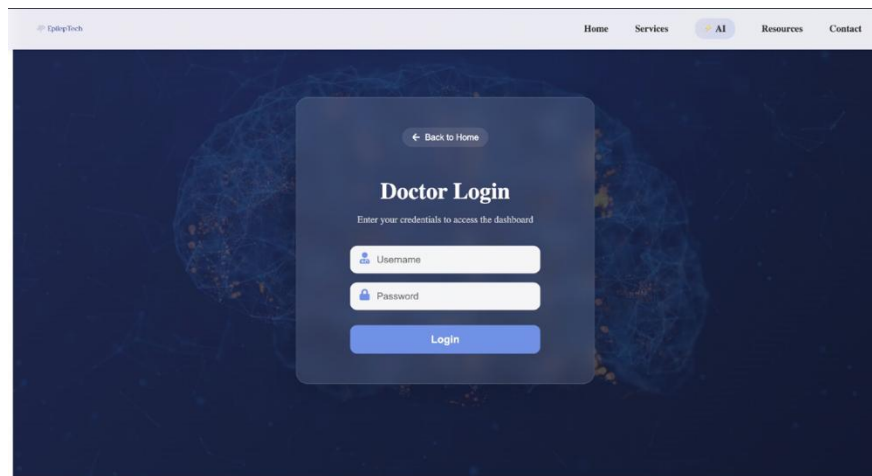


Figure 22: Doctor Login Interface.

5.3.3.1.12 Doctor Dashboard.

offers neurologists doctors an overall overview of their EEG analysis activities. It shows pending and completed analysis, provides easy access to submit new cases, and enables clinicians to analyze AI-generated EEG results.

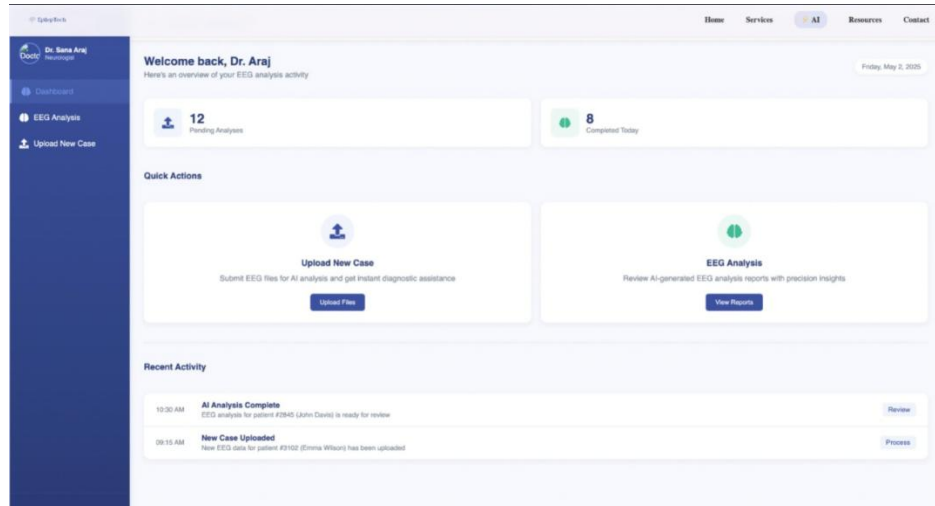


Figure 23: Doctor Dashboard Interface.

5.3.3.1.13 Doctor EEG Analysis.

enables doctors to analyze and manage AI-generated EEG results; they may choose between pending, finished, and all cases, as well as download the completed generated report. The interface is intended to facilitate case tracking and speed diagnostic evaluation.

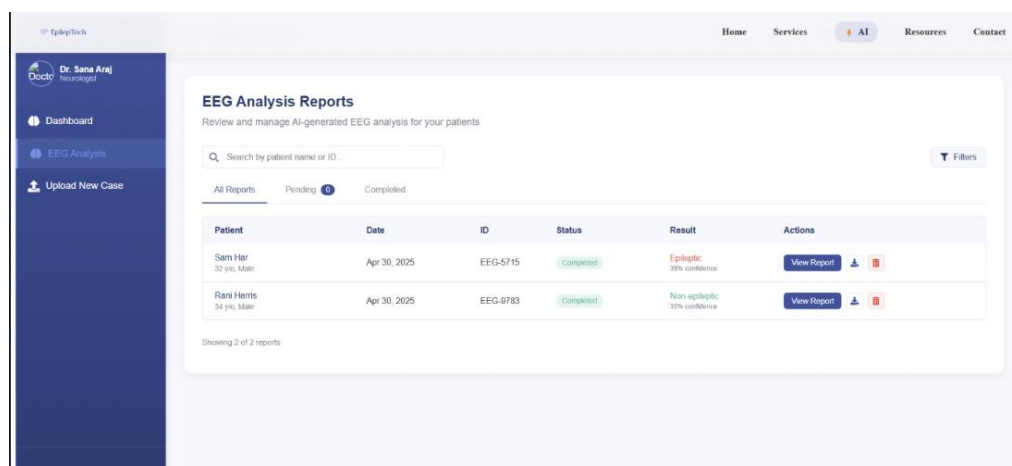


Figure 24: Doctor EEG Analysis Report.

5.3.3.1.14 Doctor Upload EEG Case.

The interface allows clinicians to upload EEG data for AI-powered analysis. It contains sections for case information, patient details, extra clinical notes, and an upload feature that accepts a variety of EEG file types.

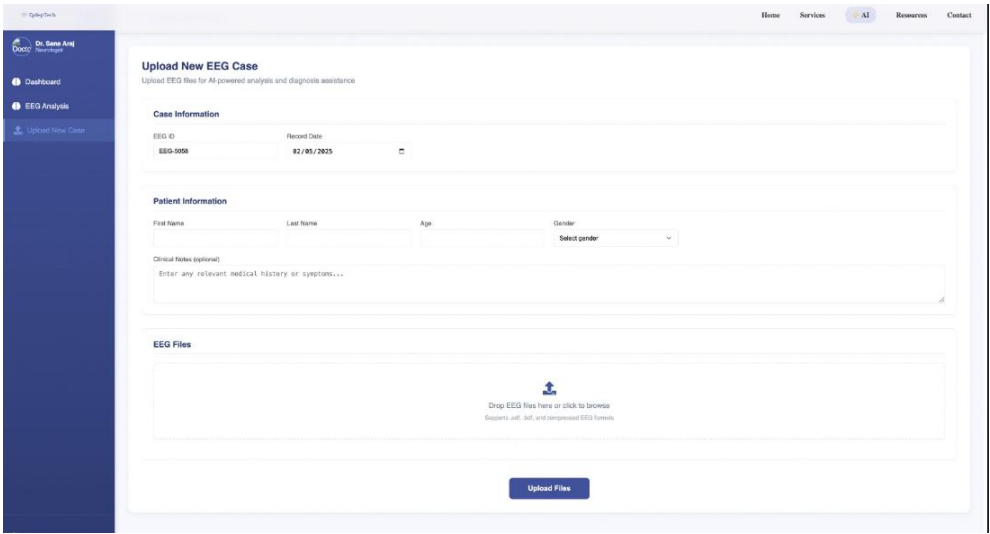


Figure 25: Upload EEG Case.

5.3.4 Screen Object Action

Table 51 shows the screen objects and actions.

Table 38: Screen Object Action.

No.	Object	Type	Action
Common Interfaces			
Home Page Interface			
1	EpilepTech	Navigation bar	Contains Home, Services, AI. Resources and Contact.
2	Doctor Portal	Button	When a user clicks, he will move to the “Doctor Login” interface.
3	Patient Portal	Button	When a user clicks, he will move to the “Patient Login” interface.
4	Learn more	Scroll down	When the user clicks it, automatically scrolling down to the feature cards.
services Interface			
5	6 Services	Service Cards	Describe the role, analysis, consultations, and overall relevance of AI in medicine.
AI Interface			
6	Learn More	Button	When the user clicks it, automatically scrolling down to the AI feature cards.
7	6 Feature Card	Feature Cards	The card explains how AI improves categorization, training, accuracy, and more.
Research & Resources Interface			

8	3 Research Card	Research Cards	The cards provide latest research in AI in epilepsy and seizure classification, as well as link to WHO epilepsy and AI research guidelines.
Contract Interface			
9	Email Us	Contact info	Support email for questions.
10	Call Us	Contact info	Contact via phone number.
11	Visit Us	Contact info	Provide physical address.
12	Send Us a Message	Contact Form	Ease the users to send custom questions via form.
Patient Interface			
Patient Login Interface			
13	Username	Text Field	To enter username.
14	Password	Text Field	To enter password.
15	Login	Button	Validates the identity and logs them.
16	Back to Home	Button	When clicks the user will return to the home interface.
Patient Dashboard Interface			
17	EEG Analysis	Card	Showcase analysis and view patient details.
18	Seizure Trends	Chart	Display trend of seizure events.
19	Current Medications	Card	Shows prescriptions levels
20	Treatment Efficacy	Graph	Track medication and seizure outcomes.
21	Upcoming Appointment	Cards	List upcoming appointments.
22	Reschedule	Button	The user can reschedule appointment.
23	Report Row	Table	Show reports details.
24	View	Button	Open detailed EEG report.
25	Download	Button	Download report PDF.
26	Book Now	Button	Open form to book appointment.
27	Reschedule	Button	The user can reschedule/ edit appointment.
28	Cancel	Button	Cancel appointment.
29	Medication	Card	Display prescription and dosage
30	Side Effect	Tags	Show side effects

EEG Report Interface			
31	Search Bar	Input Field	Allows the user to search patient names.
32	Filters	Button	Opens filter potations to sort or filter EEG report
33	Tabs	Tab Control	Switches between all, pending and all reports

34	Patient Name	Text field	Display the patient's name.
35	Date	Label	Display the date.
36	Report ID	Label	Unique identifier for EEG report
37	Status	Label	Indicates whether the report is Completed (pending , complete)
38	Result	Label	Displays classification result
39	Confidence	Label	Shows model confidence percentage
40	View Report	Button	Opens the AI generated EEG report
41	Download Report	Icon Button	Downloads the report as a PDF file
42	Delete Report	Icon Button	Deletes the report from the database
Appointment Interface			
43	Your Appointments	Section Header	Displays upcoming and past appointment tabs.
44	Upcoming	Tab	Shows all upcoming booked appointments.
45	Past Appointments	Tab	Shows history of past appointments.
46	Book Appointment	Button	Opens form to book a new appointment.
47	Book New (top right)	Button	Alternative access to open booking form.
Book Appointment Interface			
48	Select Doctor	Dropdown	User can choose a doctor from available options.
49	Appointment Type	Dropdown	User can choose type of appointment
50	Date	Calendar	User selects appointment date.
51	Preferred Time	Time	User selects appointment time.
52	Reason for Visit	Text Area	Allows patient to describe the reason for the appointment.
53	Book Appointment	Button	Confirms and submits new appointment booking.
54	Cancel	Button	Closes the form without booking.
Your Appointment Interface			
55	Scheduled Appointment	Card	Displays details of a booked appointment
56	Reason	Text Field	Displays user-entered reason for visit.
57	Notes	Info Label	Displays any doctor's notes (e.g., "bring medication list").
58	Scheduled Appointment	Card	Displays details of a booked appointment (doctor, time, location).
59	Reason	Text Field	Displays user-entered reason for visit.
60	Reschedule	Button	Allows the user to choose a new date/time for existing appointment.
61	Cancel	Button	Cancels the scheduled appointment.

AI Chatbot Interface			
62	EpilepTech AI Assistant	Header	Display the assistant name.
63	Common question	Button	The user can click common question.
64	Input Text	Text Field	User can type a custom question.
65	Submit	Button	Send the input text to the assistant.
66	Medical information & privacy notice	Link	Display legal information.
Doctor Interfaces			
Doctor Login Tab Interface			
67	Username Field	Text Input	Enter username
68	Password Field	Password Input	Enter password
69	Login Button	Button	Submit login credentials
70	Back to Home Button	Button	Return to home page

Doctor Dashboard Interface			
71	Pending Analyses Count	Display	View total pending EEG reports
72	Completed Today Count	Display	View reports completed today
73	Upload New Case	Button	Navigate to upload EEG case
74	EEG Analysis	Button	Navigate to view EEG reports
75	AI Analysis Complete Notification	Text/Label	Show recent activity.
76	Recent Activity	Display	Display past activity.
Doctor EEG Analysis Interface			
77	Search Field	Input Field	Search by patient name or ID
78	EEG Report Entry	Table Row	Display patient EEG report
79	View Report Button	Button	Open detailed EEG report
80	Download Report Icon	Button	Download report PDF
81	Filter	Button	Filter or sort EEG reports.
82	Delete Report Icon	Button	Delete EEG report
Doctor Upload Case Interface			
83	EEG ID Field	Text Input	Enter EEG case ID
84	Record Date Picker	Date Input	Select EEG recording date
85	First Name Field	Text Input	Enter patient first name
86	Last Name Field	Text Input	Enter patient last name
87	Age Field	Number Input	Enter patient age
88	Gender Dropdown	Dropdown	Select gender
89	Clinical Notes	Text Area	Enter clinical notes
90	EEG File Upload	File Input	Upload EEG data files
91	Upload Files Button	Button	Submit uploaded EEG case

5.3.5 Other Interface

In order to make the system more professional, we added extra interfaces such as error, confirmation, and information messages. This update seeks to improve professionalism and usability by ensuring that users receive notifications before performing any actions or in the event of mistakes throughout the process.

5.3.5.1 Error/ Warning Messages

This section contains samples of error messages generated by the system.

Figure 34 shows the error message for not filling the text field in the “Login” interfaces for both doctor and patient.

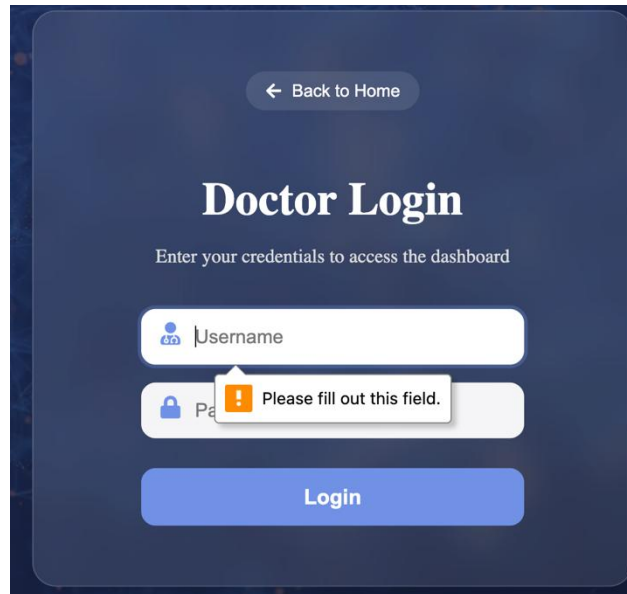


Figure 26: Not filling Text field.

Figure 35 shows the error message for invalid username and/or password in the “Login” interface for both users.

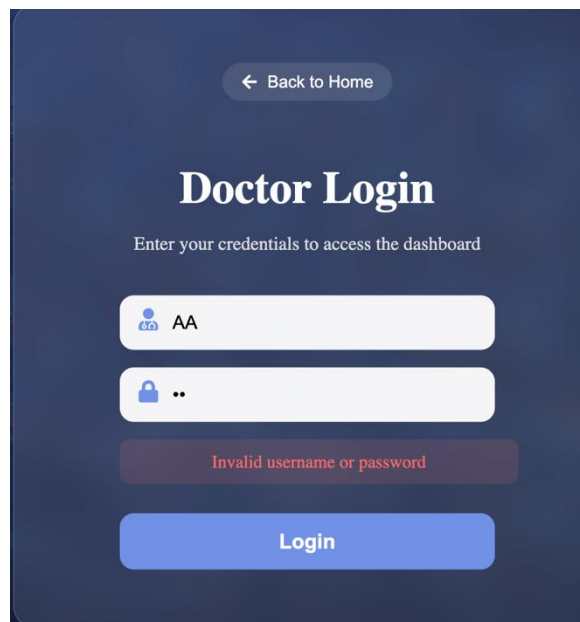


Figure 27: Invalid information.

Figure 36 The error warning for not filling patient information in the “Upload New Case” interface in the doctor portal.

Patient Information

First Name

Last Name

Age

Gender

Clinical Notes (optional)

Please fill out this field.

Figure 28: Invalid Email Error Message

Figure 37 shows the error message that appears when the doctor does not upload the edf file.

EEG Files

Drop EEG files here or click to browse

Supports .edf, .bdf, and compressed EEG formats

Please select at least one EEG file to upload

Figure 29: Not Uploading EEG File.

Figure 38 shows the error message displayed for not providing information in the “Book New Appointment” interface, in the patient portal.

Book New Appointment [Close]

Select Doctor

Appointment Type

! Please select an item in the list.

Date

Preferred Time

Reason for Visit

Figure 30: Not providing full information in Book Now Interface.

Figure 40 shows the error message that appears when the text filed in the “Contact” Interface.

Send Us a Message

Your Name

 Name is required

Email Address

 Email is required

Subject

 Subject is required

Your Message

 Message is required

Figure 31: Not providing full information in the "Contact" Interface.

5.4 System Architecture

The system architecture follows an N-tier layered approach consisting of three independent layers, which are:

- **Presentation Layer:** The presentation layer is responsible for communicating with the user interface. It can retrieve data from the data layer by communicating with the business layer. The proposed user input is transmitted from the presentation to the business layers. The business layer then manipulates the data from the data layer to generate the output.
- **Business layer:** The middle layer manages system components and acts as a link between users and application menus.
- **Data layer:** The base layer of this architecture is primarily responsible for data-related functions, such as data storage and retrieval within the system. This data is typically stored on a database or file server and can be accessed through various means without revealing the storage and retrieval methods.

The N-tier architecture can provide the system with several benefits, as listed below:

- **Performance and scalability:** N-tier architecture allows adding resources to each layer without impacting other layers. This enhances performance and scalability.
- **Security and safety:** There are fewer direct connections between layers in the N-tier architecture, which results in increased security and safety. Each layer can have a secure pathway, further enhancing security and safety.
- **Maintainability:** Updating any processes in the system without impacting the system's functionality is possible due to the support for maintainability in the N-tier architecture.
- **Flexibility:** The N-tier architecture provides system flexibility by allowing for the expansion of each layer.

5.4.1 Architectural Design

The EpileptTech system is designed using an N-tier layered architecture, consisting of three distinct layers that work collaboratively to ensure functionality, efficiency, and scalability. The structure and roles of these layers are as follows:

- **Presentation Layer:** The presentation layer acts as the user interface, enabling direct interaction with the system. It displays relevant information to users and captures their inputs, which are then sent to the business layer for processing. Additionally, this layer retrieves data from the data layer, facilitated by the business layer, to present to users. Its primary role is to provide a seamless and intuitive communication channel between users and the system.
- **Business layer:** Also referred to as the middle tier, the business layer handles all core application logic and processes user inputs. It ensures inputs are validated, processed, and appropriately routed to the data layer for storage or retrieval. Simultaneously, it manages outputs sent to the presentation layer, serving as the system's functional hub.
- **Data layer:** The data layer serves as the system's foundation, focusing on data storage and retrieval. It securely manages system data using a database server or other storage tools while protecting the underlying mechanisms. This layer ensures data integrity, privacy, and secure access.

Benefits of the N-Tier Architecture

- **Performance and scalability:** The N-tier architecture allows for resource-specific scaling within individual

layers without impacting others. This ensures that system performance can grow proportionally with expansion needs.

- **Security and safety:** By minimizing direct communication between layers, the architecture enhances security. Each layer can implement its own security measures, reducing vulnerabilities and safeguarding sensitive data.
- **Maintainability:** The modular design of the N-tier architecture enables updates or modifications to be performed within a specific layer without disrupting overall system functionality, simplifying maintenance.
- **Flexibility:** The architecture provides flexibility by allowing independent upgrades or expansions of individual layers, ensuring the system remains adaptable to future requirements.

5.4.1.1 Architectural Design Approach

As previously discussed, there are three layers of N-tier architecture. Each of these layers contains specific functionalities:

- **Presentation Layer**
 - **Web Application:**
 - ♣ Developed a responsive and intuitive user interface using modern web frameworks like React or Vue.js.
 - ♣ Designed interactive components, such as forms, tables, and charts, to enhance user engagement and streamline data visualization.
 - **Client-Side Logic:**
 - ♣ Implemented client-side functionality using JavaScript or TypeScript to validate user inputs and enable dynamic behaviors.
 - ♣ Ensured error-free user interactions with robust input validation and seamless communication with the business layer.
- **Business Layer**
 - **Application Server:**
 - Choose server-side technology (e.g., Node.js & Flask) to handle business logic and user requests.
 - Integrated middleware for efficient request processing, error handling, and communication between the presentation and data layers.
 - **Business Logic:**
 - Defined and implemented the core functionalities of the system, including EEG data analysis and diagnostic classification.
 - Organized business rules into modular code structures to ensure maintainability and scalability.
- **Data Layer:**
 - **Database Management System (DBMS):**
 - Selected a reliable DBMS (MongoDB) based on data storage and retrieval needs.
 - Defined database schemas and relationships.
 - **Databases:**
 - Created and managed databases to securely store application data.
 - Implemented database queries and transactions aligned with business logic requirements.

Figure 41 presents the architectural design diagram for the system.

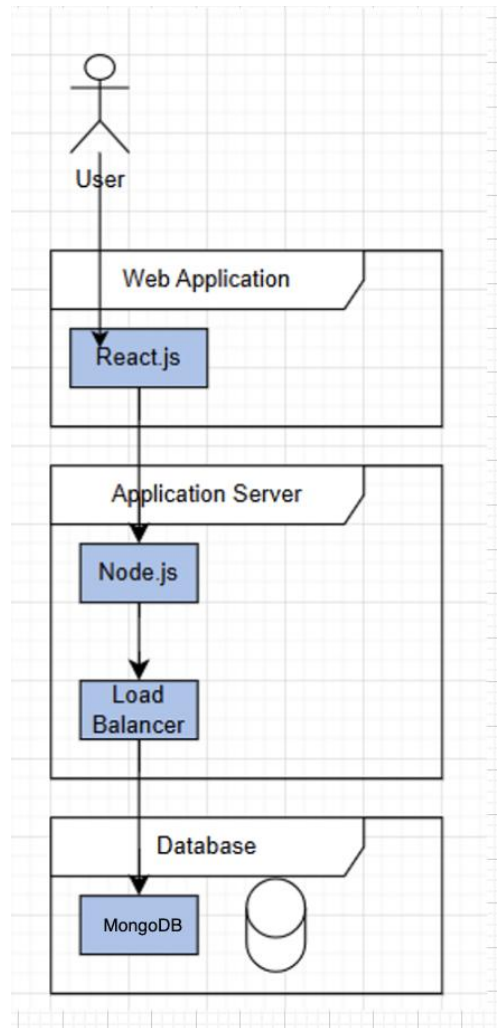


Figure 32: Architectural Design Diagram.

5.4.2 Subsystem Architecture

This section describes the decomposition of subsystems within the EpilepTech architectural design, emphasizing their individual functionalities and methods for data storage and management. To provide a clear understanding of the system's logical architecture, Data Flow Diagrams (DFDs) are included to depict the processes, data flows, and data stores across various system levels. These diagrams offer a detailed representation of how the subsystems interact, ensuring a comprehensive view of the application's overall functionality.

5.4.2.1 General View of the System

The overall perspective of the EpilepTech system offers a comprehensive understanding of its interconnected subsystems and their relationships. This perspective forms the basis for detailed descriptions of each subsystem and their specific functionalities, highlighting how they collaboratively support the system's objectives.

5.4.2.2 All Users' Subsystems

The user subsystems in the EpilepTech system are tailored to accommodate the distinct roles and interactions of its users. These subsystems are designed to optimize the user experience, streamline functionality, and facilitate seamless navigation and interaction within the system.

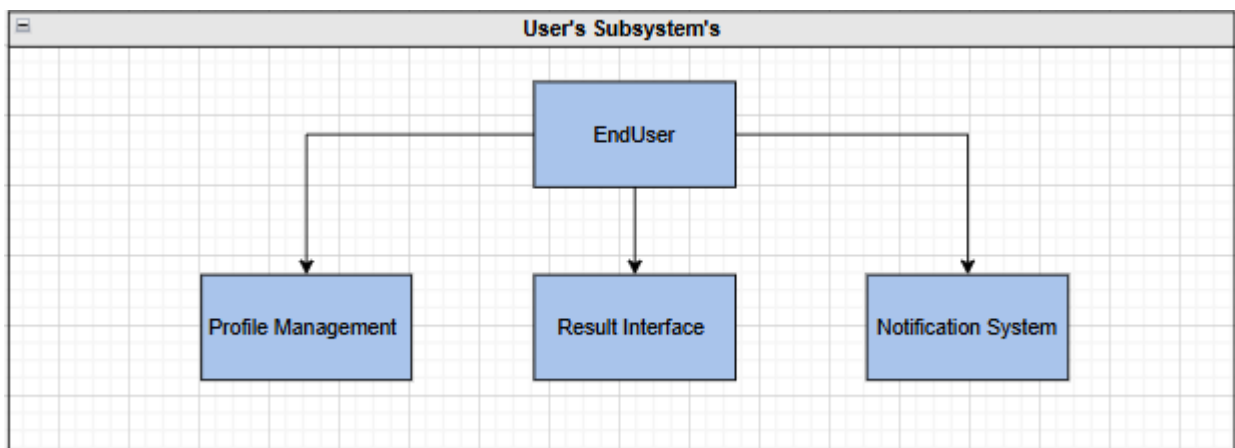


Figure 33: All users' subsystems

5.4.2.3 Doctor Subsystem

The doctor subsystem enables seamless interaction between doctors and the system, supporting tasks such as login, upload and view EEG reports, view activity logs , downloading and deleting reports.

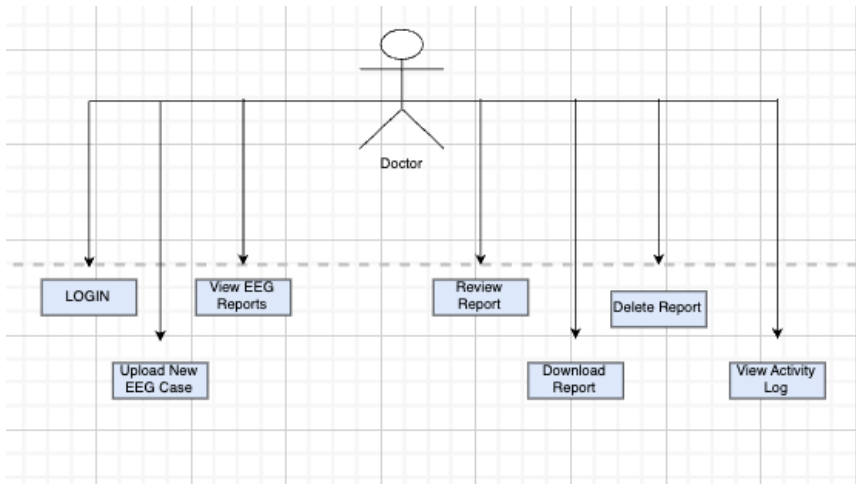


Figure 34: Doctor Subsystem

5.4.2.4 Patient Subsystem

The Patient subsystem facilitates smooth interactions between users and the EpilepTech system. It allows users to view their diagnostic history, and print diagnostic results, delivering a seamless experience and supporting personalized care.

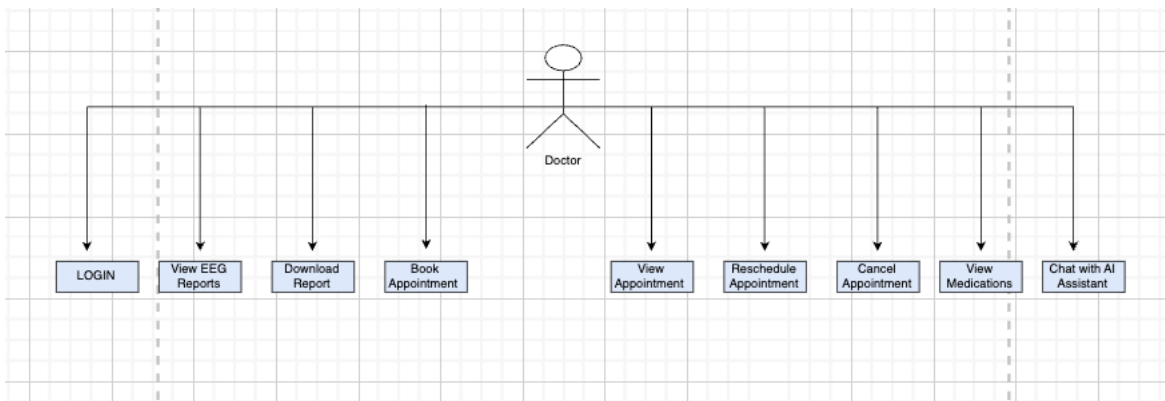


Figure 35: Patient Subsystem

5.5 Data Design

This section offers a comprehensive description of the data, including its types, required fields, and an outline of the application's database structure. It ensures that all essential data is clearly defined and systematically organized to effectively support the system's operations.

5.5.1 Data Description

The EpilepTech system securely stores all essential data required for smooth and reliable operation. Each user's role is granted access to specific data relevant to their responsibilities within the system. Data entities are structured with defined fields, types, and constraints, ensuring effective management and processing of system operations.

Table 39: The Database Entities and Fields

Entity	Field	Type	Constraints
Account	Username	VARCHAR (20)	Not null, Foreign Key
	Name	VARCHAR (45)	Not null
	Email	VARCHAR (255)	Unique, Not null
	Password	VARCHAR (35)	Not null
	Email Confirmed	ENUM ('Y', 'N')	Default 'N', Not null
	Users_Role	ENUM('Patient', 'Doctor')	Not null
	Specification	VARCHAR(45)	Optional
Patient	PID	INT (10)	Not null, Primary key, Auto increment
	Name	VARCHAR (45)	Not null
	Birth_Date	DATE	Not null
	Seizure_Type	ENUM('Focal', 'Generalized')	Not null
	Seizure_Freq	VARCHAR(45)	Optional
	Gender	ENUM ('Male', 'Female')	Not null
Doctor	Username	VARCHAR (20)	Not null, Primary key
	Name	VARCHAR (45)	Not null
	Email	VARCHAR (255)	Not null, Unique
	Password	VARCHAR (35)	Not null
Result	ResultID	INT (10)	Not null, Primary key
	Seizure_Type	ENUM(Epileptic, PNES, Non-Epileptic)	Not null
Model	Model_ID	VARCHAR (25)	Not null, Primary key
	Epileptic Seizure Detect	ENUM('Yes', 'No')	Not null
	Accuracy	DECIMAL (5,2)	Not null
	Confidence_score	INT(10)	Not null
	TrainingPercentage	DECIMAL (5,2)	Not null

5.5.2 Data Dictionary

Below is the Data Dictionary for the EpilepTech system, detailing all entities, their respective fields, and descriptions:

Table 40:Data Dictionary

Entity	Field	Description
Account	Username	Unique identifier linking to Doctor or Patient.
	Name	Name of the account holder.
	Email	Email address linked to the account.
	Password	Password for the account login.
Patient	Username	Patient ID, unique for each patient.
	Name	Patients' full name.
	Gender	Gender of the patient (Male or Female).
	Birth_Date	Patient's date of birth.
	Seizure_Type	Type of seizure (e.g., Focal, Generalized).
	Seizure_Freq	Frequency of seizures (e.g., daily, weekly).
Result	Result_ID	Unique identifier for each test result.
	Seizure_Type	Type of seizure detected during the test.
Model	Model_ID	Unique identifier for the diagnostic model.
	Epileptic Seizure Detect	Indicates if the model detects epileptic seizures (Yes or No).
	Accuracy	Accuracy rate of the diagnostic model.
	Confidince_score	Confidence scores represent the accuracy of highest class.
	TrainingPercentage	Tracks how much the dataset was used for training.

5.5.3 Database Description

The EpilepTech system uses MongoDB, a flexible NoSQL database, to manage user data, EEG uploads, and diagnosis results. Collections are structured to reflect logical relationships between patients, doctors, and AI outputs, ensuring both data accessibility and privacy. The main entities and their relationships are illustrated in the following ERD and ERD mapping diagrams, adapted for a non-relational schema.

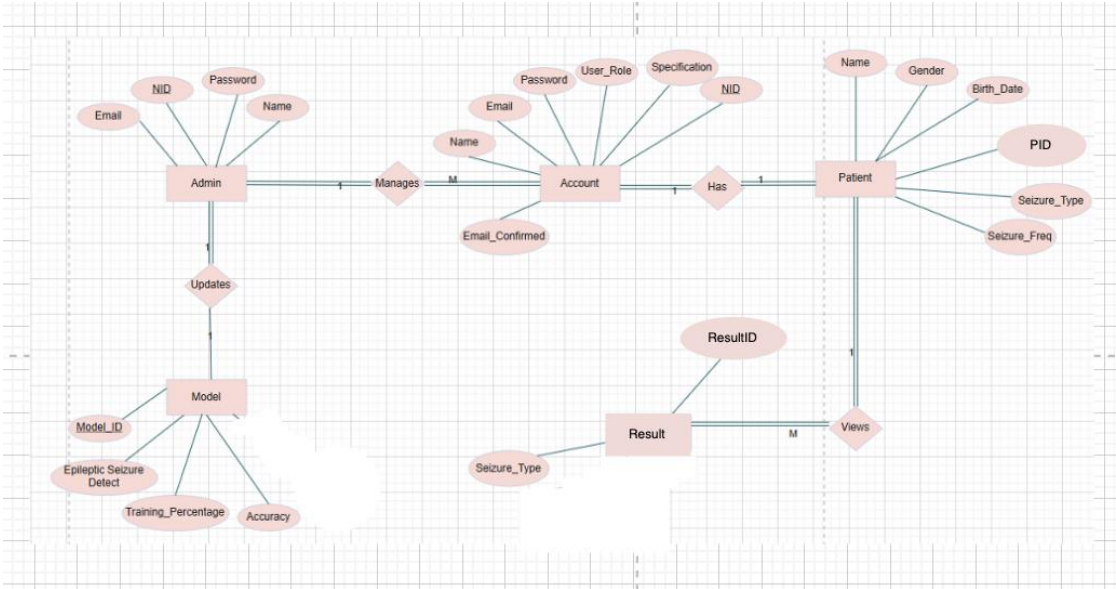


Figure 36: Entity Relationship Diagram

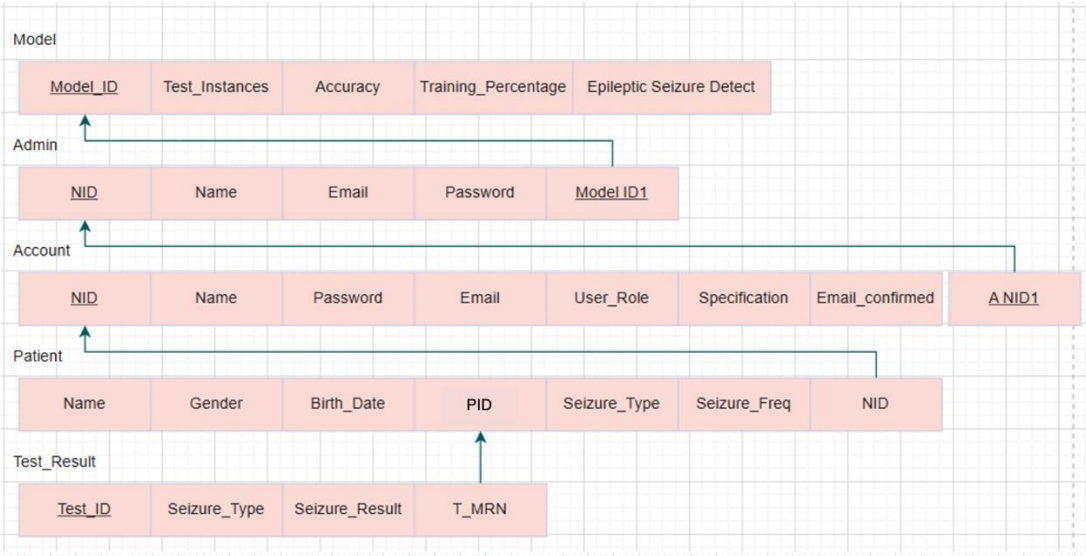


Figure 37: ERD Mapping Diagrams

5.6 Component Design

This section outlines simplified pseudocode for the main functionalities of the EpilepTech system. The logic is described in plain English to ensure that both developers and non-technical reviewers can easily understand how each part works. These pseudocode representations focus on the flow of core features as they are implemented across the backend (Flask), frontend (React), and integrations like MongoDB, Colab, and OpenAI.

5.6.1 Login/ Logout Functions

5.6.1.1 Login_With_Credentials ()

Input Username, Password

Logic:

If Username or Password is empty:

Display "Please fill in all required fields."

Else if check_user_in_database(Username, Password):

Redirect user to their dashboard (Doctor or Patient)

Else:

Display "Invalid username or password. Please try again"

5.6.1.2 Logout_User ()

Logic:

Clear the user's session or authentication token

Redirect to Login interface

Show message: "You have been logged out successfully."

5.6.2 Diagnostic Functions

5.6.2.1 Upload_EEG_File ()

Input: .edf EEG File

Logic:

If uploaded file is not in .edf format:

Return "Only .edf EEG files are supported."

Save EEG file temporarily on the server or upload to a shared location (e.g., Google Drive)

Send a request to the Colab-based diagnosis endpoint with file reference

Return "File uploaded successfully. Running diagnosis..."

5.6.2.2 Diagnosis ()

Trigger:

Flask API calls Google Colab script

Logic:

Colab receives the EEG file

Preprocess the EEG signal (e.g., segmentation, filtering)

Run the trained GNN model on the preprocessed data

Get Classification:

- Seizure
- Non-Seizure
- PNES
- + confidence scores

Send back result to Flask in structured JSON format

Store result and timestamp in MongoDB under patient's record

Return "Diagnosis complete. View result now."

5.6.2.3 . Display_Diagnosis_Result()

Frontend fetches diagnosis result from Flask API

Display:

- Diagnosis class
- Confidence scores
- Time of diagnosis

Provide button: "Generate Report"

5.6.2.4 Genrate_Diagnosis_Report

Input:

Diagnosis result, patient info (optional), date/time

Logic:

Backend prepares a prompt with:

- Diagnosis summary
- Model confidence
- Patient age/gender (if applicable)

Send prompt to OpenAI API

Receive AI-generated report in medical tone

Save report content to MongoDB

Return "Report generated successfully."

5.7 Detailed System Design

This section provides a structured overview of the EpilepTech system's core components. It includes the classification, purpose, and responsibilities of each function within the system, based on the actual deployment architecture. The platform is developed using a Flask-based backend, React frontend, and MongoDB for data storage, with OpenAI integration for generating reports. Each function described supports specific roles, ensuring seamless collaboration between patients, doctors, and system administrators. Visual modeling tools such as sequence diagrams and UML were used to illustrate interactions across subsystems.

Classification, Definition, and Responsibilities

This section describes the classification, definition, and responsibilities of each system component. The table 55 below offers essential information:

Table 41: Component, Classification, Definition and Responsibility

Component	Classification	Definition and Responsibility
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Login	Function	Allows users to securely access the system using their Username and password. User credentials are verified through the Flask backend and MongoDB.
Upload EEG Files	Function	Lets users upload EEG recordings (in EDF format) through the web interface. The uploaded files are temporarily stored in memory and then passed to the classification model.
Run Diagnosis	Function	Sends the uploaded EEG data to a pre-trained deep learning model hosted on Google Colab. The model returns the seizure classification (e.g., Seizure, PNES, or Non-Seizure) with confidence scores.
View Diagnosis Result	Function	Allows users to view their latest classification results, including class labels and confidence levels. Results are fetched from the MongoDB.
Generate Medical Report	Function	Automatically generates a human-readable medical report using OpenAI, summarizing the classification results for clinical interpretation.
AI Chat Assistant	Function	Offers users a conversational assistant powered by OpenAI to answer epilepsy-related questions, explain system features, or provide guidance.

Logout	Function	Ends the user's session, removing access tokens and redirecting them to the login page.
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5.7.1 Constraints and Composition

This section presents the constraints of each system component. Table 56 lists the pre-conditions and post-conditions for each system component.

Table 42: Component, Limitation Pre-condition, and condition

Component	Limitation	Pre-condition	Post-condition
Login	Users must have a valid account.	Enter a valid username and password.	User is successfully logged into the system.
Upload EEG File	Only supported formats (e.g., .edf) are accepted.	User is logged in	File is uploaded and ready for processing.
Run Diagnosis	File must be uploaded first.	Click the “Run Diagnosis” button.	Diagnosis result is generated by the AI model.
View Diagnosis Result	A diagnosis must have already been completed.	User has submitted an EEG file for diagnosis.	Diagnosis result is retrieved and displayed.
Generate Report	Diagnosis must exist.	Click “Generate Report.”	PDF report is generated using OpenAI.
AI Chat Assistant	User must be logged in.	Open the chat assistant from the menu.	Chatbot responds to user queries related to the system.
Logout	User must be logged in.	Click the “Logout” button.	Session ends and user is redirected to login page.

5.7.2 Uses/Interactions

This document part uses sequence diagrams to show the interactions between different system components

5.7.2.1 Login

Figure 47 shows the “Login” sequence diagram

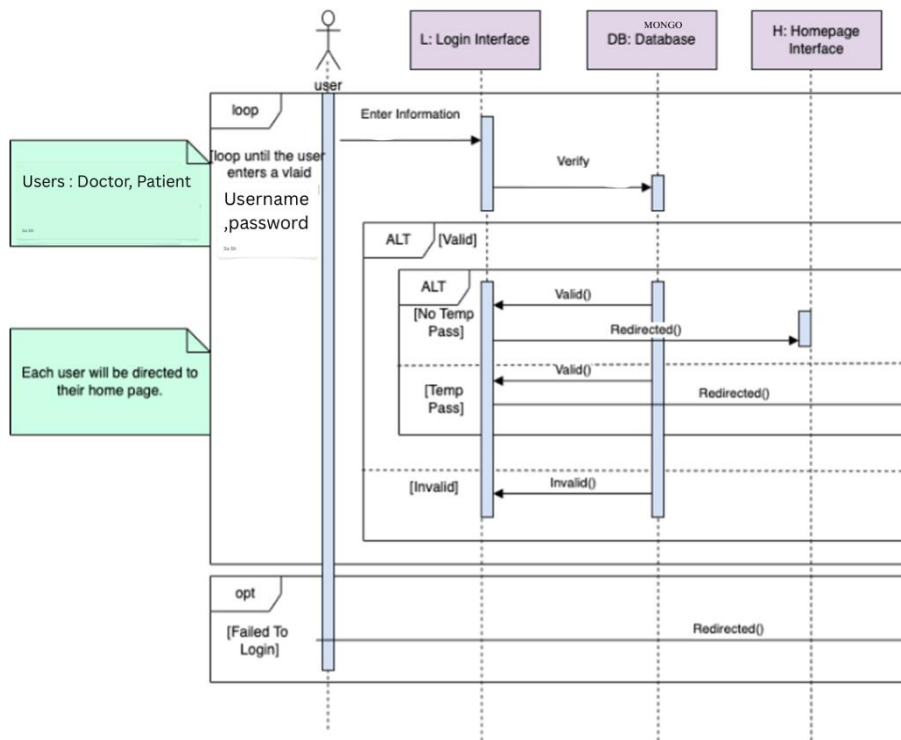


Figure 38: “Login” sequence diagram

5.7.2.2 View Diagnosis

Figure 48 shows the “View Diagnosis” sequence diagram.

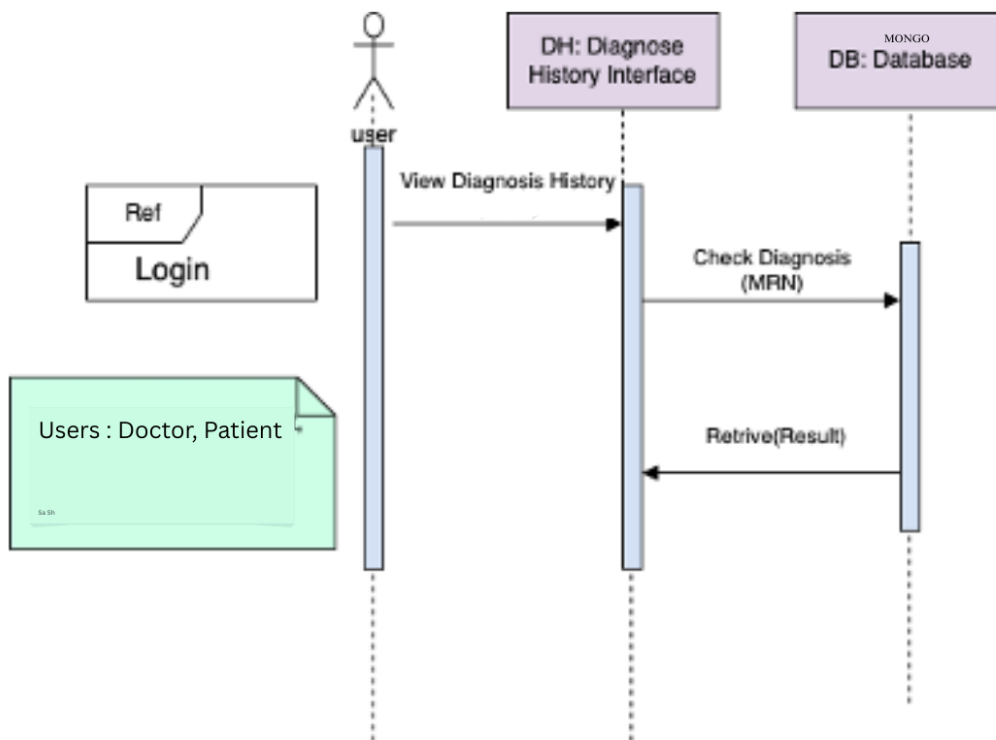


Figure 39: “View Diagnosis” sequence diagram

5.7.2.3 Print Result

Figure 49 shows the “Print Result” sequence diagram.

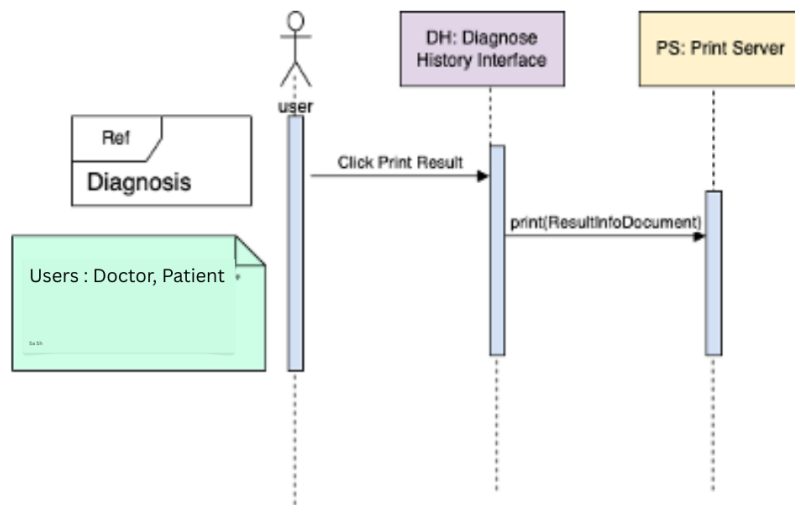


Figure 40: “Print Result” sequence diagram

5.7.2.2 Diagnose

Figure 50 shows the “Diagnose” sequence diagram.

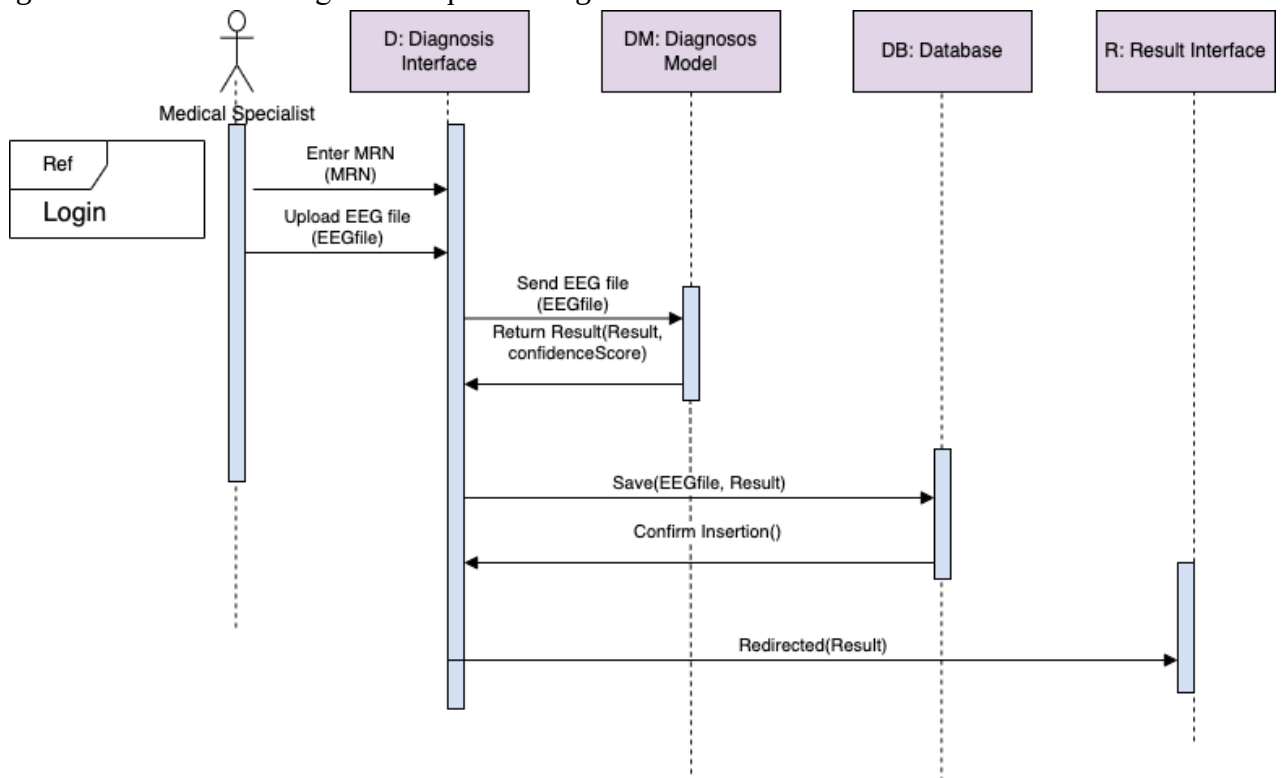


Figure 41: “Diagnose” sequence diagram

5.7.3 Processing

Each component has inputs and outputs, constraints, as well as descriptions. The sections below provide detailed descriptions of each aspect.

5.7.3.1 Login

Table 57 presents the description, inputs, outputs, and constraints of the “Login” component.

Table 43: Description of "Login" Component

Description	Users log in by entering their Username and password. The system verifies credentials and grants access to the main dashboard.
Input	• Username. • Password.
Output	- Access to the system's main interface - Error message if credentials are invalid
Constraint	- Username and password must match records in the database - Login not allowed with incorrect credentials

5.7.3.2 View Diagnosis

Table 58 presents the description inputs, outputs, and constraints of the “View Diagnosis Result” component.

Table 44:Description of " View Diagnosis Component

Description	Displays previously generated diagnosis results stored in the system, including seizure type and confidence level.
Input	• Username. • Password.
Output	- List of past diagnostic results - Message if no results are available
Constraint	- Diagnosis must have been performed earlier for the logged-in user

5.7.3.3 Diagnose

Table 59 presents the description, inputs, outputs, and constraints of the “Diagnose” component.

Table 45:Description of " Diagnose" Component

Description	Allows users to upload EEG files and trigger the AI model to classify the seizure as epileptic, non-seizure, or PNES with confidence scoring.
Input	• EEG data file (.edf)
Output	- Diagnosis result (e.g., Epileptic, PNES, Non-Seizure) - Confidence scores from the model
Constraint	- File must be valid and properly formatted - Model must be active and responsive

5.7.3.4 Print Result

Table 60 presents the description, inputs, outputs, and constraints of the “Print Result” component.

Table 46:Description of "Print Result" Component

Description	Converts the diagnosis result into a downloadable PDF medical report generated via OpenAI’s report interface.
Input	-Diagnostic results
Output	- Downloadable PDF report - Confirmation message
Constraint	- A diagnosis must be completed before the report can be generated

5.7.4 Interface/Exports

To guarantee that the system works as expected, developers must have a deep understanding of each subsystem's capabilities. They should be knowledgeable of the system's inputs, prerequisites, required procedures, postconditions, and restrictions. Section 5.7.1 describes the system's functional limits, prerequisites, and postconditions. Section 5.7.3 provides detailed explanations of the input, output, and functional requirements. Section 4.2.3 of the SRS contains activity diagrams for functional needs.

5.7.5 Detailed Subsystem Design

This section thoroughly describes how this application component works and how its control flow is managed. The related activity diagrams can be found in Section 4.2 of the SRS functional requirements.

5.8 Other Design Features

Every design element was covered in the earlier sections.

5.9 Requirements Traceability Matrix

Table 61 illustrates the requirement matrix.

Table 47:Requirement Tractability Matrix

Associated ID in SRS	Technical Assumption	Functional Requirement	User	Associated ID in SDS	System Component
4.2.3.1.2	Users must be registered to login.	Allow users to log in using their username and password.	Doctor, Patient	5.6.1.1	Login_With_Credentials()
4.2.3.1.7	Doctor must upload EEG data for diagnosis.	Enable doctors to perform seizure classification using AI model.	Doctor	5.6.2.1	Diagnose()
4.2.3.1.4	Doctors, Patients should be able to access the Results.	Allow Doctors, patients to view and download diagnosis results.	Doctor, Patient	5.6.2.3	Display_Diagnosis_Result()

Chapter 6: Implementation Progress

6.1 Changes from the Proposal Phase and Justification

The project's initial idea was to categorize seizures into two groups: epileptic and non-epileptic. But as the study progressed, it became clear how important it was to differentiate between psychogenic non-epileptic seizures (PNES), therefore the scope was expanded to include a trinary categorization system. To better represent actual clinical demands, this modification enabled the system to categorize EEG data into three groups: epileptic, non-epileptic, and psychogenic non-epileptic seizures.

Significant modifications were made to the model creation procedure concurrent with the scope extension. More complex designs including Bidirectional Long Short-Term Memory (BLSTM) networks, CNN-LSTM hybrids, PGM-CNN models, and Graph Neural Networks (GNNs) have been added to the original strategy, which mostly used simple convolutional neural networks (CNNs). These improvements raised the system's classification accuracy in every category by enhancing its capacity to recognize intricate spatial-temporal patterns in EEG data.

The data preparation approach was further improved. For the models to identify more precise seizure patterns, dynamic segmentation approaches with overlapping windows eventually took the role of fixed window segmentation. Techniques for frequency domain analysis and noise reduction were also established to increase the quality of feature extraction and signal clarity.

The system's use of Large Language Models (LLMs) was another significant development. Although the project's original focus was just on classifying EEG signals, LLMs were eventually included to automatically produce comprehensive diagnosis reports for physicians based on the model outputs. Furthermore, LLM-driven chatbots were created to improve the platform's accessibility and usefulness by facilitating communication between patients and clinicians.

A web-based framework was created to facilitate deployment in the real world. Doctors may use this platform to submit EEG data, get classification findings, view reports produced by AI, and communicate via a sophisticated chatbot interface. Together, these modifications have greatly expanded the project's potential and made it more user-centered, scalable, and therapeutically relevant.

6.2 Empirical Study

6.2.1 Data Description

Our project's dataset focuses on electroencephalogram (EEG) signals that are gathered in order to classify seizures. Both locally collected clinical data and publically accessible databases are used as our data sources. Our research utilized the University of Siena EEG Seizure Dataset, which is accessible via Kaggle, for both epileptic and non-epileptic instances. A wide range of patient demographics, including differences in age, gender, and clinical history, are covered by this dataset, which offers high-quality EEG recordings annotated by clinical specialists.

However, accessible EEG datasets do not adequately capture psychogenic non-epileptic seizures (PNES). We worked with King Fahd University Hospital to fill this gap by gathering local EEG data tailored to PNES cases. Specialized neurologists have meticulously analyzed these recordings, guaranteed clinical correctness and increased the project's applicability to actual healthcare environments.

Essential metadata such as patient age, gender, and diagnostic details are included with every EEG recording, giving our models crucial context for training and assessment. The identification of subtle neurological patterns required for accurate categorization is made possible by high-resolution EEG data. The datasets are separated into training, validation, and testing sets to facilitate efficient model building. This ensures that the three goal categories—epileptic, non-epileptic, and psychogenic non-epileptic seizures—are fairly represented.

Table 48: Data Description

Dataset Component	Description
Total EEG	Over 10,000 5 second segmentation from seizure and non-seizure.
Patient Demographics	Including both adult genders.
Annotation Quality	Annotations provided by experienced neurologist.
Seizure Labels	Epilepsy seizure, Psychogenic non epileptic seizure and non-epilepsy
Signal Quality	High-quality clean EEG recordings
Annotation Details	Multi-class annotations, event markers, seizure start and offset information are supplied.
Training Set	Used for training.
Validation Set	Employed validation during training
Test Set	Applied for final performance evaluation

6.2.2 Data Separation

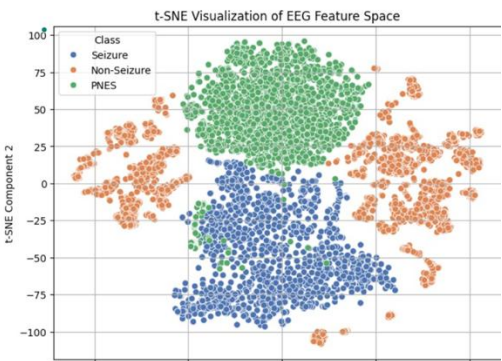


Figure 42: Data Class Separation

6.2.3 Experimental Setup

1. Data Collection and Preprocessing:

- Obtain the EDF data from KFUEH and <https://www.kaggle.com/datasets/abhishekinnvonix/epilepsy-seizure-dataset-seina-scalp-complete/data>
- Remove artifacts
- Extract the seizure event from the data and moving it to separate folder
- Segment the data to 5 seconds.
- Generate synthetic data and augmentation methods to increase the data size.

2. Model Selection and Configuration:

- Selecting (ShallowFBCSPNet, DeepEEGNet and GNN) for binary classification (seizure and non-seizure)
- Selecting (BLSTM, CNN-PGM and GNN) for trinary classification (seizure, PNES and non-seizure)
- Choose the appropriate pre-trained weights for each model.
- Configure the architectures by adjusting the output layers to match the number classes, regularisation techniques, early stopping and class weights

3. Training Procedure and Evaluating Metrics:

- All models were trained with cross-entropy-based loss functions that are proper for binary and trinary classification.
- Monitoring training and evaluating loss during model optimization.
- Evaluating based on (Accuracy, Loss, F1 and Confusion matrix)

4. Hardware and Software Environment:

Hardware:

- Google Colab T4 GPU.
- NVIDIA RTX GPU
- Apples M2 chips using MPS (Metal Performance Shaders)

Software:

- Programming Language: Python 3.11, JavaScript
- Deep Learning Libraries: PyTorch
- Other Libraries: MNE, Scikit-learn, Matplotlib, Pandas and Numpy.
- Deployment: Flask.
- Frontend: React.js.
- Database: MongoDB.
- Language Model:
 - o OpenAI
 - o PMC_LLaMA
- Development Environments: Jupyter Notebook, Google Colab, VSCode
- Operating System: Windows and MacOS

5. Ethical Considerations

- Anonymized all EEG dataset
- Used authorized or publicly accessible datasets while keeping to ethical guidelines.
- No patient interactions in real time.
- Clinically authorized data was used.

6. Documentation and Reporting:

- Documenting project stages for future reference.
- Producing performance reports using visuals.
- presented results in a way that was clear and scientifically sound through organized reports.

6.2.4 Description of the Proposed Techniques:

6.2.4.1 Synthetic Data

- Generating synthetic data to address the data imbalance issue.
- Using GANs and VAE to create a high-quality synthetic sample particularly for rare cases such as PNES seizure.
- The samples were carefully validate using T-SNE visualizing to make sure they were very similar to actual EEG signals.

6.2.4.2 Training Model

All mode were trained with Cross-entropy loss function, and evaluated based on the accuracy, loss, F1-score and confusion matrices. Also, adding early stopping, dropout and data augmentation to prevent overfitting.

Binary:

ShallowFBCSPNet, DeepEEGNet, and GNN were trained to classify between Seizure and Non-Seizure

Trinary:

BLSTM, CNN-PGM, and GNN were trained to classify between seizure, Non-Seizure and PNES

6.2.4.2.3 Large Language Model

Included language models to improve the clinical accessibility of the system:

Chatbot (For Patients):

Using OpenAI API to provide users with medical related information.

Generating Reports (For Doctors):

Using OpenAI language model to use the classification result and generate a report.

Doctors and patients are guaranteed to obtain precise forecasts and clear medical explanations due to this combination of diagnostic AI models and language creation modules.

6.2.5 Preprocessing:

6.2.5.1 Data Cleaning

The raw data were cleaned by resampling all the signals to 128 HZ, using bandpass filter to remove noise. This step is crucial to make sure only important neural activity remains.

6.2.5.2 Seizure Event Extraction:

Automated methods were used to extract seizure events from the continuous EEG data. To make processing the dataset easier and increase training effectiveness, extracted seizure segments were then placed in special files and arranged apart from non-seizure segments.

6.2.5.3 Segmentation:

All data were segmented into fixed 5-seconds windows, to ensure this window size balances capturing enough temporal information while maintaining reasonable input sizes for deep learning models.

6.2.5.4 Synthetic Data & Augmentation:

To solve the imbalance problem, generating synthetic data using GANs and VAEs as well as using augmentation methods such as amplitude scaling, adding noise, time wrap .. etc.

6.2.5.5 Normalization and Padding:

The segments were normalized to provide consistency throughout the dataset. Also, padding was used to ensure all the segments (generated or not) to standardize segment lengths and ensure compliance with model input requirements.

6.2.5.6 Data Splitting:

The final step was to split the data to (train, validation and test) sets. Class balance had been preserved across all classes, ensuring that performance metrics accurately reflected real-world generalization.

6.2.6 Optimization Strategy

6.2.6.1 Hyperparameter Tuning:

Learning Rate: To maximize convergence speed and model stability, various learning rates were tested. To avoid overfitting and increase classification accuracy across the epileptic, non-epileptic, and PNES categories, strategies such as learning rate decay and adaptive optimizers (such as the Adam optimizer) were used.

Batch Size: To balance GPU memory consumption and model generalization, several batch sizes were evaluated. While intermediate batch sizes sped up training without sacrificing model performance, smaller batch sizes increased gradient accuracy.

Number of Epochs: Early stopping was used to avoid overfitting during extended training, and the

proper number of epochs was established by tracking validation loss and classification accuracy.

6.2.6.2 Model Architecture Adjustments:

Hybrid Model Incorporation: To better capture the spatial-temporal properties of EEG data, the model complexity grew to incorporate BLSTM, CNN-LSTM hybrids, PGM-CNN, and GNNs after basic CNN architectures were first taken into consideration.

Temporal Feature Extraction: By using recurrent layers (LSTM and BLSTM), the models were able to recognize sequential patterns in EEG data that are essential for distinguishing between various kinds of seizures.

Graph-Based Learning: By modeling the intricate, non-linear interactions between EEG channels, GNNs greatly increased the sensitivity of PNES detection.

6.2.6.3 Data Augmentation Techniques:

Window Overlapping: To provide more training samples and more accurately depict transition phases between seizure and non-seizure events, dynamic segmentation with 20% overlap was used.

Noise Injection: To increase the model's resistance to artifacts, controlled Gaussian noise was introduced to EEG data.

Synthetic Data Generation: In order to supplement underrepresented PNES instances, it was investigated to create synthetic EEG data using Variational Autoencoders (VAE) and Generative Adversarial Networks (GANs).

6.2.6.4 Regularization Techniques:

Dropout layers and batch normalization were two regularization approaches used in the model designs to improve model generalization and lessen overfitting. In order to avoid overfitting and maximize training efficiency, Early Stopping was also used during training by tracking validation loss.

6.2.6.5 Class Imbalance Handling:

Class Weights: To guarantee that minority classes—particularly PNES—were not underrepresented in model learning, customized class weights were included during training.

6.2.6.6 Transfer Learning and Fine-Tuning:

Use of Pretrained Models: Some models that had previously been trained on extensive time-series datasets were modified for the EEG domain by freezing early feature-extraction layers and fine-tuning higher-level layers.

Progressive Unfreezing: To enable previous levels to gradually adjust to EEG-specific patterns,

network layers were gradually unfrozen after training.

6.2.6.7 Performance Monitoring and Early Stopping:

Validation Metrics: Across all three classes—Epileptic, Non-Epileptic, and Psychogenic Non-Epileptic Seizures (PNES)—model training was continually tracked using metrics including validation loss, classification accuracy, and F1-score.

Early Stopping: When overfitting symptoms emerged, training was automatically stopped by using patience thresholds.

6.2.6.8 Cross-Validation:

k-Fold Cross-Validation: To verify model resilience and guarantee generalizable performance across various patient subgroups, a 5-fold cross-validation technique was employed.

6.2.6.9 Ensemble Methods:

By combining predictions from many trained models (BLSTM, CNN-LSTM, PGM-CNN, and GNN) through the use of soft voting approaches, the final classification accuracy and stability were increased.

6.2.6.10 Hyperparameter Optimization Tools:

Grid Search and Random Search: To methodically investigate and improve hyperparameters such batch sizes, learning rates, and dropout rates, grid search and random search were used.

6.2.6.11 Post-Processing Techniques:

Confidence Thresholding: To provide doctors with only extremely dependable classification findings, post-processing entailed screening out predictions with low confidence.

LLM-Assisted Report Generation: To improve the system's usability for physicians, large language models (LLMs) were used to automatically produce clinical-style reports using the classification outputs and patient metadata.

6.2.7 Results and Discussion

6.2.7.1 Overall Model Performance

To identify and distinguish between epileptic, non-epileptic, and psychogenic non-epileptic seizures (PNES), our

study used deep learning-based classification models, namely CNN-PGM, BLSTM, and GNN. Accuracy, loss, and the F1 score were used to assess each model. The outcomes show that all models perform well, especially when it comes to trinary classification tasks.

Table 49 Overall Trinary Classification Performance

Model	F1 Score	Accuracy	Loss	Epochs
GNN	0.96	0.96	0.52	800
CNN-PGM	0.9	0.96	0.34	20
BLSTM	0.92	0.92	0.13	30

These numbers show how well the models can balance memory and accuracy, which results in strong F1 scores and overall classification efficacy. Even with fewer training epochs, CNN-PGM demonstrated effective learning as it obtained the lowest loss.

6.2.7.2 Disease-Specific Performance (Synthetic Data)

We used GAN and VAE approaches to generate synthetic data to address class imbalance, namely the lack of PNES data. The metrics from the data creation process are listed below:

Table 50 Synthetic Data Generation Performance

Data Type	Generator Loss	Discriminator Loss	Validation Loss	Epochs
Seizure (GAN)	0.57	0.51	0.66	100
PNES (VAE)	0.10	2.22	0.00	16

6.2.7.3 Results of the Binary Models

We experimented with binary classification models prior to applying the trinary technique. Despite its great accuracy, they had trouble distinguishing PNES from other kinds of seizures.

Table 51 Binary Model Performance

Model	F1 Score	Accuracy
GNN	0.99	0.99
ShallowFBCSPNet	0.99	0.99
DeepEEGNet	0.95	0.95

Despite its high accuracy, DeepEEGNet's incredibly low F1 score suggests that it handles class imbalance poorly or misclassifies minority classes. The necessity for a more complex trinary categorization was reaffirmed by this discovery.

6.2.7.4 Discussion

Our findings support the efficacy of hybrid deep learning models for seizure categorization, such as CNN-PGM and GNN. The model's resilience for underrepresented seizure categories was enhanced by the addition of synthetic data. Because they produced more predictions that may be used in clinical settings, trinary models fared noticeably better than binary models.

Strengths: Strong performance utilizing fewer training epochs in some models, dependable PNES production using VAE, and high F1 and accuracy ratings.

Difficulties: VAE discriminator loss was substantial; PNES generation remained noisy. In binary tasks, DeepEEGNet did not perform well.

Lessons Learned: Classifying uncommon diseases requires the use of hybrid architectures and synthetic data.

Future Work: Model reliability and deployment readiness will be further improved by integrating clinical trials, growing datasets, refining GAN-VAE pipelines, and expand our classification with other types of seizures.

6.3 System Interface Overview

The EpilepTech platform provides a thorough and intuitive user interface designed for physicians and patients involved in the diagnosis and treatment of epilepsy.

6.3.1 Home Page

The platform's webpage presents its goal of using AI-powered solutions to transform the treatment of epilepsy. In addition to highlighting three key features—AI Classification, Doctor Dashboard, and Patient Portal—it also emphasizes two entry points: Doctor Portal and Patient Portal. To help consumers understand what the platform has to offer, each function is depicted with clear iconography and succinct explanations.

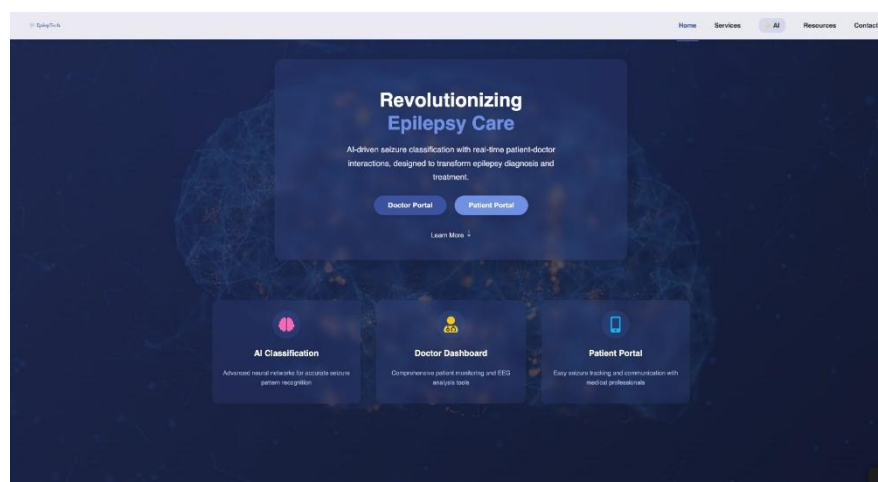


Figure 43: Home Page

6.3.2 Service Page

Numerous epileptic care options are described on the services page, such as automated epilepsy diagnosis, doctor-patient consultations, seizure classification, medical report generation, EEG analysis with AI, and treatment effectiveness analysis. Every service has a matching symbol and a brief description that explains its importance and how it may be used for continuing treatment or diagnosis.

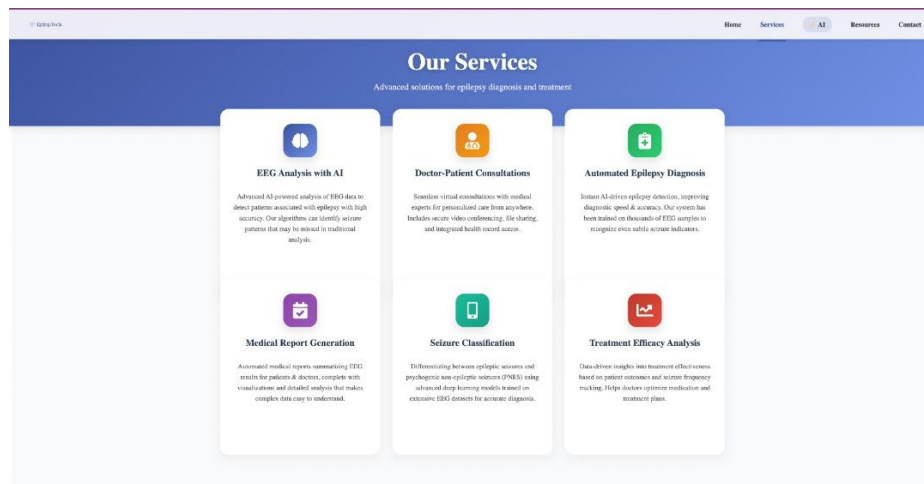


Figure 44: Service Page

6.3.3 AI Page

With descriptions of Deep Learning Analysis, AI Chatbot Support, Neural Networks, Extensive Training, High Accuracy, and Human-AI Collaboration, this section highlights the platform's technological capabilities. These arguments support the dependability and intelligence of the AI algorithms that are employed across the platform.

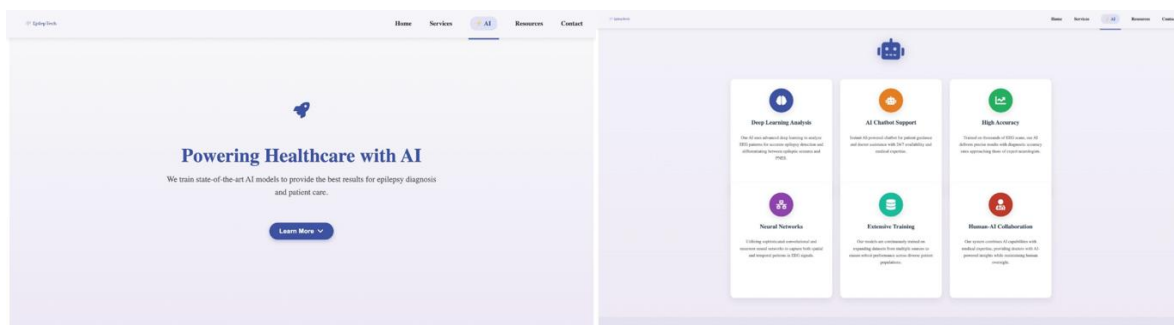


Figure 45: AI Page

6.3.4 Resources Page

Users may access important materials and studies, including official WHO guidelines, the influence of deep learning on EEG, and research on AI in epilepsy diagnosis, under the resources page. For visual clarity, each object is presented in the form of a card with a symbol and a brief description.

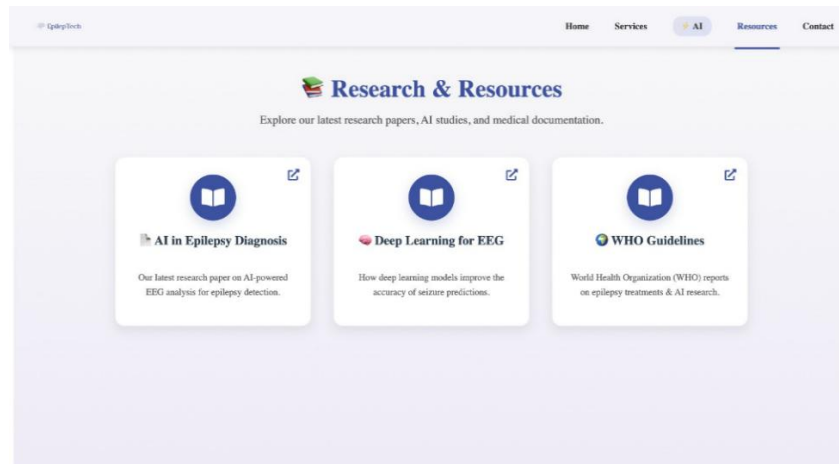


Figure 46: Research & Resources Page

6.3.5 Contact Page

All of the channels of communication required for assistance and questions are available in the contact section. Users can contact the mentioned institution in Dammam, Saudi Arabia, via phone, email, or in person. Additionally, there is a contact form that people may utilize to send messages straight from the website.

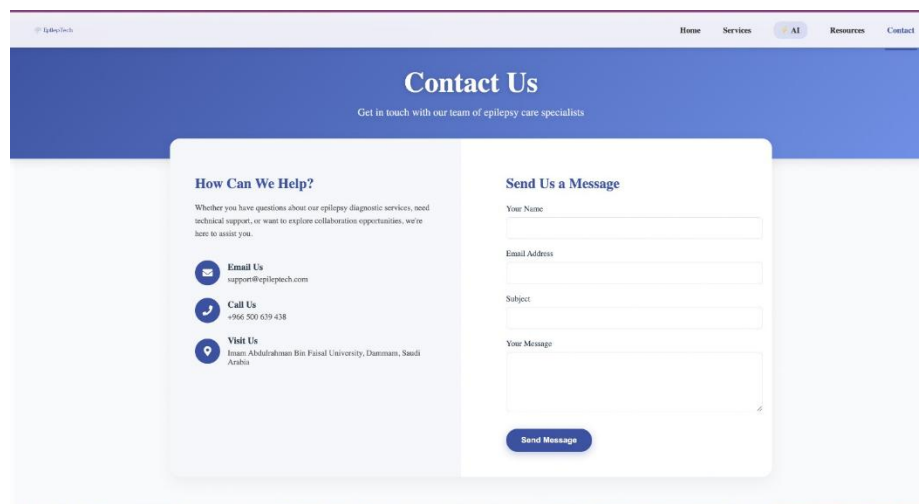


Figure 47: Contact Page

6.3.6 Patient Page

To access their customized dashboard, patients are sent to a secure login screen where they must enter their username and password.

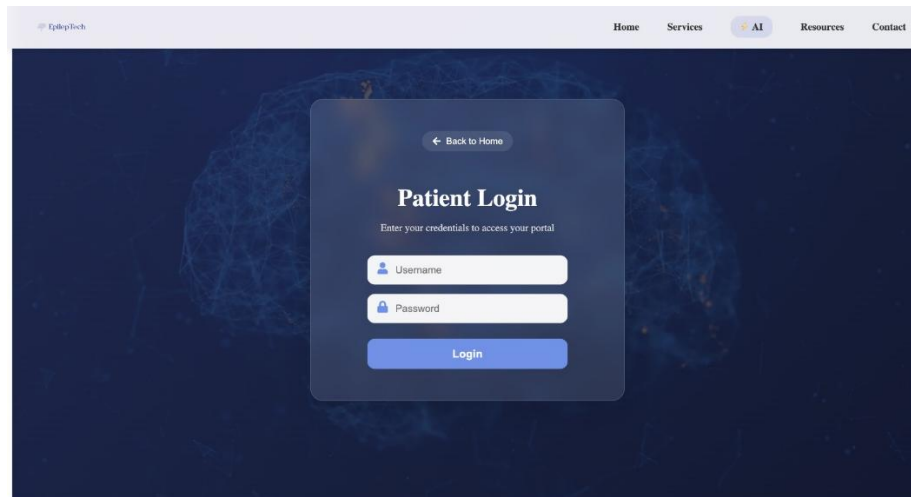


Figure 48: Login Page

6.3.7 Dashboard Overview

Overview of the Dashboard Patients are presented with a customized dashboard that summarizes their health information after logging in. In addition to visual trends of seizure frequency over time, medication adherence percentages, and forthcoming visits, the summary displays their most recent EEG study findings with interpretative confidence (e.g., “non-epileptic – 92%”). Additionally, patients may see their recent activity, which includes completed EEG evaluations, medicine refills, and any new communications from their neurologist. Their treatment progress and interactions with care professionals may be quickly and thoroughly understood thanks to the layout.

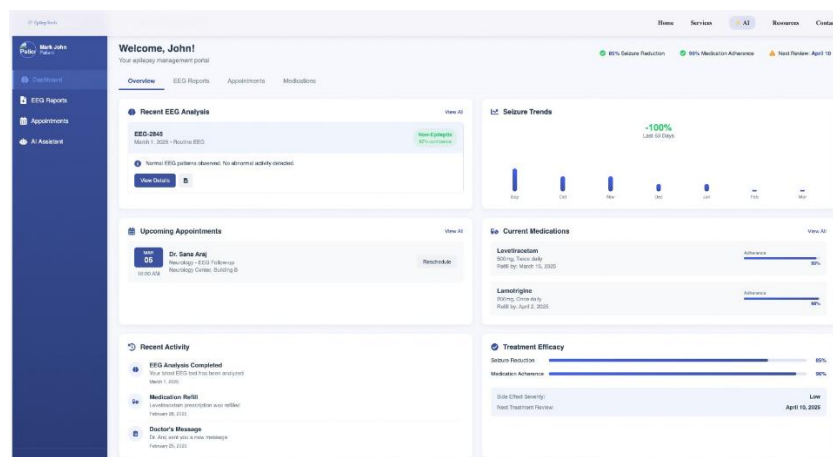


Figure 49: Dashboard

6.3.8 EEG Reports

Patients can obtain their past EEG test results in an orderly manner through the EEG Reports area. The report ID, test date, EEG type, doctor assigned, diagnostic outcome, and AI confidence level for the diagnosis are all included in each report entry. Patients can get a copy of the report or examine specific results by clicking on action options. Easy navigation and an educated patient assessment of medical insights are the goals of this area.

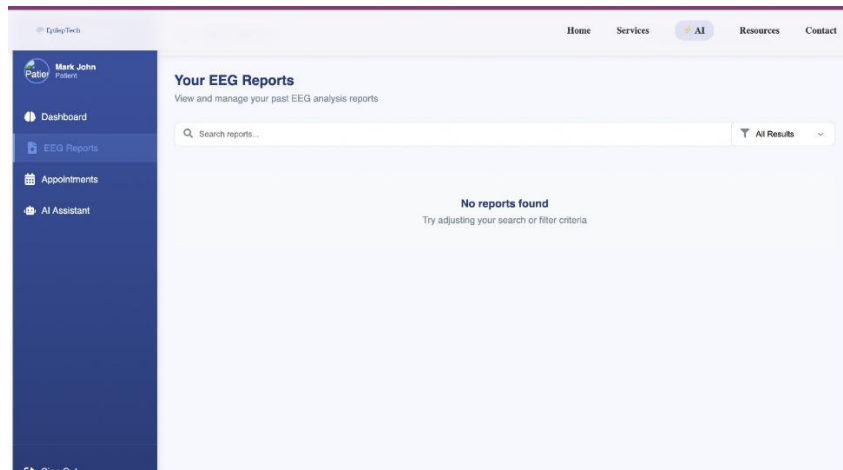


Figure 50: EEG Report

6.3.9 Appointments Module

Patients may easily manage their clinical appointments with the help of the appointment section. All planned appointments are shown, along with pertinent information such as the date, time, doctor, and clinic location. With only one click, patients may postpone or cancel. If there are no appointments available, a booking interface allows you to make one by choosing the doctor, visit type, date, time, and consultation reason. To make navigating simple, tabs are also included for switching between previous and forthcoming appointments.

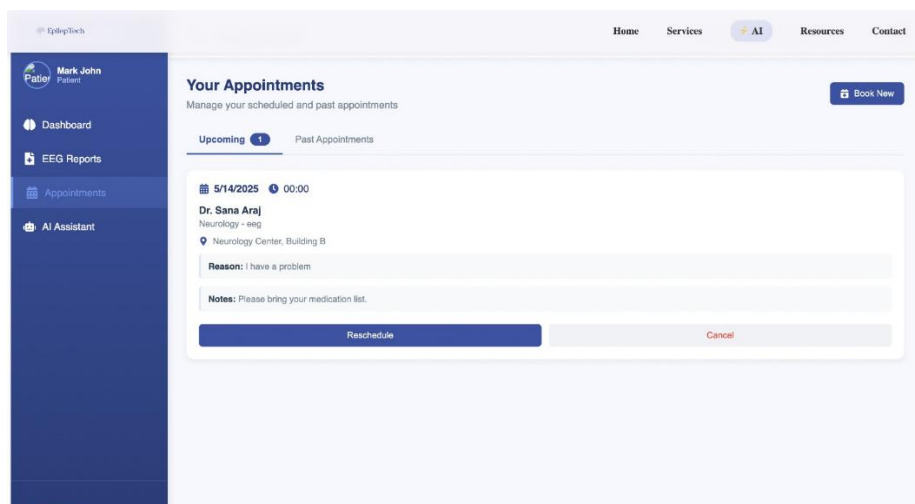


Figure 51: Appointments Module

6.3.10 Medications Panel

A list of currently prescribed medications, together with quantity guidelines and refill due dates, is shown in this section. Additionally, it displays drug adherence as a percentage, visually represented by circular progress indicators, and includes reported adverse effects, such as weariness or dizziness.

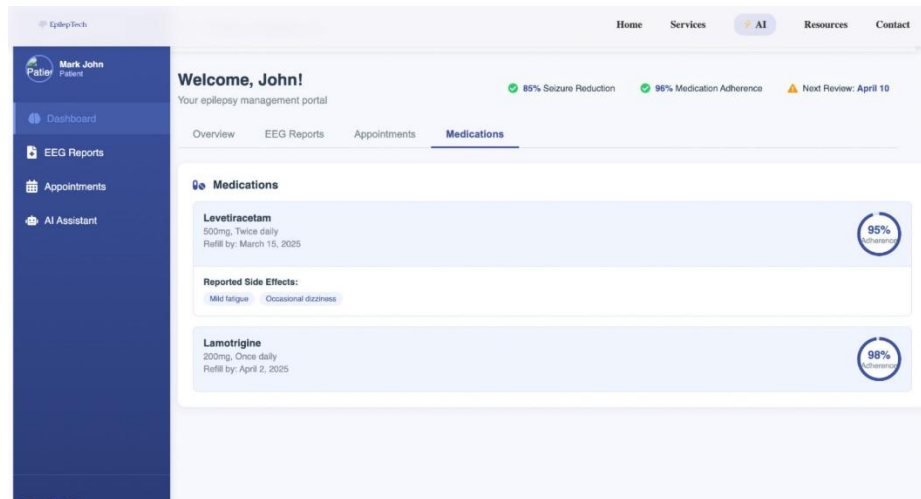


Figure 52: Medication Panel

6.3.11 Book Appointments Interface

Using a form on the booking system, patients may pick a doctor, indicate the sort of appointment, choose a desired day and time, and explain why they are there. After finishing, the request for an appointment can be sent in for verification.

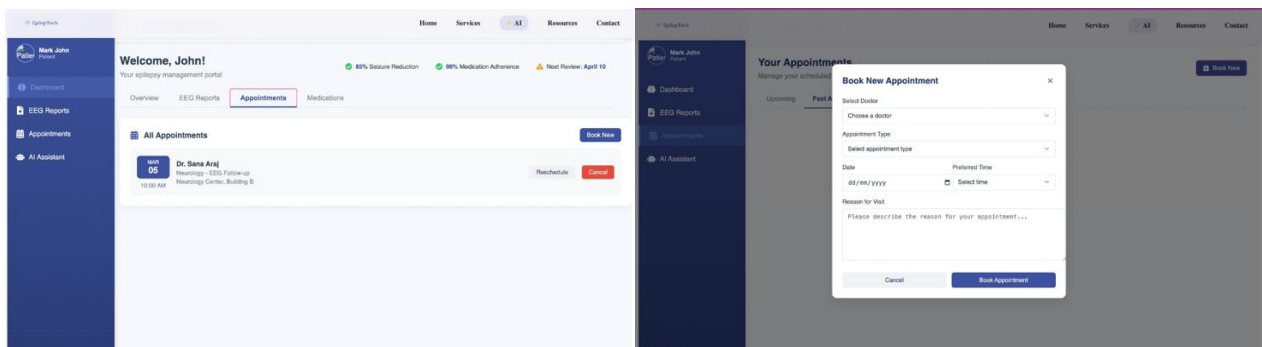


Figure 53: Book Appointment

6.3.12 AI Chatbot

The AI-powered assistant provides round-the-clock assistance by responding to often asked queries about prescription drugs, scheduling appointments, seizure types and lifestyle advice. Access to medical advice is improved by allowing patients to write their own questions or choose from pre-suggested subjects to get prompt answers.

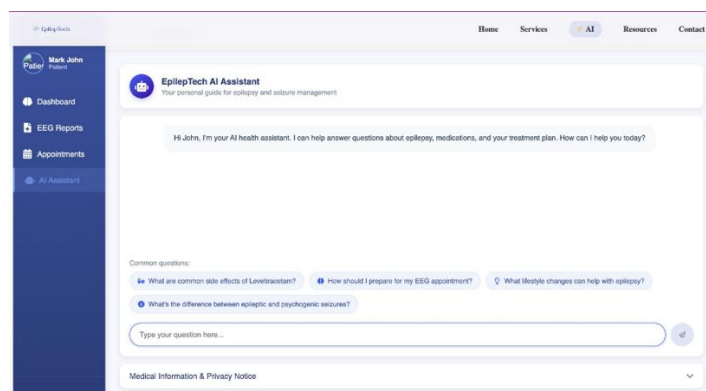


Figure 54: AI Chatbot

6.3.13 Doctor Page

Doctors enter their professional dashboard by using their username and password to access their secure login page.

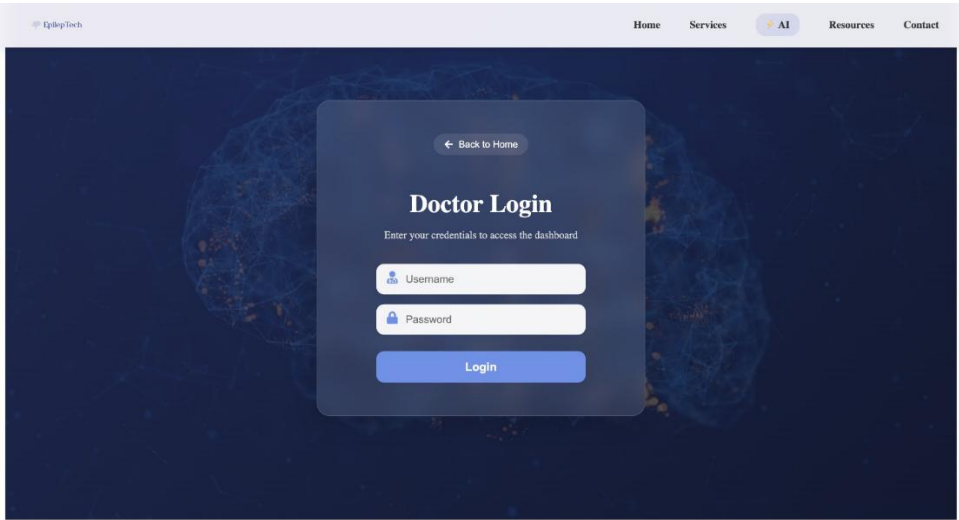


Figure 55: Doctor Login

6.3.14 Doctor EEG Analysis Reports

Physicians have access to any EEG results that patients have filed or that are connected to them in this area. To help clinicians prioritize cases swiftly, the interface separates reports into three tabs: All Reports, Pending, and Completed. It also offers search and filter features. Each report has links to read all the details, along with the patient's name, ID, date, outcome, and status.

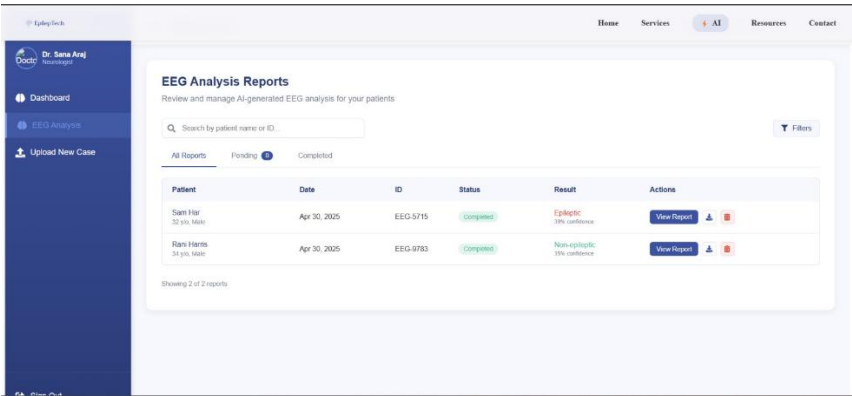


Figure 56: Doctor EEG Report Page

6.3.15 Doctor Dashboard

A customized dashboard that provides a summary of their recent EEG analysis activity is shown to doctors upon arrival. In addition to providing easy action buttons for reading reports and uploading new cases, it displays the number of outstanding analyses as well as the number of analyses finished today. A recent activity stream also provides timestamps for events like finished AI analysis and the uploading of new cases. Doctors can effectively

manage diagnostic workflows and keep an eye on their caseload thanks to this simplified interface.

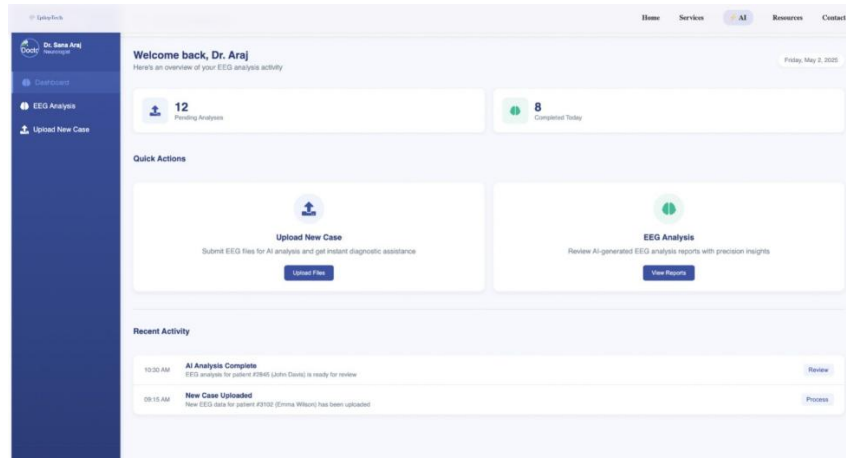


Figure 57: Doctor Dashboard

6.3.16 Doctor Case Upload

Upload of a Doctor Case by using the upload interface, doctors may quickly upload EEG data for AI analysis by dragging and dropping or browsing files. This lessens the administrative load and enables speedy diagnostic turnarounds.

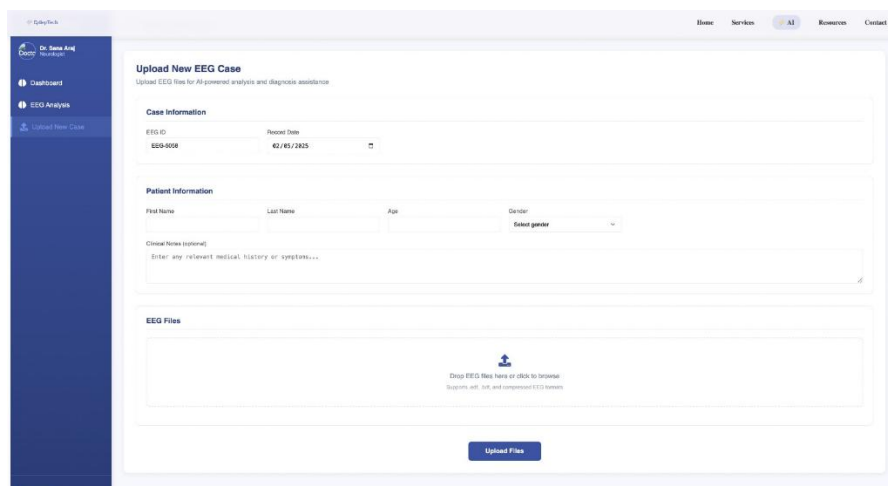


Figure 58: Doctor Case Upload Page

Chapter 7: Software Testing (Software Test Plan)

7.1 Test Plan

EpilepTech is designed as an intelligent, user-focused platform for analyzing EEG data and assisting in the classification of seizures, including epileptic, non-epileptic, and psychogenic (PNES) types. To ensure its functionality and reliability, we conducted thorough testing across both the frontend (web interfaces for doctors and patients) and backend (AI classification engine, database, and APIs). Testing was guided by the system's Software Requirements Specification (SRS) and Software Design Specification (SDS). We verified that core features, from secure login and EEG file uploads to AI-based analysis, chatbot responses, and automated PDF report generation, worked as expected. Given the medical nature of the platform, particular attention was given to data accuracy, consistency, and security. Our aim was to deliver a smooth and dependable experience for both healthcare professionals and patients.

7.2 Features to be Tested.

Testing covered all major features outlined in the SRS and SDS documents. These include:

- EEG data upload and preprocessing
- AI-driven classification of seizure types
- Confidence score visualization
- Display and download of diagnostic reports in PDF
- Patient and doctor login/authentication
- Doctor dashboard with patient-specific report access
- Patient-side chatbot integration for health-related queries
- Appointment booking functionality
- MongoDB-based data persistence and retrieval

Each feature was carefully tested to ensure correct operation, user clarity, and consistent results across real usage scenarios.

7.3 Test Report (Approach)

To validate the system's robustness, usability, and readiness for deployment, a comprehensive mix of testing techniques was applied throughout development and post-integration stages. The following methods were used:

- **Data and Database Integrity Testing**
- **Component (Unit) Testing**
- **Integration Testing** (AI model + web + DB)
- **User Interface (UI) Testing**
- **Interface/API Testing** (between frontend and Flask backend)
- **Validation and Verification Testing**
- **Security Testing** (account protection, access control)
- **Performance Testing** (latency in prediction/report generation)
- **Constraint Testing**
- **Beta Testing** with selected users (doctors and patients)

This multi-layered testing approach allowed us to detect and resolve issues across different parts of the platform while ensuring the system performs consistently under various conditions.

7.3.1 Data and Database Integrity Testing

As EpilepTech handles confidential health data and AI-generated medical insights, ensuring database reliability was a top priority. MongoDB was used to store user details, EEG session metadata, results, and system logs.

Techniques Used:

- Simulated standard DB operations (insert, update, delete) using realistic and edge-case data to observe system behavior.
- Checked referential integrity and ensured atomic updates across linked collections (e.g., linking reports to users).
- Validated that data retrieved for patient history, report download, and dashboard analytics was always accurate and complete.

Completion Criteria:

- All database operations must execute without triggering errors, data corruption, or inconsistencies.
- Data should remain reliable even after multiple read-write cycles and concurrent access during test sessions.

Special Focus:

- Testing emphasized real-world scenarios, such as retrieving historical EEG reports for a given patient or matching classification results with record IDs.
- This ensured both user trust and backend accuracy, critical for medical systems.

7.3.1.1 Test Cases

Table 66 shows the test cases with expected results.

Table 52: Test Cases with Expected Results

Test ID	Test Description	Expected Result	Verified (Yes/No)
1	Update a patient's EEG report metadata	The record is updated in the MongoDB collection and correctly reflected in the UI	Yes
2	Insert a new EEG session record after file upload	A new document is created in the database, linked to the correct user, and accessible via the dashboard	Yes
3	Delete an EEG report	The selected record is removed, and all linked references (e.g., PDF files, patient dashboard view) are appropriately handled	Yes
4	Attempt to insert a duplicate _id (MongoDB ObjectId)	Operation is rejected due to unique constraint violation	Yes

5	Retrieve past EEG reports for a specific user	The query returns only the records that match the user's ID and search filters	Yes
6	Insert a null value in a required field	Document insertion is rejected, or validation fails if the backend enforces schema	Yes
7	Insert a record with incorrect data type	Insertion is rejected or flagged by schema validation layer in backend logic	Yes
8	Insert an overly long string in a limited field	Insertion is blocked or trimmed to respect length validation	Yes
9	Insert a report with a user reference that doesn't exist	Operation is denied or flagged since no corresponding user is found	Yes

7.3.2 Component Testing

Component testing in EpilepTech focused on validating each functional unit of the system independently before combining them into the complete workflow. This allowed us to catch bugs at the individual level and ensure that each part behaves as expected in isolation.

Technique Used:

- Each core module, such as EEG file upload, AI classification response handling, report generation, user login, and chatbot interaction, was tested separately based on the scenarios defined in the Software Requirements Specification (SRS).
- We tested each component using both valid and invalid inputs:
 - For valid inputs, the component was expected to return the correct output and proceed through its normal flow.
 - For invalid inputs, appropriate error messages or validation prompts were expected, and the function was halted without performing unintended actions.

Completion Criteria:

- Every tested component had to operate reliably without crashing, freezing, or producing inconsistent results.
- Any error conditions had to be properly caught and handled, ensuring graceful recovery or feedback.

Special Consideration:

- Complex components were further broken down into smaller units. For example, the EEG upload feature was split into file selection, backend transmission, and post-upload status rendering, each tested individually to ensure consistent behavior.

7.3.2.1 Test Cases

7.3.2.1.1 Login Functionality (Doctor & Patient)

The following Table 67 shows test cases for the 'Login' function.

Table 53: Test Cases of 'Login' feature.

Test ID	Login_001
Prerequisite	User (doctor or patient) account exists in the database

Test Procedure	The login feature was tested by entering: 1. Correct username and password 2. Correct username with different capitalization 3. Empty username and/or password fields 4. Correct username with incorrect password 5. Correct credentials with incorrect capitalization in the password
Expected Result	The result of each of the previously mentioned test procedures: 1. User logs in and is redirected to the dashboard 2. Login is successful 3. Error message: "Invalid Entries"; fields are cleared 4. Error message: "Invalid Username or Password"; fields are cleared 5. Error message: "Invalid Username or Password"; fields are cleared
Actual Result	Matching the expected results.
Verified (Yes/No)	Yes

7.3.2.1.2 EEG Upload and Classification (Doctor)

The following Table 68 shows test cases for the 'EEG Upload and Classification' function.

Table 54: Test Cases of 'EEG Upload and Classification' feature.

Test ID	EEG_Upload_002
Prerequisite	Doctor is logged in
Test Procedure	Upload. edf EEG file through the frontend
Expected Result	File is uploaded , AI classification is triggered
Actual Result	EEG classified and result saved
Verified (Yes/No)	Yes

7.3.2.1.3 View EEG Diagnosis (Doctor)

The following Table 69 shows test cases for the 'View EEG Diagnosis' function.

Table 55: Test Cases of 'View EEG Diagnosis' feature.

Test ID	EEG_View_003
Prerequisite	EEG classification completed
Test Procedure	Navigate to "EEG Analysis" → click "View Report"

Expected Result	Diagnosis modal shows prediction and confidence scores
Actual Result	Matching the expected results.
Verified (Yes/No)	Yes

7.3.2.1.4 Generate PDF Report (Doctor)

The following Table 70 shows the ‘Generate Report’ feature test cases. Test cases include, but are not limited to, the following items in the table.

Table 56: Test Cases of ‘Generate Report’ feature.

Test ID	Report_Gen_004
Prerequisite	EEG has diagnosis result
Test Procedure	Click "Download" PDF
Expected Result	Full medical report with patient info and AI-generated insights
Actual Result	Matching the expected results.
Verified (Yes/No)	Yes

7.3.2.1.5 Chatbot Response (Patient)

Table 71 shows the ‘Chatbot Response’ feature test cases. Test cases include, but are not limited to, the following items in the table.

Table 57: Test Cases of ‘Chatbot Response’ feature

Test ID	Chatbot_005
Prerequisite	Patient is logged in
Test Procedure	Open "AI Assistant" tab, type a health-related question
Expected Result	Chatbot provides OpenAI-generated medical advice
Actual Result	Response shown in UI
Verified (Yes/No)	Yes

7.3.2.1.6 Doctor Case Overview

The following Table 72 shows test cases for the ‘Doctor Case Overview’ function.

Table 58: Test Cases of ‘Doctor Case Overview’ feature

Test ID	Doctor_Review_006
Prerequisite	Cases exist
Test Procedure	Open "EEG Analysis Reports"
Expected Result	List displays patient name, ID, result, date, and actions
Actual Result	Matches backend database
Verified (Yes/No)	Yes

7.3.2.1.7 Patient Diagnosis History

The following Table 73 shows the test cases of ‘Patient Diagnosis History’ functionality.

Table 59: Test Cases of ‘Patient Diagnosis History’ feature.

Test ID	Patient_View_007
Prerequisite	Patient has prior reports
Test Procedure	View “EEG Reports” in patient portal
Expected Result	Diagnosis results, confidence levels, and actions shown
Actual Result	Works as expected
Verified (Yes/No)	Yes

7.3.2.1.8 Appointment Booking

The following Table 74 shows test cases for the ‘Appointment Booking’ function.

Table 60: Test Cases of ‘Appointment Booking’ feature.

Test ID	Book_Appointment_008
Prerequisite	Patient logged in
Test Procedure	Click “Book Appointment” → fill form → submit

Expected Result	Appointment saved and listed
Actual Result	Appointment stored in MongoDB
Verified (Yes/No)	Yes

7.3.3 Integration Testing

After completing component-level testing, we carried out integration testing to ensure that the various parts of the EpilepTech system work together smoothly. This stage focused on verifying those interactions between the frontend (React), backend (Flask), database (MongoDB), and AI services (OpenAI) were functioning correctly as a complete system. The main objective was to confirm that full user workflows, such as uploading an EEG, viewing the diagnosis, and downloading a report, could be completed without issues.

Testing Approach:

We began by testing the system from its main access points, such as the login pages for doctors and patients. From there, we followed typical user journeys to make sure data was passed correctly between the frontend, backend, and database. For example, when a doctor uploads an EEG file, the system should process the file, generate a result using the AI model, and return a clear diagnosis to be displayed and downloadable. Each step in the process was tested individually, and then in sequence, to ensure proper integration.

If any issues appeared during testing, we isolated the affected components, resolved the issue, and retested the flow to ensure stability.

Completion Criteria:

Integration testing was considered successful once all key user workflows ran without errors. This included correct data exchange between components, real-time responsiveness, and consistent display of results across both interfaces (doctor and patient). No information should be lost, delayed, or incorrectly presented at any stage.

Key Scenarios Tested:

- A doctor logs in, uploads an EEG file, receives the diagnosis, and downloads the report.
- A patient logs in and interacts with the AI chatbot to receive health-related guidance.
- A patient accesses their EEG report history and views/downloads previous results.
- A doctor accesses their dashboard and views a full list of patients with status, predictions, and available actions.

7.3.3.1 Test Cases

7.3.3.1.1 Doctor Interface Integration Test Case

Table 72 illustrates the integration among components from the Doctor's home interface.

Table 61: Integration Test Case of 'Doctor Home' Interface

Test ID	Doctor_Home_Interface
Prerequisite	The doctor is successfully logged into the system.
Test Procedure	Attempt each of the following individually: 1. Access the Upload New Case page. 2. Upload an EEG file and verify diagnosis appears. 3. View the AI-generated report and download the PDF. 4. Check Diagnosis History . 5. Log out and confirm redirection to the login page.
Expected Result	1. Each interface should open correctly and display its content. 2. EEG data should flow seamlessly to the backend and return an accurate prediction. 3. The correct doctor identity should persist across pages during the session. 4. All confirmation messages and error messages (if any) should be clear and functional.
Actual Result	Matching the expected results.
Verified (Yes/No)	Yes

7.3.3.1.2 Patient Interface Integration Test Case

Table 76 illustrates the integration among components from the Patient' home interface.

Table 62: Integration Test Case of ' Patient Home ' Interface

Test ID	Patient_Home_Interface
Prerequisite	The patient is successfully logged into the system.
Test Procedure	Attempt each of the following individually: 1. Access the EEG Reports page to view previous reports. 2. Open and download a specific report. 3. Navigate to Appointments and try booking a new appointment. 4. Interact with the AI Assistant chatbot. 5. Log out and confirm redirection to the login page.
Expected Result	1. All pages should load with personalized content. 2. Previously generated reports should display correctly and be downloadable. 3. Appointment booking should confirm and appear in the records. 4. Chatbot should respond to queries appropriately. 5. The session should maintain patient identity and data throughout.
Actual Result	Matching the expected results.
Verified (Yes/No)	Yes

7.3.4 User Interface Testing

User Interface testing was carried out to ensure that the EpilepTech platform provides a smooth and intuitive experience for both doctors and patients. This testing focused on making sure the design elements looked and behaved consistently across pages, devices, and screen sizes.

Technique:

- We checked that all components (buttons, forms, navigation bars, etc.) are clearly visible, aligned, and easy to use.
- Interfaces were tested on various screen resolutions to ensure the design adjusts properly on both desktops and laptops.
- Font sizes, colors, and messages were evaluated for clarity and professionalism.
- Error messages, buttons, and links were reviewed to ensure correct spelling and user guidance.
- We confirmed that responsive design holds across zooming and resizing, without overlapping or hidden elements.
- Hyperlinks and buttons were tested to ensure they always lead to the correct destination.

Completion Criteria:

- The platform must be visually consistent, accessible, and usable without confusion or layout issues.

7.3.5 Interface Testing

Interface testing focused on verifying that all system parts, especially the backend, database, and third-party services, communicate properly. This is especially important in EpilepTech, where real-time interactions occur between the frontend (React), backend (Flask API), MongoDB (database), and external APIs (OpenAI for reports and chatbot).

Technique:

- We tested API endpoints responsible for uploading EEG files, retrieving results, storing user data, and generating reports.
- Backend to MongoDB connections were tested to ensure all data insertion, update, and retrieval work correctly.
- The mail service integration for account registration and password reset was tested to confirm delivery of verification messages.
- OpenAI API integration was verified for report generation and chatbot responses.

Completion Criteria:

- All connected services and APIs must respond correctly with accurate data, without delay or failure.

7.3.6 Validation and Verification Testing

To ensure our system is both technically correct and practically useful, we conducted both verification and validation testing.

Technique:

- **Verification:** During development, we reviewed code, architecture, and system designs to ensure we followed the requirements in the SRS and SDS.
- **Validation:** After development, we tested the full system with real scenarios to confirm that doctors and patients could easily use the system and receive meaningful outputs.

Completion Criteria:

- The system is verified against technical specifications and validated through realistic user interactions.

7.3.7 Security Testing

Security is a priority in EpilepTech, especially given the sensitivity of medical data. Security testing was done to ensure that only authorized users can access their data.

Technique:

- We checked that only registered users with correct usernames and passwords could log in.
- Sessions were monitored to ensure no data leakage or unintended access.
- Invalid login attempts were correctly handled with error messages.

Completion Criteria:

- Only authenticated users are allowed to access their respective dashboards.

7.3.8 Performance Testing

Performance testing was conducted to evaluate how well the EpilepTech platform performs under different usage conditions. Since the system is expected to handle real-time EEG uploads, AI-based classifications, and interactive report generation, it was important to assess responsiveness, stability, and reliability across the core functionalities.

Technique:

- Simulated multiple users accessing the platform simultaneously to mimic real-world usage.
- Tested the end-to-end process of uploading EEG data, classifying results, generating AI reports, and using the chatbot.
- Observed how the system handles increased load on both the backend (Flask API and MongoDB) and frontend (React interface).
- Verified that the system provides timely feedback to users and does not hang or crash under expected usage conditions.

Completion Criteria:

- The platform remains responsive during peak usage.
- All essential tasks, including login, file upload, classification, and report viewing, complete successfully without failure.
- Users receive appropriate feedback and validation messages even when submitting large or invalid input.

7.3.9 Constraints

All development and testing tasks were completed within the timeframes outlined in the Software Project Management Plan (SPMP). Meeting these deadlines was essential to ensure a steady and manageable workflow throughout the project lifecycle.

7.3.10 Beta Testing

Before finalizing the system, a beta version of EpilepTech was shared with a small group of users, for real-world testing. Their feedback provided valuable insight into usability, interface clarity, and system flow.

7.3.11 Acceptance Testing

Since EpilepTech was developed as part of an academic and research-focused project, a formal acceptance release process was not required. However, thorough internal testing and review sessions were conducted with the development team and faculty supervisor and the neurology consultant, to ensure that all core goals, as defined in the original requirements, were fully achieved.

7.4 Pass/Fail Testing

To be considered successful, the EpilepTech system needed to meet the following criteria:

- All key features functioned as described in the Software Requirements Specification (SRS).
- All defined test scenarios were executed and passed without major issues.
- Critical features such as EEG data upload, AI-based classification, PDF report generation, and chatbot support worked reliably and accurately.

Only after verifying these results was the system approved for demonstration and further use.

7.5 Testing Process

This section provides a summary of the key testing activities, responsibilities, and supporting materials used during the testing phase.

7.5.1 Test Deliverables

The following documents and materials were prepared to support the testing process:

- **Software Test Plan (STP):** A detailed outline of all test cases, methods, and expected results used throughout the system's evaluation.
- **User Guide:** A concise manual designed to help doctors and patients understand how to navigate the system and use its core features effectively.

7.5.2 Testing Tasks

The following table 77 outlines the core testing-related tasks carried out during EpilepTech's development, along with the required skills and task dependencies:

Table 63: Testing Tasks

Task #	Task Name	Skills Required	Dependencies
1	Prepare SRS and SDS documents	Writing and designing	-
2	Develop test cases for all key functionalities	Analytical Thinking, Testing	1

3	Prepare STP document	Writing & Planning	2
4	Set up test environment (Flask, MongoDB, React)	Technical & Deployment	3
5	Executing all the test scenarios	General System Use	4
6	Debug and fix any issues	Programming	5
7	Maintain the system and monitor logs	Backend + Frontend Skills	6

7.5.3 Responsibilities

All members of the EpilepTech development team shared responsibility for the preparation and execution of testing activities. This included writing the Software Test Plan (STP), running component and integration test cases, and logging results. When bugs were encountered, the team collaborated to debug and apply fixes. Each phase of testing was documented and reviewed before moving forward.

7.5.4 Resources

The resources used throughout testing are listed below. These tools and platforms supported development, collaboration, and real-time testing across the system:

Table 64: Testing Resources

Resource	Description
Hardware	Laptops running on both Windows and Mac operating systems. A high-performance PC (used by the supervisor) was also utilized to handle memory-intensive tasks during model testing and deployment.
Software	Google Colab (for running AI code), MongoDB Atlas (cloud database), Google Drive, Notion, Chrome (for browser testing), and Stack Overflow (as a technical reference).
Human	Project team members. Oversight was provided by our supervisor Dr. Mustafa and a neurology consultant from King Fahad University Hospital Dr. Modhi, ensuring both technical soundness and clinical relevance.

7.5.5 Schedule

Table 79 outlines the schedule for the testing tasks.

Table 65: Testing Tasks Schedule

Task	Date
Finalize Software Requirements Specification (SRS)	25 th December 2024
Finalize Software Design Specification (SDS)	30 th November 2024
Develop comprehensive test cases	1 st April 2025
Complete the Software Test Plan (STP)	17 th April 2025

Configure the test environment (Flask, MongoDB, etc.)	20 th April 2025
Executing all the test activities	27 th April 2025
Debug and resolve remaining issues	30 th April 2025

7.5.6 Environmental Requirements

This section outlines the required testing environment, including the hardware and software configurations, server-related needs, and associated risks and assumptions encountered throughout the testing process.

7.5.6.1 Hardware

Testing and model execution for EpilepTech were primarily conducted on personal laptops and PCs. However, due to the system's dependence on AI models and resource-intensive classification tasks, we frequently encountered system crashes, especially due to limited RAM on standard devices.

To address this, we utilized the high-performance workstation of our supervisor. This machine, equipped with advanced GPU capabilities and significantly greater memory, allowed us to execute the most demanding tests and AI inferences smoothly.

Hardware used includes:

- MacBooks and Windows laptops (with varying performance levels).
- Standard PC lab machines (limited memory caused frequent application crashes).
- A supervisor-provided high-end PC with large RAM and GPU resources (used for training/testing large models).

7.5.6.2 Software

The software tools we used during the development and testing phases were selected to match the needs of an AI-driven web platform while keeping collaboration and efficiency in mind. Most of the Python-based AI components were developed and executed using **Google Colab**, which simplified environment setup.

Software tools include:

- **Google Colab:** primary environment for running Python-based AI code and model testing.
- **MongoDB Atlas:** cloud-based database used for storing user records and classification results.
- **Google Drive:** used for dataset storage, shared outputs, and report drafts.
- **Notion:** for documentation tracking, planning, and task management.
- **Google Chrome:** used for testing user interface compatibility and performance.
- **Stack Overflow:** for referencing solutions to technical implementation and debugging issues.

7.5.6.3 Server

While EpilepTech does not rely on a dedicated external server, the testing setup had internal requirements for optimal performance:

- Low-latency response for database read/write operations.
- High memory allocation (especially when running OpenAI integrations and AI classification simultaneously).
- Consistent performance under local Flask API requests and database queries.

7.5.6.4 Risks and assumptions

During testing, several risks and constraints were anticipated. The following sections explain how the team planned for and adapted to such challenges:

7.5.6.4.1 Test Item Availability

At various stages, incomplete or changing components delayed full integration testing. We assumed the development team would continue testing available units and circle back to the incomplete ones once ready.

7.5.6.4.2 Test Resource Availability

Frequent RAM exhaustion and browser lag, especially during heavy AI inference or PDF generation, posed technical hurdles. Our assumption was to always escalate serious crashes to the supervisor's machine or re-run tests on lighter hardware with minimal loads.

7.5.6.4.3 Time Constraint

Due to unexpected delays (hardware limitations, model failures), the team occasionally needed to work outside regular hours to meet deadlines. We assumed flexibility in testing schedules, with extended work sessions and rescheduling where necessary.

7.5.6.4.4 Change Management Procedure

Any proposed change to the testing process followed a structured approval method:

1. The suggestion is presented to the team and supervisor (in meetings).
2. The group discusses and decides on the proposal.
3. Once consensus is reached, the change is documented and officially applied to the test plan.

Chapter 8: Conclusion

8.1 Introduction

The *EpilepTech* project was developed with the goal of creating a modern AI-powered diagnostic platform that helps identify and differentiate between three key neurological EEG patterns: epileptic seizures, psychogenic non-epileptic seizures (PNES), and normal non-seizure signals. At the heart of this system is a blend of deep learning models, synthetic EEG data generation, and language processing tools, all brought together into an accessible web-based platform designed with clinicians and neurologists in mind.

To handle classification, we utilized specialized neural networks like SallowFBCSPNet and DeepEEGNet for binary decisions. For more complex trinary classifications, we used Bidirectional LSTM (BLSTM), CNN-PGM, and Graph Neural Networks (GNN). Since PNES data is typically scarce, we applied Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) to generate additional high-quality synthetic EEG data.

We also integrated the OpenAI API to enable automatic medical report generation and used GPT-3.5 to support a conversational chatbot interface. Technically, the platform is built using Flask for backend operations, React for the user interface, and MongoDB for database management. *EpilepTech* reflects a meaningful step forward in combining AI with clinical practice, contributing to Saudi Arabia's Vision 2030 goals by aligning technology with real-world healthcare needs.

8.2 Key Contributions and Outcomes

Throughout the development of *EpilepTech*, several important outcomes were achieved. Our trinary classification models — including BLSTM, CNN-PGM, and the multi-layer GNN — showed strong performance and marked improvements in detecting PNES, a type of seizure that is notoriously difficult to diagnose. The synthetic EEG data generated using GANs and VAEs helped balance our datasets and improved our models' ability to generalize to new data.

By bringing together multiple AI techniques — from time-series deep learning to graph-based models and natural language processing — we were able to build a system that offers broad support in neurological diagnosis. The automated reporting feature, powered by the OpenAI API, produced clear and useful summaries that clinicians can use for documentation. Our use of both local data from King Fahd University Hospital and publicly available EEG data from Kaggle added depth and variety to the training process.

The platform was also designed with scalability and modularity in mind, making it well-suited for integration into clinical environments. All of these factors combined reflect meaningful progress in applying machine learning to the field of clinical neuroscience.

8.3 Entrepreneurial Potential

While *EpilepTech* began as a research-focused project, its real-world potential goes far beyond the academic setting. It fills an important clinical gap by offering a tool that can assist with the diagnosis of seizure-related conditions, especially in places where access to trained neurologists is limited. The user-friendly interface and automated features enable healthcare providers to make faster and more confident decisions.

Its modular architecture also opens the door for several commercialization paths. Whether deployed as a cloud-based service or as part of an on-site hospital system, *EpilepTech* is flexible enough to adapt to various settings. This kind of digital health innovation supports Saudi Arabia's growing entrepreneurial landscape and offers a strong model for startups looking to leverage AI for national healthcare challenges in line with Vision 2030.

8.4 Challenges Encountered

Like most complex projects, the development of *EpilepTech* came with its fair share of challenges. One of the biggest issues we faced was the limited availability of PNES data. This was addressed through the use of generative models, which helped us produce realistic synthetic EEG signals and enrich the dataset. Integrating different types of deep learning models into a single pipeline was also technically demanding and required careful backend development and interface design.

Another major challenge was balancing performance and system responsiveness. We initially experimented with LLaMA 3 for report generation, which worked well in terms of output quality but required more than 30GB of resources and caused noticeable slowdowns on the frontend. To solve this, we switched to the OpenAI API, which offered fast, reliable performance without compromising the quality of generated reports.

Ensuring the medical accuracy of generated text also required attention. With limited EEG-specific training material, we had to fine-tune our prompt design and establish new evaluation methods to ensure the language model outputs were both clinically useful and accurate. Through consistent iteration and collaboration, we were able to overcome these challenges and deliver a stable and effective clinical tool.

8.5 Lessons Learned

This project has been a significant learning journey for our entire team. On the technical side, we gained hands-on experience working with advanced deep learning architectures, such as GNNs, BLSTM, and CNNs. We also became proficient in EEG data preprocessing techniques, including filtering, resampling, normalization, and Independent Component Analysis (ICA).

Collaborating with medical professionals gave us insight into how clinical problems can be translated into AI-based solutions. This experience taught us the importance of interdisciplinary communication, project management, and the ability to explain complex technical ideas in a way that's understandable to non-technical stakeholders. We also improved our skills in documentation, team coordination, and agile development. These experiences have equipped us to contribute meaningfully to future projects that combine AI with healthcare.

8.6 Future Directions

Looking ahead, there are several exciting directions that could enhance *EpilepTech* even further. One area of potential development is expanding the classification system to distinguish between different types of epileptic seizures, such as focal versus generalized. Another important feature would be incorporating seizure prediction models, which could turn the platform from a diagnostic tool into a proactive intervention system.

To improve generalizability and minimize data bias, we also recommend expanding the dataset by collaborating with hospitals and research institutions across different regions. Implementing federated learning would allow such collaborations while protecting patient privacy.

Additionally, improving the language model's medical vocabulary by training it on domain-specific neurological literature could further enhance the quality and precision of automatically generated clinical reports. These future steps would help *EpilepTech* remain a cutting-edge tool in both medical practice and research, with long-term potential to shape the future of AI in neurology.

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APPENDIX

10.1 Appendix: response to comments

The table below should contain all suggestions and comments from reviewers and responses to each comment.

Department	Computer Engineering Department	Term and Year	First Semester 2024
Course Title	Project Proposal	Catalog Code	ARTI 511
Group No.	ARTI 10F7	Class Number	4007
Names of students and IDs	Name	ID	
	Sana Majed Araj	2210009067	
	Deem Jameel Alrashidi	2210002913	
	Shahad Adel Alsharedh	2210002856	
	Sarah Thaer	2200006457	
	Sahar Ali	2210002718	
Name of Supervisor	Dr. Mustafa Youldash		
Title of Project	From Binary to Trinary: Enhancing Seizure Classification with AI for Epileptic, Non-Epileptic, and Psychogenic Detection		

1- Comment	Comment was by
Formatting for pages 8 and 9	Dr. Mehwash Farooqui
Response	
Format Updated.	

2- Comment	Comment was by
Repeated Content (Header)	Dr. Mehwash Farooqui
Response	
Delete Repeated content	

3- Comment	Comment was by
Blank Page	Dr. Mehwash Farooqui
Response	
Blank Page Removed	

Posters Submitted



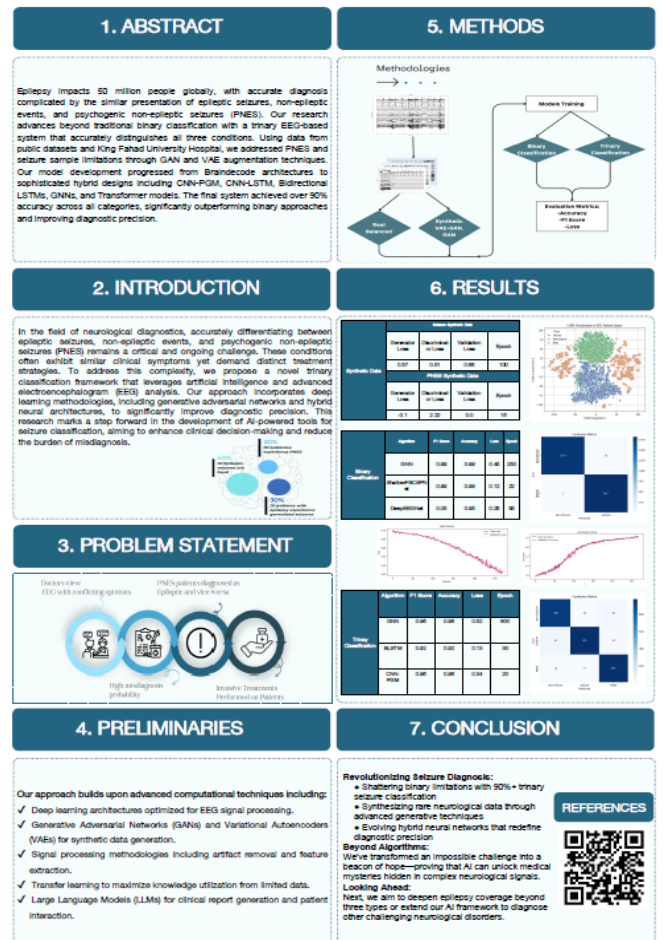
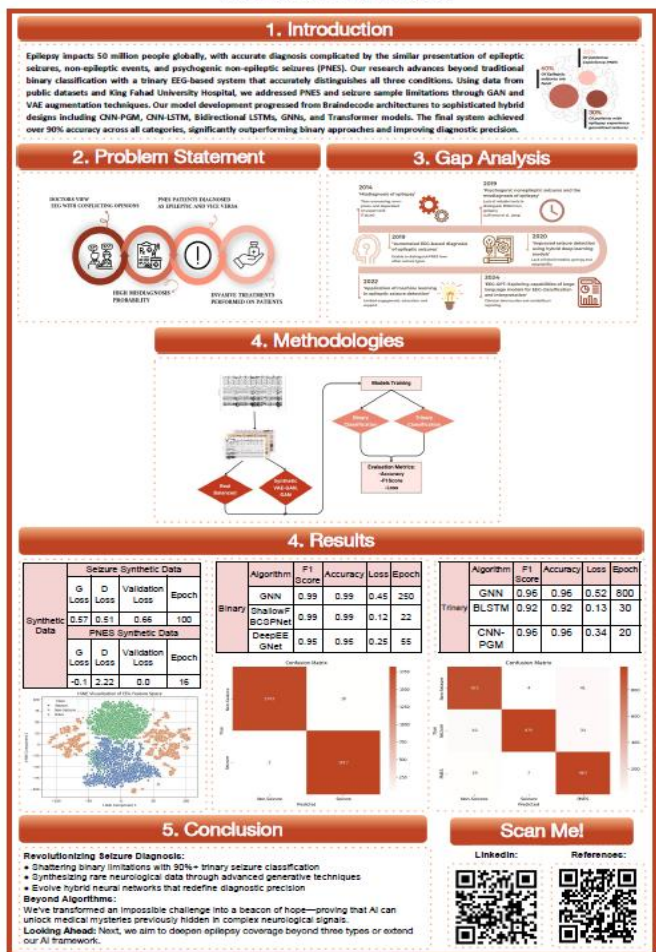
From Binary to Trinary: Enhancing Seizure Classification with AI for Epileptic, Non-Epileptic, and Psychogenic Detection

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From Binary to Trinary: Enhancing Seizure Classification with AI for Epileptic, Non-Epileptic, and Psychogenic Detection

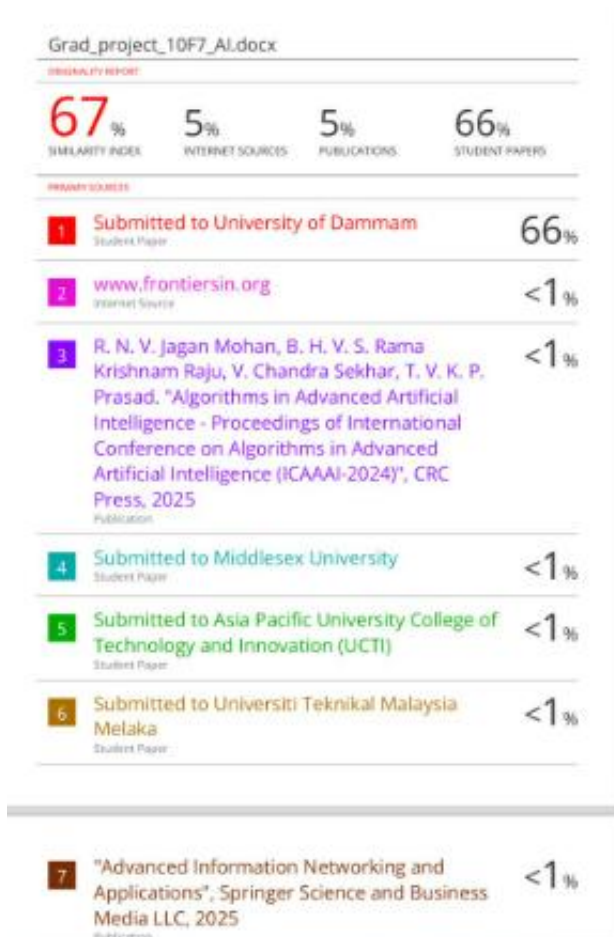
Sana Majed Araj¹, Deem Jameel Alrashidi², Shahad Adel Alsharad³, Sarah Thaeer Abuhashash⁴, Sahar Ali Alshehri⁵
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APPENDIX-B

10.2 Appendix: plagiarism report

The figure below illustrates the plagiarism report.



Note : since we used the same link it highlights chapters 1 to 5 as submitted previously.